appliedbiosystems

RapidHIT™ ID System v1.3.3

USER GUIDE

for use with:

RapidLINK™ Software v1.3.3

Publication Number MAN0025746

Revision C.0



For Forensic, Human Identification, or Paternity/Kinship Use Only. Not for use in diagnostic or therapeutic applications.

ThermoFisher SCIENTIFIC

Life Technologies Corporation | 6055 Sunol Blvd | Pleasanton, California 94566 USA

Revision history: Pub. No. MAN0025746

Revision	Date	Description
C.0	18 August 2022	Updated for RapidHIT™ ID System v1.3.3: • Added images of the sample cartridge ("Sample cartridge components" on page 10). • Added a procedure for selecting a specimen category ("(If needed) Select a specimen category" on page 27). • Updated images of the settings screen throughout the user guide. The **Run Name On Lock Screen** checkbox was removed. • Clarified the backup and restore functions ("Back up run data for all runs (administrator only)" on page 36 and "Restore run data (administrator only)" on page 57). • For the administrator user role, added the ability to edit a user account, which includes password reset ("Edit a user account (includes password reset)" on page 41). • Clarified routine maintenance tasks ("Routine maintenance" on page 45). • Added information about what to do if an allelic ladder run fails ("Insert the prepared primary cartridge" on page 54). • Noted that the recovery function can be used to remove a cartridge ("Recover the instrument (administrator or supervisor only)" on page 57). • Added Appendix F, "Software validation (RapidHIT™ ID System v1.3.3)".
B.0	4 May 2022	Added Appendix D, "System modification study (RapidHIT™ ID System v1.3.2)" and Appendix E, "Software validation (RapidHIT™ ID System v1.3.2)". Added the United Kingdom Conformity Assessment mark to "Symbols on this instrument" on page 118.
A.0	21 September 2021	New document for RapidHIT™ ID System v1.3.2.

The information in this guide is subject to change without notice.

DISCLAIMER: TO THE EXTENT ALLOWED BY LAW, THERMO FISHER SCIENTIFIC INC. AND/OR ITS AFFILIATE(S) WILL NOT BE LIABLE FOR SPECIAL, INCIDENTAL, INDIRECT, PUNITIVE, MULTIPLE, OR CONSEQUENTIAL DAMAGES IN CONNECTION WITH OR ARISING FROM THIS DOCUMENT, INCLUDING YOUR USE OF IT.

Important Licensing Information: These products may be covered by one or more Limited Use Label Licenses. By use of these products, you accept the terms and conditions of all applicable Limited Use Label Licenses.

TRADEMARKS: GeneMarker is a trademark of SoftGenetics LLC. Puritan is a trademark of Puritan Medical Products. All trademarks are the property of Thermo Fisher Scientific and its subsidiaries unless otherwise specified.

©2022 Thermo Fisher Scientific Inc. All rights reserved.

Contents

CHAPTER 1 Product information 7

Product description 8
Sample cartridge types 9
Sample cartridge components 10
Parts of the instrument 11
Required materials not supplied 12
Primary cartridge kits 12
Sample cartridge kits 14
Accessories 15
(Optional) Control and utility cartridges 15
Network and password security requirements 16
Network configuration and security 16
Password security 16
Workflow 17

CHAPTER 2 Collect and prepare the sample 19

Sample collection: GlobalFiler™ Express and NGM SElect™ Express sample cartridges . 19
Sample collection: RapidINTEL™ sample cartridges 19

CHAPTER 3 Run the instrument to generate a DNA profile 20

Power on the internal computer 21
Sign in to the instrument 22
Menu screen 25
Enter the sample identification number 26
Scan a barcode with the camera 26
Enter a sample ID or barcode 26
(If needed) Select a specimen category 27
Insert the sample cartridge into the instrument (automatically starts the run) 28
Run times for different sample cartridges 29
Remove the sample cartridge from the instrument 30
Check the run result 31
View the run results 33
Sign out 34

CHAPTER 4 Manage the instrument and software 35

View the run results for all runs (administrator only) 35
Back up run data for all runs (administrator only) 36
Configure instrument settings (administrator or supervisor only) 36
Manage users (administrator only) 38
User roles and permissions 38
Add a user account 38
Edit a user account (includes password reset) 41
Remove authorization for an instrument 44

CHAPTER 5 Maintain the instrument 45

Routine maintenance 45
View primary cartridge information 47
Replace the primary cartridge (administrator only) 48
Parts of the primary cartridge 48
Prepare a new primary cartridge 49
Insert the prepared primary cartridge 54
Run a control cartridge 56
Restore run data (administrator only) 57
Recover the instrument (administrator or supervisor only) 57
Prepare the instrument for shipping 57

APPENDIX A Troubleshooting 58

Check the RapidLINK™ Software connection 58
Troubleshooting: Symptoms, causes, and actions 58
Error codes 61
Perform the SCCL recovery procedure 63

APPENDIX B Analysis settings 66

Threshold settings 66
Display analysis settings in the GeneMarker™ HID STR Human Identity Software 68
Analysis settings: GlobalFiler™ Express sample cartridges 69
Analysis settings: RapidINTEL™ sample cartridges 70
Analysis settings: NGM SElect™ *Express* sample cartridges 72

APPENDIX C Instrument specifications 74

Dimensions, clearance, and weight 74
Instrument layout and connections 75
Environmental requirements 77
Electrical requirements 78

APPENDIX D System modification study (RapidHIT™ ID System v1.3.2) 80

About the system modification studies 80
Gel polymer priming study 80
Primary cartridge stability testing 83
Systematic Allelic Ladder Library 85
Automated maintenance check 91
Conclusion 92

APPENDIX E Software validation (RapidHIT™ ID System v1.3.2) 93

Software description 93
Materials and methods 93
Studies, standards, and samples 94
Functionality testing results 95
Regression testing results 101
Accuracy testing results 101
Reliability testing results 105
Conclusion 106

APPENDIX F Software validation (RapidHIT™ ID System v1.3.3) 107

Software description 107
Cartridge type abbreviations 108
Validation overview 108
Functionality testing results 110
Regression testing results 114
Accuracy testing results 115
Reliability testing results 116
Amelogenin verification 117
Conclusion 117

APPENDIX G Safety 118

Symbols on this instrument 118
Standard safety symbols 118
Location of safety labels 119
Control and connection symbols 119
Conformity symbols 120
Safety information for instruments not manufactured by Thermo Fisher Scientific 121
Instrument safety 121
General 121
Physical injury 121
Electrical safety 121
Cleaning and decontamination 122
Instrument component and accessory disposal 122
Laser safety 122
Safety and electromagnetic compatibility (EMC) standards 123
Safety standards 123
EMC standards 123
Environmental design standards 124
Radio compliance standards 124
Chemical safety 126
Biological hazard safety 127

APPENDIX H Documentation and support 128

Related documentation 128
Customer and technical support 128
Limited product warranty 129

1 Product information

- Product description 8
- Sample cartridge types 9
- Parts of the instrument 11
- Required materials not supplied 12
- Network and password security requirements 16
- Workflow 17

IMPORTANT! Before using this product, read and understand the information in the “Safety” appendix in this document.

Product description

Applied Biosystems™ RapidHIT™ ID System v1.3.3 uses the following components.



- **RapidHIT™ ID sample cartridges**—Self-contained sample cartridges contain reagents and size standard for Applied Biosystems™ STR (short tandem repeat) chemistry.

  The sample is added to the cartridge, then the sample cartridge is inserted into the instrument.

- **RapidHIT™ ID v1.3.3 instrument**—The instrument provides an intuitive user interface that guides users through routine use.

  The instrument generates a DNA profile in 90–110 minutes.

- **RapidLINK™ Software v1.3.3**—When an instrument run is complete, the DNA profile is sent to the RapidLINK™ Software. In the RapidLINK™ Software, users can perform the following functions:
  - Review DNA profiles, view electropherograms, and update DNA profiles.
  - Monitor all instruments that contribute DNA profiles to the RapidLINK™ Software.
  - *(Optional)* Activate one or more of the RapidLINK™ applications (apps):
    - **Matched Runs**—Calculates the likelihood that two DNA profiles in the Match database are from the same individual.
    - **Familial Search**—Searches a selected DNA profile for all likely familial matches in the Match database.
    - **Kinship**—Calculates the likelihood that two DNA profiles in the Match database are related.
    - **Staff Elimination Database (SED)**—Compares DNA profiles to the SED. The SED is a repository for DNA profiles of employees who handle the DNA samples.

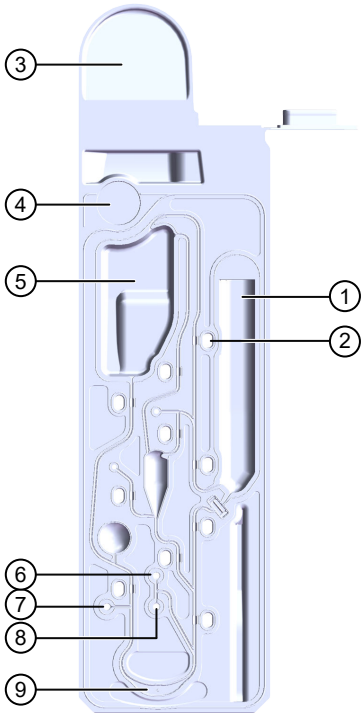
For information on RapidLINK™ Software v1.3.3, see the *RapidLINK™ Software v1.3.3 User Guide* (Pub. No. MAN0018939).

Sample cartridge types

Abbreviation on labels / in software	Sample type	STR chemistry	Features	For more information
RapidHIT™ ID ACE GlobalFiler™ Express Sample Cartridge				
ACE GFE / GFE	Single-source, buccal swab	GlobalFiler™ Express	• Systematic allelic ladder library • Optimized data analysis thresholds (based on v1.3.1)	• Appendix D, "System modification study (RapidHIT™ ID System v1.3.2)" • Appendix E, "Software validation (RapidHIT™ ID System v1.3.2)" • *Applied Biosystems RapidHIT™ ID System v1.3 with RapidHIT™ ID ACE GlobalFiler™ Express Sample Cartridges Validation Technical Note* (see "Related documentation" on page 128)
RapidINTEL™ Sample Cartridge				
AB RI	Casework, single-source, blood and saliva samples	GlobalFiler™ Express	Increased sensitivity for investigative leads samples	• Appendix D, "System modification study (RapidHIT™ ID System v1.3.2)" • Appendix E, "Software validation (RapidHIT™ ID System v1.3.2)" • *RapidINTEL™ Sample Cartridge for Blood and Saliva Samples Validation User Bulletin* (Pub. No. MAN0018979)
RapidHIT™ ID ACE NGM SElect™ *Express* Sample Cartridge				
ACE NGM / NGM	Single-source, buccal swab	NGM SElect™ Express	Systematic allelic ladder library	• Appendix D, "System modification study (RapidHIT™ ID System v1.3.2)" • Appendix E, "Software validation (RapidHIT™ ID System v1.3.2)" • *RapidHIT™ ID System v1.3.1 User Guide* (Pub. No. MAN0018938)

Sample cartridge components

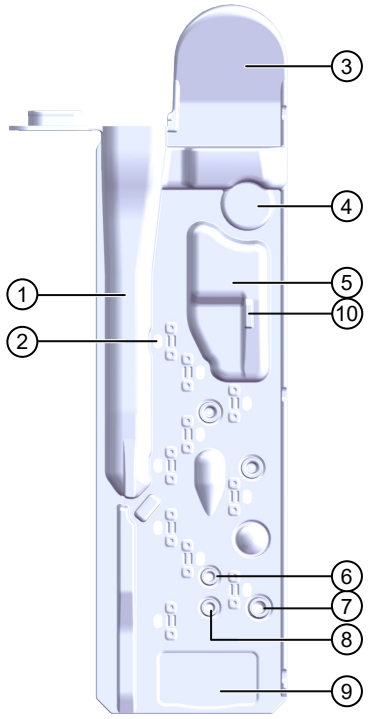
The sample cartridges include the following components.



1. Sample swab chamber
2. Crosshatch cuts in the cartridge film (× 10)
3. Cartridge label
4. RFID tag
5. Waste sponge
6. Pre-mix (clear)
7. Size standard
8. Primer mix (pink)
9. FTA paper

Note: The insertion flag is not visible in this view.

Figure 1 Sample cartridge—front view



1. Sample swab chamber
2. Crosshatch cuts in the cartridge film (× 10)
3. Cartridge label
4. RFID tag
5. Waste sponge
6. Pre-mix (clear)
7. Size standard
8. Primer mix (pink)
9. FTA paper
10. Insertion flag (contains an opaque silver coating on the tab)

Figure 2 Sample cartridge—back view

Parts of the instrument



1. Camera
2. Sample cartridge port
3. Touchscreen (display screen)
4. Fingerprint reader
5. Primary cartridge (contains gel cartridge)
6. Power button (controls the internal computer)
7. USB port
8. Rear panel with main power switch, USB, and computer connections

The instrument also includes an internal environmental sensor that monitors temperature and humidity.

Do not lean on any part of the instrument. This action could cause damage or breakage, especially to the display screen.

Required materials not supplied

Unless otherwise indicated, all materials are available through **thermofisher.com**. "MLS" indicates that the material is available from **fisherscientific.com** or another major laboratory supplier.

Catalog numbers that appear as links open the web pages for those products.

Primary cartridge kits

Note: For all primary cartridge kits, the allelic ladder control, positive control, and utility cartridges can be ordered separately through your local Thermo Fisher Scientific sales representative. See "(Optional) Control and utility cartridges" on page 15.

Table 1 RapidHIT™ ID Primary Cartridge GlobalFiler™ Express 150 Kit (Cat. No. A53084)

Contents	Amount	Storage
Box 1		
RapidHIT™ ID Primary Cartridge—150 runs	1 primary cartridge for 150 runs	Room temperature (15–30°C)
RapidHIT™ ID Utility Cartridge (for installation or primary cartridge replacement)	2 utility cartridges	
Box 2		
RapidHIT™ ID Gel Cartridge (for installation or primary cartridge replacement)	1 gel cartridge	2–8°C
RapidHIT™ ID ACE GlobalFiler™ Express Control Cartridge (for installation or primary cartridge replacement)	1 allelic ladder control cartridge	
RapidHIT™ ID ACE GlobalFiler™ Express Positive Control Cartridge (for installation or primary cartridge replacement)	1 positive control cartridge	

Table 2 RapidHIT™ ID Primary Cartridge GlobalFiler™ Express 100 Kit (Cat. No. A41841)

Contents	Amount	Storage
Box 1		
RapidHIT™ ID Primary Cartridge—100 runs	1 primary cartridge for 100 runs	Room temperature (15–30°C)
RapidHIT™ ID Utility Cartridge (for installation or primary cartridge replacement)	2 utility cartridges	
Box 2		
RapidHIT™ ID Gel Cartridge (for installation or primary cartridge replacement)	1 gel cartridge	2–8°C
RapidHIT™ ID ACE GlobalFiler™ Express Control Cartridge (for installation or primary cartridge replacement)	1 allelic ladder control cartridge	
RapidHIT™ ID ACE GlobalFiler™ Express Positive Control Cartridge (for installation or primary cartridge replacement)	1 positive control cartridge	

Table 3 RapidHIT™ ID Primary Cartridge NGM SElect™ 150 Kit (Cat. No. A53085)

Contents	Amount	Storage
Box 1		
RapidHIT™ ID Primary Cartridge—150 runs	1 primary cartridge for 150 runs	Room temperature (15–30°C)
RapidHIT™ ID Utility Cartridge (for installation or primary cartridge replacement)	2 utility cartridges	
Box 2		
RapidHIT™ ID Gel Cartridge (for installation or primary cartridge replacement)	1 gel cartridge	2–8°C
RapidHIT™ ID ACE NGM SElect™ Control Cartridge (for installation or primary cartridge replacement)	1 allelic ladder control cartridge	
NGM SElect™ Positive Control Cartridge (for installation or primary cartridge replacement)	1 positive control cartridge	

Table 4 RapidHIT™ ID Primary Cartridge NGM SElect™ 100 Kit (Cat. No. A41847)

Contents	Amount	Storage
Box 1		
RapidHIT™ ID Primary Cartridge—100 runs	1 primary cartridge for 100 runs	Room temperature (15–30°C)
RapidHIT™ ID Utility Cartridge (for installation or primary cartridge replacement)	2 utility cartridges	
Box 2		
RapidHIT™ ID Gel Cartridge (for installation or primary cartridge replacement)	1 gel cartridge	2–8°C
RapidHIT™ ID ACE NGM SElect™ Control Cartridge (for installation or primary cartridge replacement)	1 allelic ladder control cartridge	
NGM SElect™ Positive Control Cartridge (for installation or primary cartridge replacement)	1 positive control cartridge	

Sample cartridge kits

Table 5 Sample cartridge kits

Kit	Cat. No.	Contents	Storage
RapidHIT™ ID ACE GlobalFiler™ Express 50 Sample Kit	A41831	• 50 sample cartridges • 2 positive control cartridges • 2 negative control cartridges	15–25°C for ≤2 months, or 4–10°C until the expiration date on the label
RapidHIT™ ID ACE NGM SElect™ *Express* 50 Sample Cartridge Kit	A41838	• 50 sample cartridges • 2 positive control cartridges • 2 negative control cartridges	
RapidINTEL™ Sample Cartridge Kit	A43942	• 50 sample cartridges • 2 positive control cartridges • 2 negative control cartridges	

Accessories

Table 6 Accessories

Component	Cat. No.
(Optional) Barcode scanner compatible with one of the following formats: GS1-128, Industrial 2 of 5, Interleaved 2 of 5, Code 128, Code 39 **Note:** The instrument must be configured by a service representative to enable barcode scanning. If you want to add a barcode scanner to the instrument after initial installation, contact Thermo Fisher Scientific to request an update to your configuration.	MLS
Puritan™ 3″ Sterile Standard Cotton Swab with Semi-Flexible Polystyrene Handle	25-803 2PC

(Optional) Control and utility cartridges

Table 7 GlobalFiler™ Express control and utility cartridges

Contents	Cat. No.	Amount	Storage
RapidHIT™ ID ACE GlobalFiler™ Express Positive Control Cartridge	A42469	1 positive control cartridge	2–8°C
RapidHIT™ ID ACE GlobalFiler™ Express Negative Control Cartridge	A42470	1 negative control cartridge	
RapidHIT™ ID ACE GlobalFiler™ Express Control Cartridge	400096RH	5 allelic ladder control cartridges	
RapidHIT™ ID Utility Cartridge	A42471	1 utility cartridge	Room temperature (15–30°C)

Table 8 NGM SElect™ control and utility cartridges

Contents	Cat. No.	Amount	Storage
NGM SElect™ Positive Control Cartridge	A43582	1 positive control cartridge	2–8°C
RapidHIT™ ID ACE NGM SElect™ Negative Control Cartridge	A43583	1 negative control cartridge	
RapidHIT™ ID ACE NGM SElect™ Control Cartridge	400097RH	5 allelic ladder control cartridges	
RapidHIT™ ID Utility Cartridge	A42471	1 utility cartridge	Room temperature (15–30°C)

Network and password security requirements

Network configuration and security

The network configuration and security settings of your laboratory or facility (such as firewalls, anti-virus software, network passwords) are the sole responsibility of your facility administrator, IT, and security personnel. This product does not provide any network or security configuration files, utilities, or instructions.

If external or network drives are connected to the software, it is the responsibility of your IT personnel to ensure that such drives are configured and secured correctly to prevent data corruption or loss. It is the responsibility of your facility administrator, IT, and security personnel to prevent the use of any unsecured ports (such as USB, Ethernet) and ensure that the system security is maintained.

Password security

Thermo Fisher Scientific strongly recommends that you maintain unique passwords for all accounts in use on this product. All passwords should be reset upon first sign in to the product. Change passwords according to your organization's password policy.

It is the sole responsibility of your IT personnel to develop and enforce secure use of passwords.

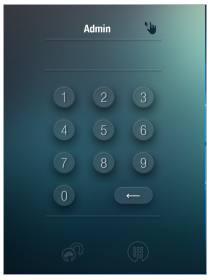
Workflow

Obtain a swab or substrate sample according to your standard operating procedure.

Insert the swab or substrate sample into a sample cartridge.

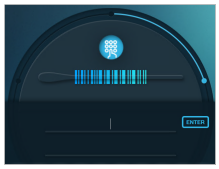


Sign in.



Scan the sample cartridge barcode or enter the sample identification number.

IMPORTANT! Do not include the following in sample IDs: "ladder", "posctrl", or "negctrl" (case does not matter). The software processes any sample IDs that contain these names as allelic ladders or controls.

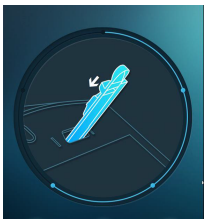


(If needed) Select a specimen category

If the RapidLINK™ server connected to the instrument is configured for integration with an external database, you will be prompted to select a specimen category.

Insert the sample cartridge into the instrument.

The run starts automatically.



The countdown timer starts at 110 minutes. The typical run time is ~90 minutes.

Note: The timer stops when the run is complete. The timer does not decrease to 0 minutes. For more information, see “Run times for different sample cartridges” on page 29.



Remove the sample cartridge from the instrument, then view the result.



Review results and run applications in the RapidLINK™ Software.

RapidLINK™

2 Collect and prepare the sample

Sample collection: GlobalFiler™ Express and NGM SElect™ Express sample cartridges

Sample cartridges were validated using the following swab types, but other swab types have been used successfully with both cartridges.

- **GlobalFiler™ Express**—Puritan 3" Sterile Standard Cotton Swab w/Semi-Flexible Polystyrene Handle (Puritan Cat. No. 25-8032 PC)
- **NGM SElect™ Express**—Whatman™ OmniSwab, (Fisher Scientific™ Cat. No. 09-923-376 [WB100035])

IMPORTANT! When you handle buccal swabs, follow your standard operating protocol for the use of safety glasses, lab coats, and chemical-resistant, disposable gloves (powder-free).

Sample collection: RapidINTEL™ sample cartridges

Blood and saliva samples have been validated for use with RapidINTEL™ sample cartridges. Samples were processed for validation as described in *RapidINTEL™ Sample Cartridge for Blood and Saliva Samples Validation User Bulletin* (Pub. No. MAN0018979).

IMPORTANT! When you handle samples, follow your standard operating protocol for the use of safety glasses, lab coats, and chemical-resistant, disposable gloves (powder-free).

3 Run the instrument to generate a DNA profile

- Power on the internal computer 21
- Sign in to the instrument 22
- Enter the sample identification number 26
- (If needed) Select a specimen category 27
- Insert the sample cartridge into the instrument (automatically starts the run) 28
- Run times for different sample cartridges 29
- Remove the sample cartridge from the instrument 30
- Check the run result 31
- View the run results 33
- Sign out 34

Power on the internal computer

IMPORTANT! Leave the main power switch (on the back panel) set to On at all times.

Power is necessary to maintain the gel temperature.

Press the power button on the front lower-right of the instrument.

The power button changes from green to blue.

The startup screen is displayed until the instrument finishes the following operations.

- System check
- System prime, if needed (≤15 minutes)

The lock screen is displayed after system startup is complete.

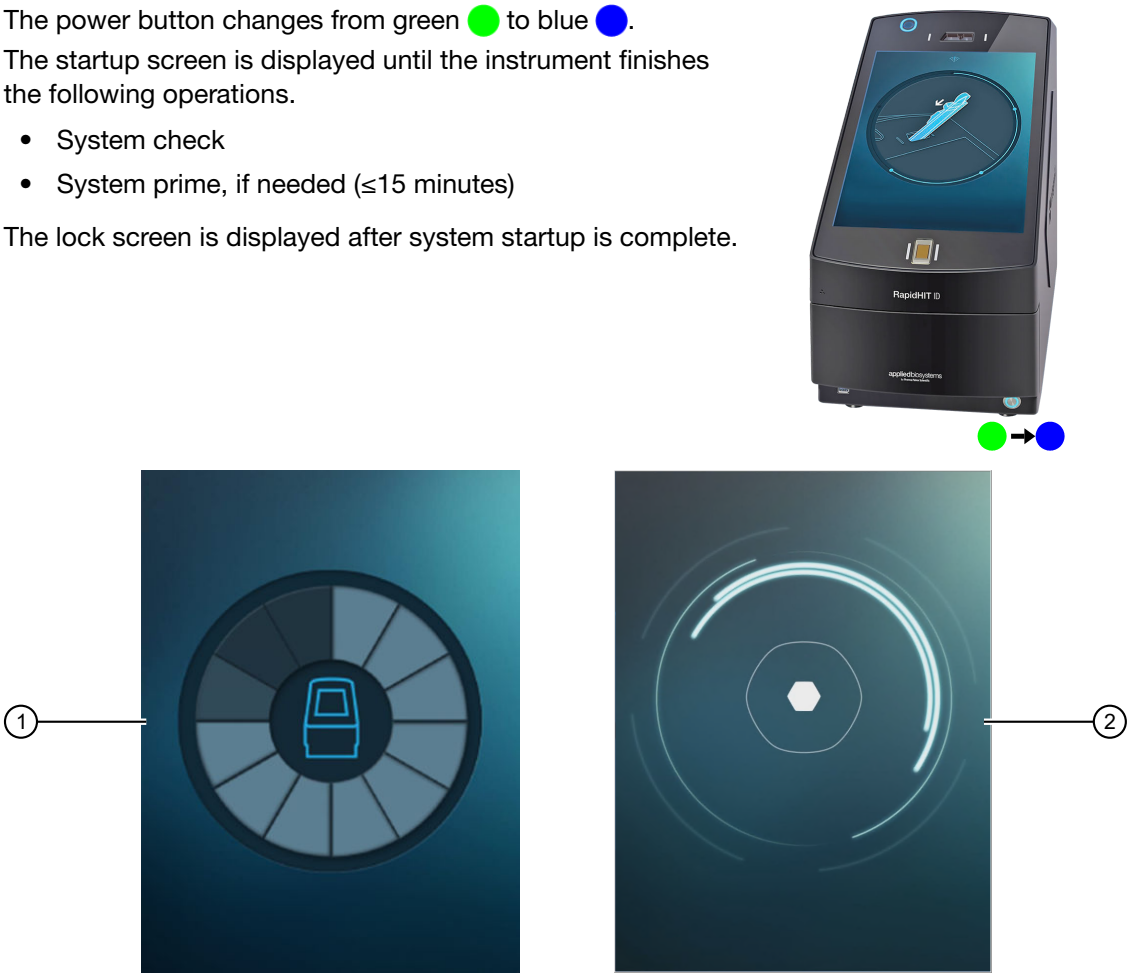


Figure 3 Startup and lock screens

1. Startup screen
2. Lock screen

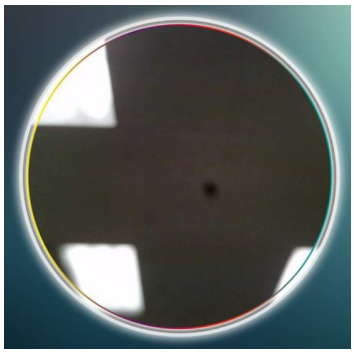
Sign in to the instrument

Sign in using any of the authentication methods that are configured for your profile.

- **Fingerprint**—Touch the fingerprint reader.



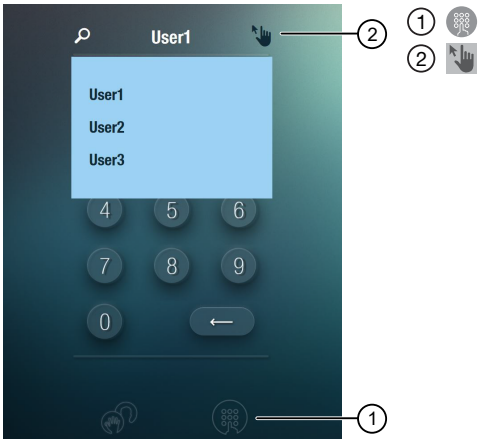
- **Facial recognition**—Position your face in front of the camera.



- **PIN**—Perform the following steps.
  a. Tap the lock screen.



  b. Tap , tap , then select your user name.



Alternatively, you can search for your user name: Tap , enter all or part of your user name with the keypad, then tap **ENTER**. A list of matching names is displayed.



1.
2. Enter all or part of your user name with the keypad, then tap **ENTER**

c. Enter your PIN with the keypad.

The sample identification screen is displayed (see Figure 4).

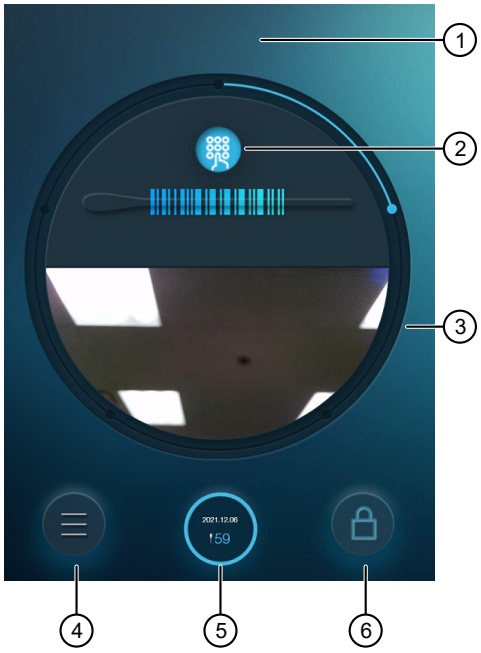


Figure 4 Parts of the sample identification screen

① No icon is displayed if the instrument is connected to the RapidLINK™ Software.
Indicates that the instrument is not connected to the RapidLINK™ Software
See "Check the RapidLINK™ Software connection" on page 58.

② Tap to display the keypad to enter a sample identification number.

③ Camera.

④ Tap to display the menu screen. See "Menu screen" on page 25.

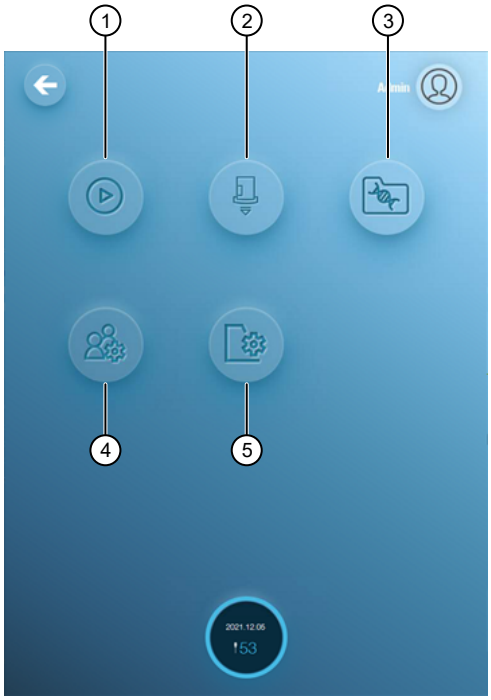
⑤ Displays the expiration date and run count for the primary (reagent) cartridge. The run count indicates the number of runs that have been performed, not the number of runs that are remaining (for example, "65" indicates that 65 runs have been performed). For more information, see "View primary cartridge information" on page 47.

⑥ Tap to sign out and lock the touchscreen (displays the lock screen).

Note: If an error code is displayed when you sign in, see page 59.

Proceed to "Enter the sample identification number" on page 26.

Menu screen



Icon	Item	User role	Description
1	**Rapid DNA Review Video**	All user roles	View a quick video on how to collect a buccal swab sample and run the sample on the instrument.
2	**Eject/Install Primary Cartridge**	Administrator and supervisor	Install a new primary cartridge.
3	**DNA Run Folder**	Operator	View all runs that you have performed on the instrument.
		Administrator and supervisor	View all runs that all users have performed on the instrument.
4	**Manage Users**	Supervisor and operator	Edit your user information.
		Administrator	Add or edit user accounts (includes password reset).
5	**RapidHIT ID Settings**	Supervisor and operator	View the instrument settings.
		Administrator	View or edit the instrument settings.

Enter the sample identification number

Scan a barcode with the camera

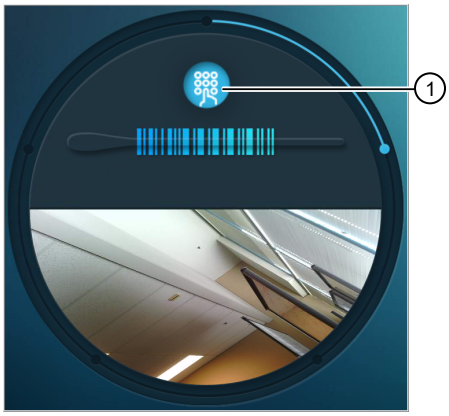
Note: You can also use a compatible handheld barcode scanner. For information on the compatible scanner, see “Required materials not supplied” on page 12.

1. If the sample identification screen is not displayed, tap ← until it is displayed.
2. Hold the barcode in front of the camera at the top left of the instrument.
   When the camera reads the barcode, the sample identification number is displayed.



Enter a sample ID or barcode

1. If the sample identification screen is not displayed, tap ← until it is displayed.
2. Tap the sample identification screen.
3. Tap ⊛ to display the keypad.



① Keypad

4. Enter the sample ID or barcode, then tap **Enter**.

   Note: If you enter the sample ID, the software allows you to include spaces and the following special characters: /\?%*:|'.<>. However, the software changes the special characters to underscore(_) when it creates the data file for the sample. The sample ID with underscores is displayed in the RapidLINK™ Software.

   Note: To create a valid CMF file according to the CODIS CMF file interface specifications, limit the sample ID to 9 characters. The specifications are published by the Federal Bureau of Investigation (FBI).

   IMPORTANT! Do not include the following in sample IDs: "ladder", "posctrl", or "negctrl" (case does not matter). The software processes any sample IDs that contain these names as allelic ladders or controls.

   The sample identification number is displayed.



(If needed) Select a specimen category

If the RapidLINK™ server connected to the instrument is configured for integration with an external database, you will be prompted to select a specimen category for all sample cartridges.

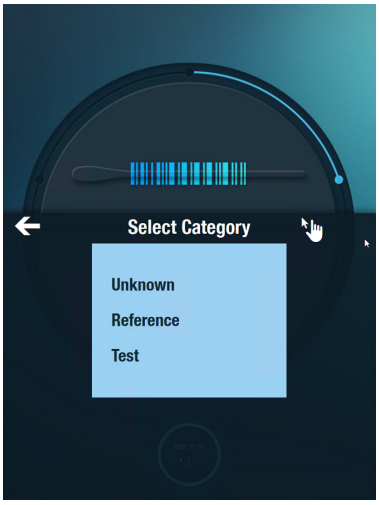
Note: Specimen categories are not required for allelic ladder, negative control, positive control, or utility cartridges.

Note: Specimen categories are managed in the RapidLINK™ Software. The same categories are used for all instruments that are connected to the same RapidLINK™ network. For more information, see the *RapidLINK™ Software v1.3.3 User Guide* (Pub. No. MAN0018939).

1. At the **Select Category** prompt, select a specimen category from the dropdown list. The list is sorted by the date and time that the category was used—the most recently used is at the top.

   An example of the **Select Category** dropdown list is shown below; the specimen categories for your laboratory may be different.

   Note: If your laboratory includes only one specimen category, the category is automatically selected.



2. Tap **Enter**.

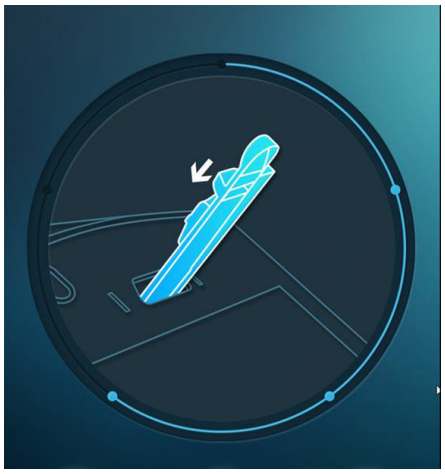
Insert the sample cartridge into the instrument (automatically starts the run)

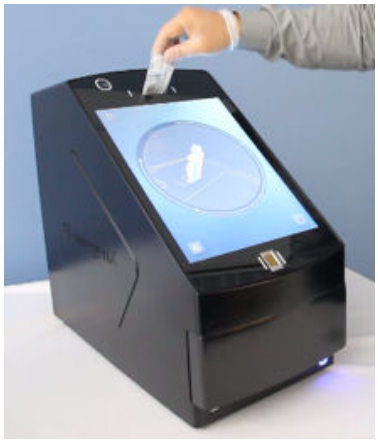
A sample cartridge run takes ~90–110 minutes to complete.

1. Insert the sample swab into the sample cartridge.



2. When the insert sample cartridge screen is displayed, insert the sample cartridge (containing the sample swab) into the sample cartridge port. See the following figures.





The run starts automatically after you insert the sample cartridge. For more information, see “Run times for different sample cartridges” on page 29.

IMPORTANT! Do not remove the sample cartridge until you are prompted to do so (see “Remove the sample cartridge from the instrument” on page 30).

When the run is complete, the remove sample cartridge screen is displayed for ~30 seconds, then the currently signed-in user is automatically signed out.

Run times for different sample cartridges

- The countdown timer starts at 110 minutes. The typical run time is ~90 minutes.
- The timer stops when the run is complete. The timer does not decrease to 0 minutes.

Cartridge	Run timer stops at
RapidHIT™ ID ACE GlobalFiler™ Express Sample Cartridge	~19 minutes
RapidHIT™ ID ACE NGM SElect™ *Express* Sample Cartridge	~19 minutes
RapidINTEL™ Sample Cartridge	~14 minutes

- If the instrument requires priming, the run time can extend up to 110 minutes.

The figure below shows a countdown timer example for an NGM SElect™ Express sample cartridge.

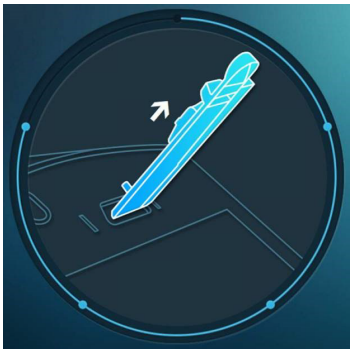


Remove the sample cartridge from the instrument

1. If the lock screen is displayed, tap the lock screen, then sign in.
2. When the remove sample cartridge screen is displayed, remove the sample cartridge from the instrument.

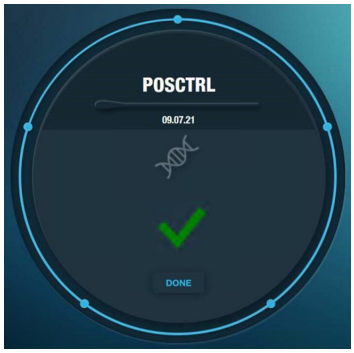
IMPORTANT! Remove the cartridge only when the remove sample cartridge screen is displayed.

If you remove the cartridge before you are prompted to do so, the results screen is not displayed. To display the results screen, insert the cartridge again, wait until you are prompted to remove the cartridge, then remove the cartridge again.



Note: If you cannot easily remove the sample cartridge, see “The sample cartridge cannot be easily removed from the instrument” on page 59.

When the cartridge is successfully removed, the result screen is displayed.

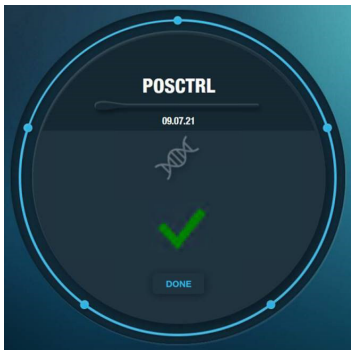


3. Discard the sample cartridge. Use proper precautions for disposal. Follow local, state, provincial and federal regulations for disposal.

Proceed immediately to "Check the run result" on page 31.

Check the run result

After you remove the sample cartridge from the instrument, the result screen is displayed.



1. Review the status.

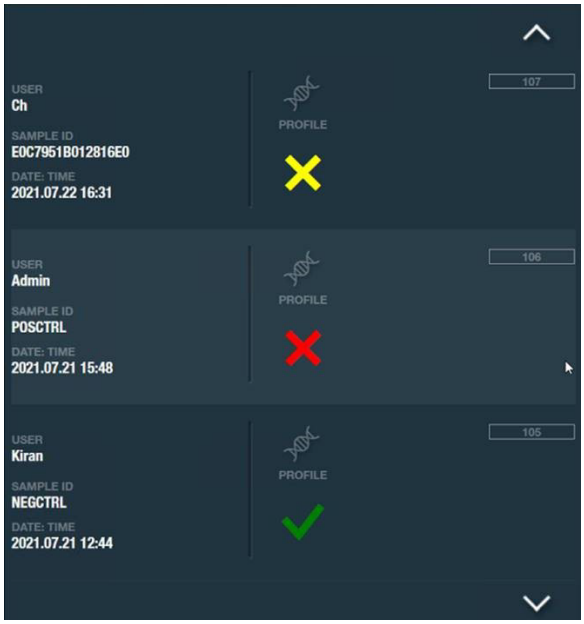
Status	DNA profile is generated	Meaning	Action
Green 	Yes	The DNA profile does not contain quality score flags.	The DNA profile is ready for analysis by the RapidLINK™ Software. No further action is needed on the instrument.
Yellow 	Yes	• The DNA profile contains quality score flags, or • The run result contains only size standard peaks. It does not contain sample peaks.	The DNA profile is available for review in the RapidLINK™ Software. Proceed according to your standard operating protocol.
Red 	No	The DNA profile was not generated.	A DNA profile is *not* available for review in the RapidLINK™ Software. Proceed according to your standard operating protocol.

2. Tap **Done** to dismiss the status.
   The instrument automatically signs out the user and displays the lock screen.

View the run results

This function lists the results for the runs performed by the signed-in user.

1. Tap **(Menu)**.
2. Tap **(Run Data)**.
   The screen displays the status from the results screen: Green, Yellow, Red.
   The following figure is an example of the run log for one user that shows each status.



Sign out

IMPORTANT! Leave the main power switch (on the back panel) set to On at all times.

Power is necessary to maintain the gel temperature.

The signed-in user is automatically signed out when a run is complete.

You can also sign out manually.

1. If the sample identification screen is not displayed, tap ← until it is displayed.
2. Tap 🔒.

The lock screen is displayed.

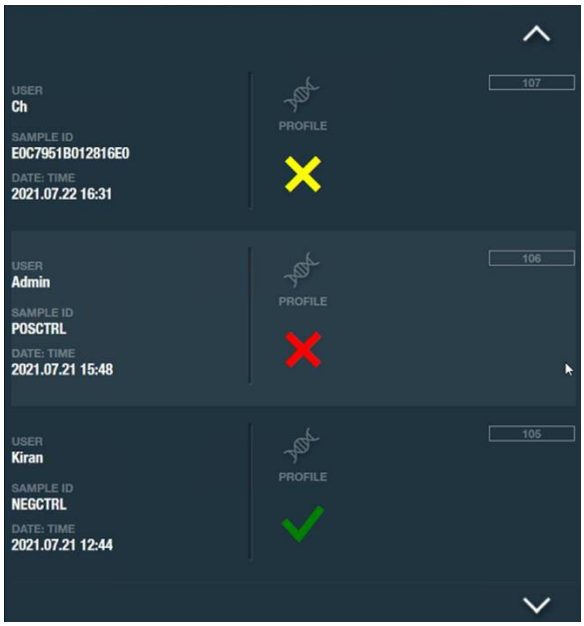


4 Manage the instrument and software

- View the run results for all runs (administrator only) 35
- Back up run data for all runs (administrator only) 36
- Configure instrument settings (administrator or supervisor only) 36
- Manage users (administrator only) 38

View the run results for all runs (administrator only)

1. Sign in as an administrator.
2. Tap **(Menu)**.
3. Tap **(Run Data)**.
   The screen displays the status from the results screen: Green, Yellow, Red.
   The following figure is an example of the run log for one user that shows each status.



Back up run data for all runs (administrator only)

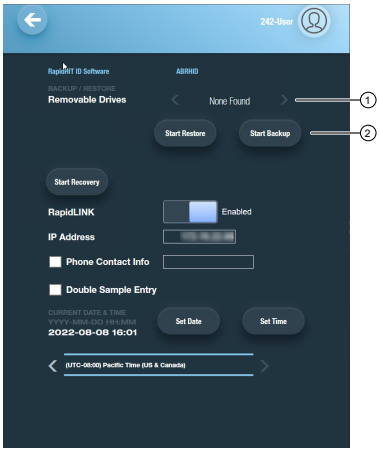
The backup function exports a copy of the run data from the instrument to a USB device. The run data is also retained on the instrument.

Note: The backup function does not copy user profiles or instrument settings.

1. Sign in as an administrator.
2. Insert a USB device into the USB port on the front of the instrument.
   The **Removable Drives** field displays the name of the USB device.
3. Tap **(Menu)**.
4. Tap **(Settings)**.
5. In the settings screen, select the USB drive, then tap **Start Backup**.

   Note: You will not be able to export run data if:
   - The RapidHIT™ ID Instrument is configured to delete the run data after it is transferred to the RapidLINK™ Software.
   - There is no USB device inserted into a USB port.

   The button changes to **Abort** and a progress bar is displayed. When the procedure is complete, a message is displayed.

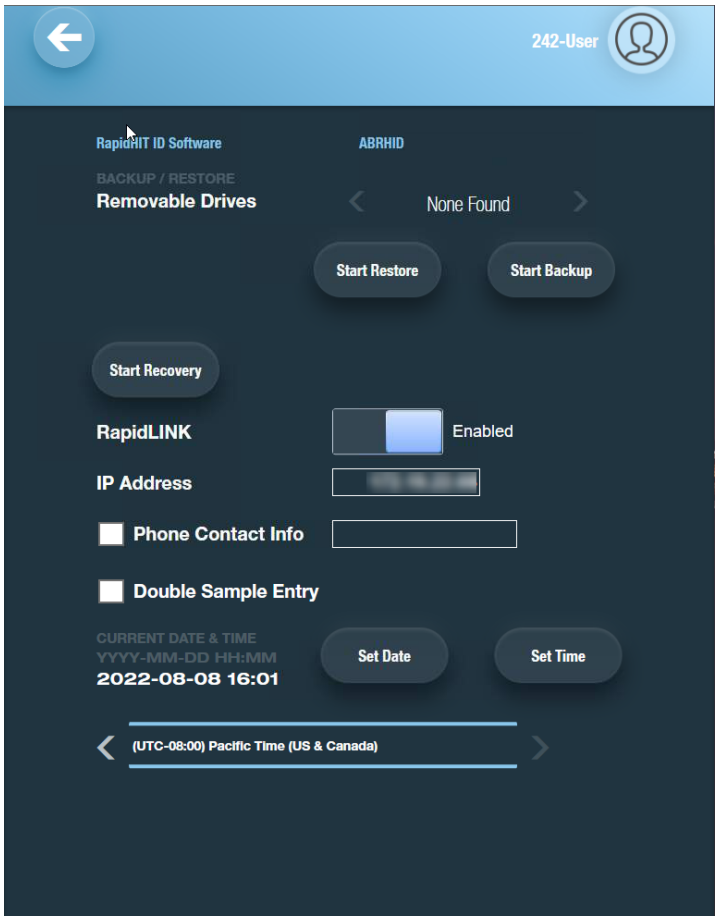


① Drive selection; if there is no USB device in the USB port, the screen displays **None Found**
② **Start Backup** button

Configure instrument settings (administrator or supervisor only)

1. Sign in as an administrator or a supervisor.
2. Tap **(Menu)**.
3. Tap **(Settings)**.

4. In the settings screen, configure the instrument settings as needed.



Setting	Description	User access
Start Backup	See “Back up run data for all runs (administrator only)” on page 36.	Administrator
Start Restore	See “Restore run data (administrator only)” on page 57.	Administrator
Start Recovery	Reset the instrument hardware. See “Recover the instrument (administrator or supervisor only)” on page 57.	Administrator or supervisor
RapidLINK	Make the instrument available for selection in the RapidLINK™ Software.	Administrator
IP Address	The IP address of the RapidLINK™ Software. The address is displayed after the RapidLINK™ Software connects to the instrument.	Read-only, can be seen by anyone
Phone Contact Info	The phone number of the primary contact for the instrument. This number is displayed in diagnostic code screens.	Administrator
Double Sample Entry	This function cannot be disabled in this version of the software.	Administrator or supervisor
Set Date and **Set Time**	Set the instrument date and time.	Administrator

Manage users (administrator only)

User roles and permissions

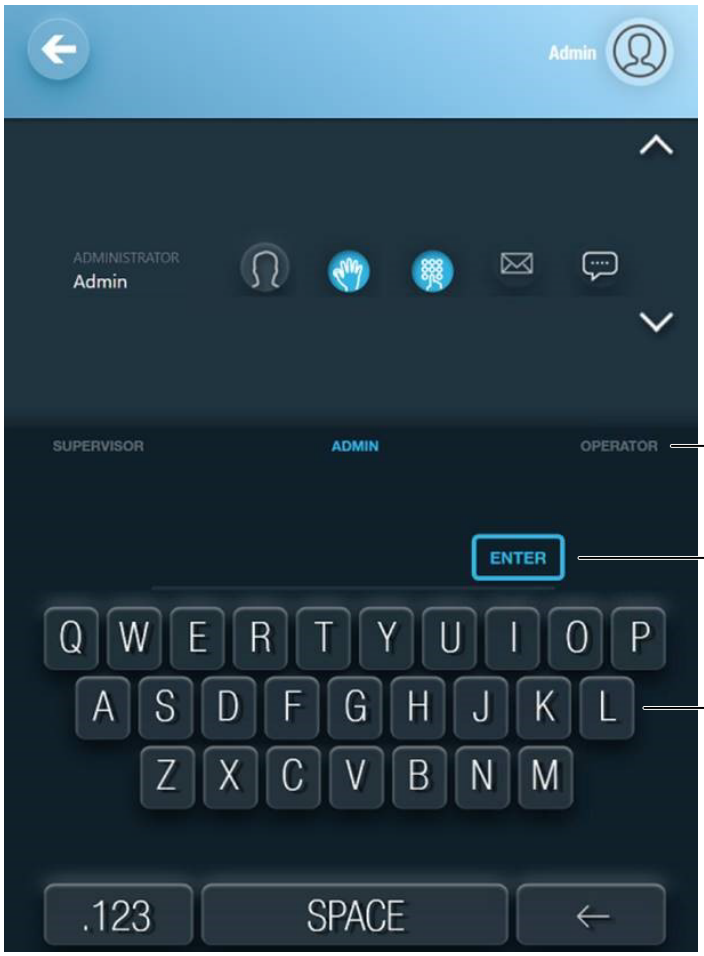
User role	Permission to perform the following tasks
Operator	• Perform a run • View the run log for the signed-in user
Supervisor	Operator tasks, plus: • Replace a primary cartridge • View the run log for all users • Recover (reset) the instrument • Export run data to a USB device
Admin (administrator)	Operator and supervisor tasks, plus: • Add new user accounts • Edit existing user accounts (includes password reset) • Back up and restore run data • Enable/disable RapidLINK™ connection • Change the date, time, and time zone

Add a user account

Note: By default, when a user account is created on one RapidHIT™ ID instrument, that same user account is authorized for all instruments on the same network. If needed, an administrator can remove the authorization for individual instruments (see “Remove authorization for an instrument” on page 44).

1. Sign in as an administrator.
2. Tap **(Menu)**.
3. Tap **(Manage users)**.
4. Tap **(Add user)**.
5. Tap a user role: **SUPERVISOR**, **ADMIN**, or **OPERATOR**.
   For a description of user roles, see “User roles and permissions” on page 38.

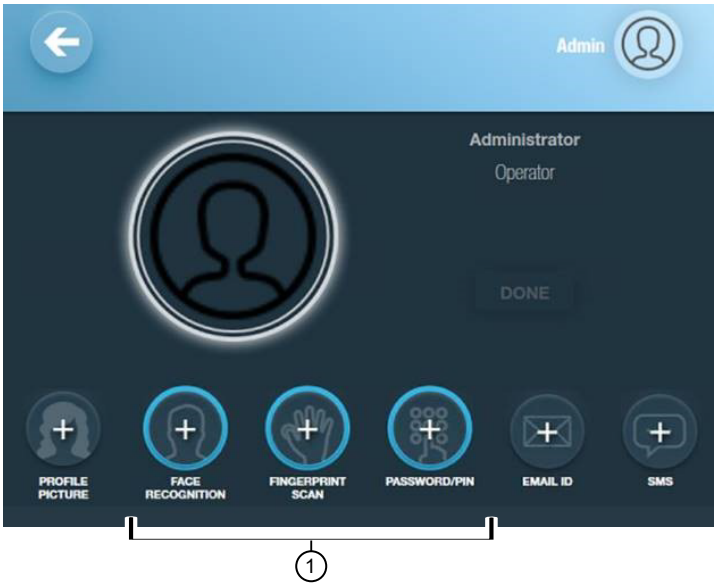
6. Enter a user name with the keypad, then tap **ENTER**.



① Select a user role
② Enter a user name with the keypad
③ Tap **ENTER**

7. Tap one or more user authentication methods.

Note: More than one method can be activated for a single user.



1 User authentication methods

Option	Procedure
FACE RECOGNITION 	1. Adjust the user position so that the face is centered in the live camera view. 2. Tap **Start Face Scan**.
FINGERPRINT SCAN 	1. Place any of the user's fingers on the fingerprint pad on the front of the instrument. 2. Remove and touch as instructed by the software. **Note:** The fingerprint scan function may be disabled for your laboratory.
PASSWORD/PIN 	Enter a PIN (6 characters) two times.

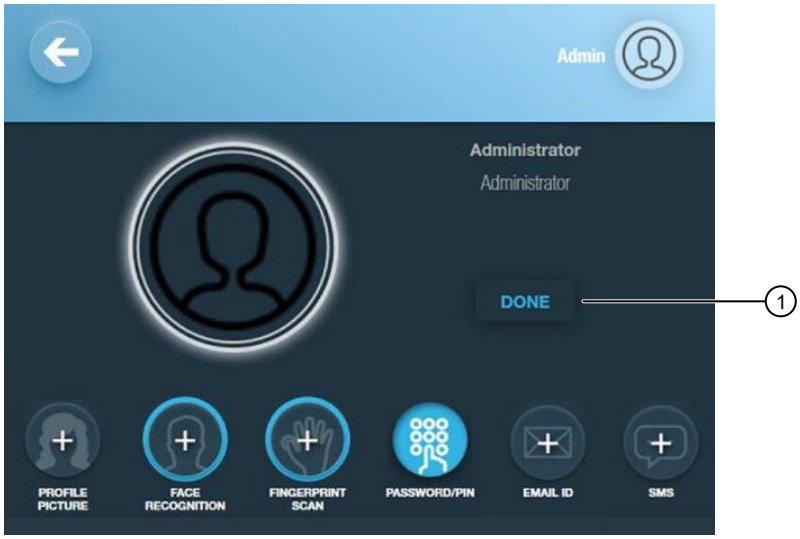
8. *(Optional)* Add an email address or SMS mobile phone number.

   Note: The profile picture function is not supported in this version of software.

9. Tap .
   A success message screen is displayed.



10. Tap the success message to close it, then tap **DONE**.

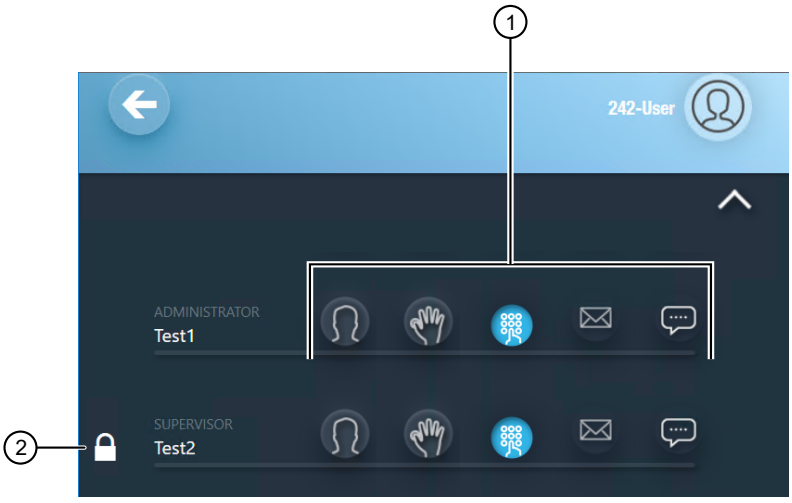


① Tap **DONE**

Edit a user account (includes password reset)

1. Sign in as an administrator.
2. Tap **(Menu)**.
3. Tap **(Manage users)**.
4. For the user account to edit, tap any of the icons to open the user profile.

Note: If is displayed next to the user account, the account cannot be edited.

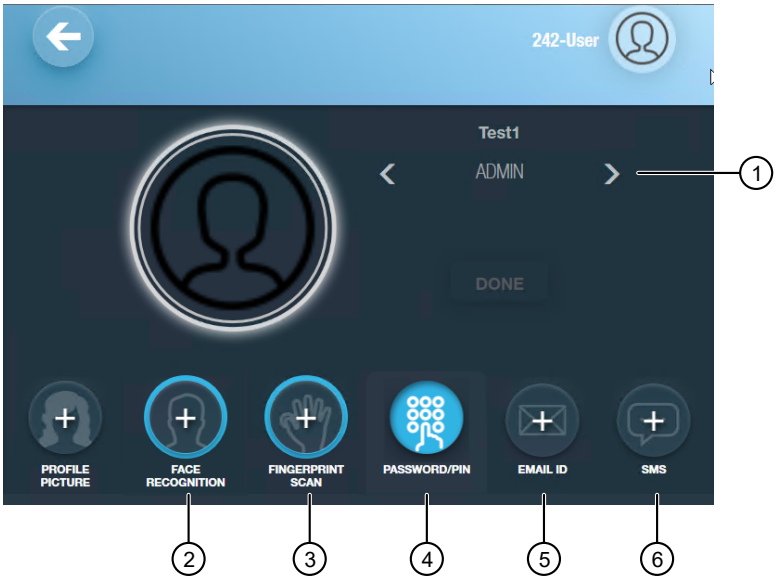


① User account icons
② Locked user account

5. Edit the user account as needed. Tap the setting to edit, then update the information. (For detailed procedures, see “Add a user account” on page 38.)

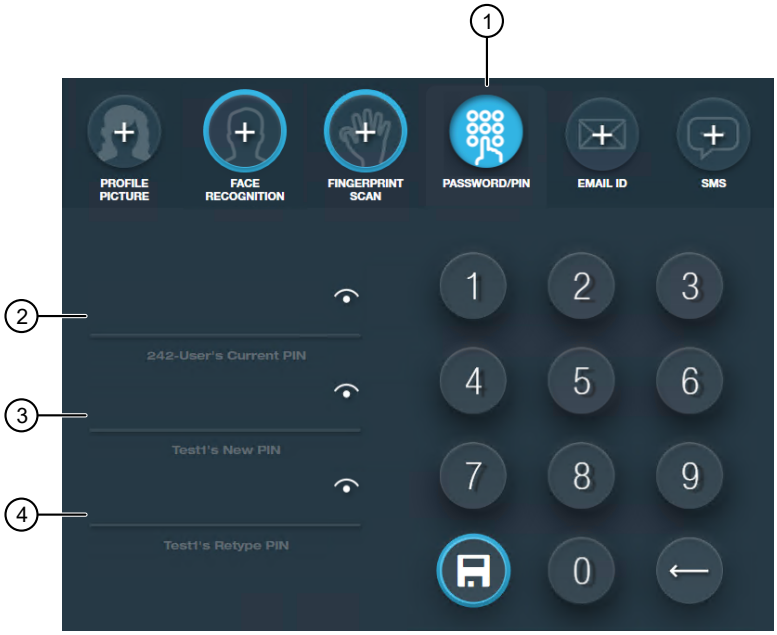
Note: You cannot edit the user name for an existing user account.

Note: The profile picture function is not supported in this version of software.



① User role
② **FACE RECOGNITION**
③ **FINGERPRINT SCAN**
④ **PASSWORD/PIN**
⑤ **EMAIL ID**
⑥ **SMS**

If you reset the **PASSWORD/PIN**, you must first enter your administrator PIN.



① **PASSWORD/PIN**
② Administrator account PIN
③ New PIN for the user account
④ Confirm PIN for the user account

6. Tap .
   A success message screen is displayed.

7. Tap the success message to close it, then tap **DONE**.



① Tap **DONE**

Remove authorization for an instrument

When you remove authorization, you prevent the user from signing in to an instrument.

In the RapidLINK™ Software, remove the instrument from the **Authorized Instruments** list for the user. For a detailed procedure, see the *RapidLINK™ Software v1.3.3 User Guide*.

Note: Removing authorization applies only to a single instrument. If you want to remove authorization for multiple instruments, repeat this procedure for each instrument.

Note: If an instrument is disconnected from the network during this process, the user authorization will not be removed for that instrument when the instrument is reconnected.

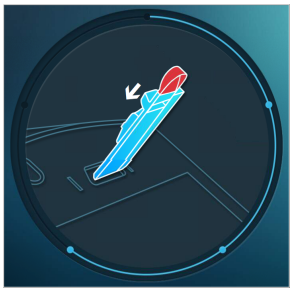
5 Maintain the instrument

- Routine maintenance 45
- View primary cartridge information 47
- Replace the primary cartridge (administrator only) 48
- Run a control cartridge 56
- Restore run data (administrator only) 57
- Recover the instrument (administrator or supervisor only) 57
- Prepare the instrument for shipping 57

Routine maintenance

Task	Action	Frequency
Maintenance check	If there is no instrument interaction for 17 hours, the instrument automatically performs a maintenance check. • If the check determines that maintenance is not needed, no user action is required. • If the check determines that maintenance is needed, the insert utility cartridge screen is displayed—a new utility cartridge is required. See "Maintenance (new utility cartridge required)" in the next row. • If the instrument is powered off, the hour count is reset when the instrument is powered on.	Every 17 hours

(continued)

Task	Action	Frequency
Maintenance (new utility cartridge required)	When maintenance is required, the instrument automatically displays the insert utility cartridge screen. **Note:** Utility cartridges can be ordered separately through your local Thermo Fisher Scientific sales representative. See “(Optional) Control and utility cartridges” on page 15. **1.** When the insert utility cartridge screen is displayed, insert a new utility cartridge into the instrument. The utility cartridge has a red label.  The instrument automatically performs the required maintenance, then displays the lock screen when the maintenance is complete. **2.** Sign in (see “Sign in to the instrument” on page 22). **3.** When the remove utility cartridge screen is displayed, remove the utility cartridge from the instrument.  **Note:** Although not recommended, you can bypass the maintenance if you are about to start a run: Tap ← to display the lock screen. Sign in, then perform the run according to Chapter 3, “Run the instrument to generate a DNA profile”. The instrument automatically performs a system prime at the start of the run.	Every 15 days[1]
Clean the touchscreen	**1.** Power off the internal computer. **2.** Spray with a non-abrasive glass cleaner, then gently wipe the screen with lint-free lab tissues.	As needed

[1] You may need to perform the maintenance sooner, if a maintenance check determines that the instrument requires maintenance before 15 days.

View primary cartridge information

If the sample identification screen is not displayed, tap ← until it is displayed. The screen displays the following information for the primary cartridge: Expiration date, number of runs performed, and gel volume indicator.



1. Expiration date—Indicates the expiration date of the gel or the primary cartridge, whichever is closest to expiration. The primary cartridge, including gel, must be changed when the expiration date is reached.
2. Number of runs—The run count for the primary cartridge. The run count indicates the number of runs that have been performed, not the number of runs that are remaining (for example, "70" indicates that 70 runs have been performed).
3. Gel volume indicator—The color of the outer ring indicates the amount of gel remaining in the primary cartridge.
   - **Blue ring**—The number of runs remaining is >30 and the amount of gel remaining is >30%.
   - **Yellow ring**—The number of runs remaining is ≤30 or the amount of gel remaining is 16–30%.
   - **Red ring**—The number of runs remaining is ≤15 or the amount of gel remaining is ≤15%.

Replace the primary cartridge (administrator only)

This procedure requires ~2 hours to complete.

Note: For RapidINTEL™ Sample Cartridge applications, use a primary cartridge from the RapidHIT™ ID Primary Cartridge GlobalFiler™ Express 150 Kit or 100 Kit.

Parts of the primary cartridge

The following figures illustrate the parts of the primary cartridge. The parts are listed in the order of removal when you prepare a new primary cartridge.

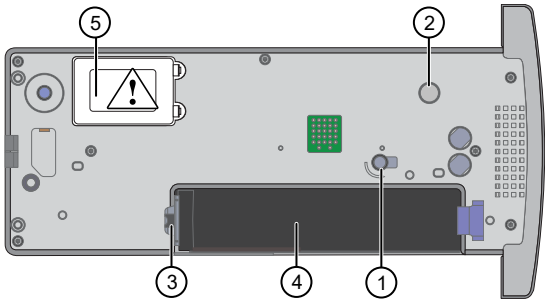


Figure 5 Parts of the primary cartridge before preparation

1. Shipping plug on check valve
2. Shipping plug on cathode block
3. Gel cartridge inlet
4. Gel cartridge slot
5. Shipping cover on capillary

CAUTION! The capillary is fragile. Handle the primary cartridge with care after you remove the capillary cover.

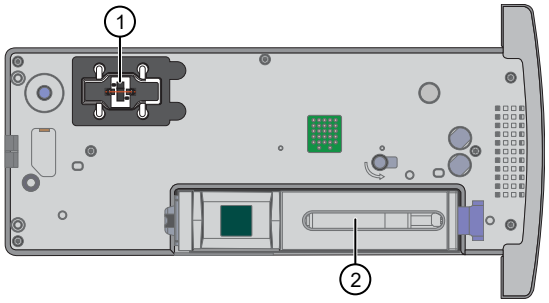


Figure 6 Primary cartridge after preparation

1. Capillary
2. Gel cartridge

Prepare a new primary cartridge

The following components are required for this procedure:

- New primary cartridge
- Gel cartridge that is provided with the new primary cartridge
- Utility cartridge that is provided with the new primary cartridge

For the location of the primary cartridge parts, see Figure 5 on page 48.

1. Sign in as an administrator.
2. Tap **(Menu)**.
3. Tap **(Primary cartridge)**.



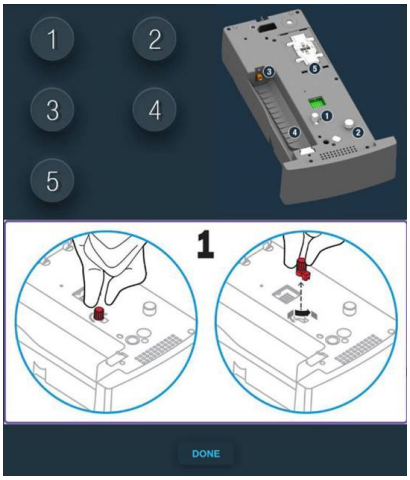
① Primary cartridge icon

4. Tap **Yes** to confirm that you want to remove the primary cartridge.

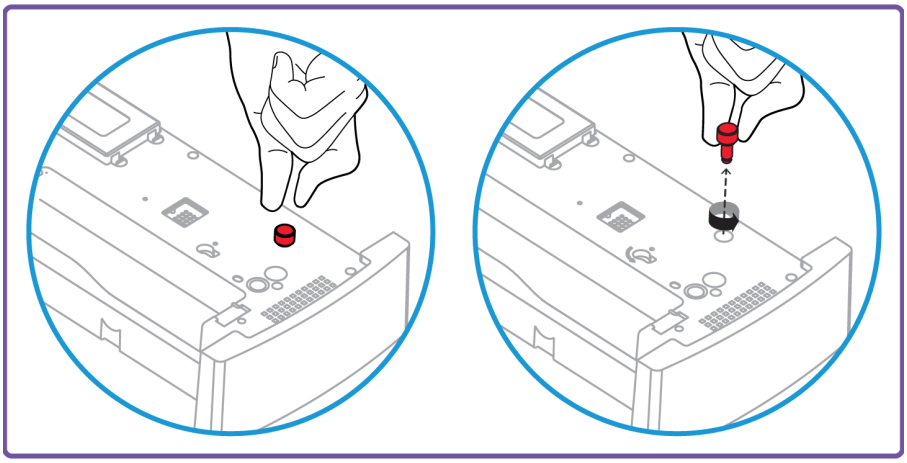


The first primary cartridge screen is displayed.

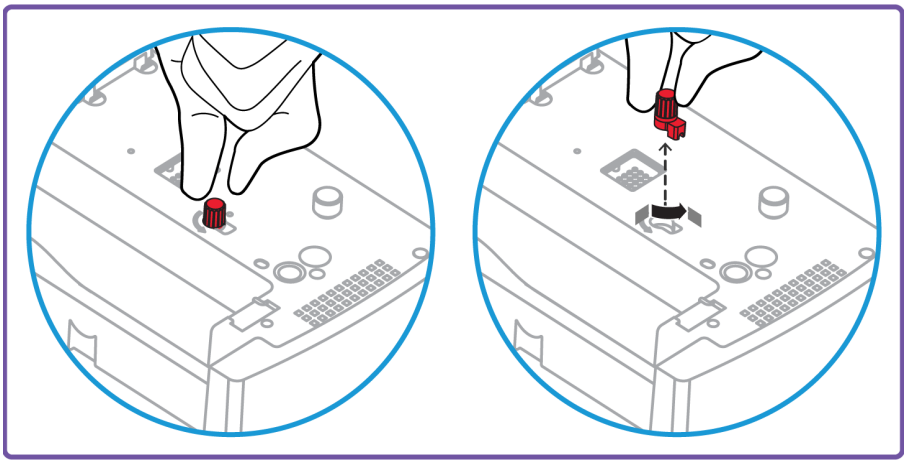
Note: To display illustrations for the remaining steps, tap the numbered buttons.



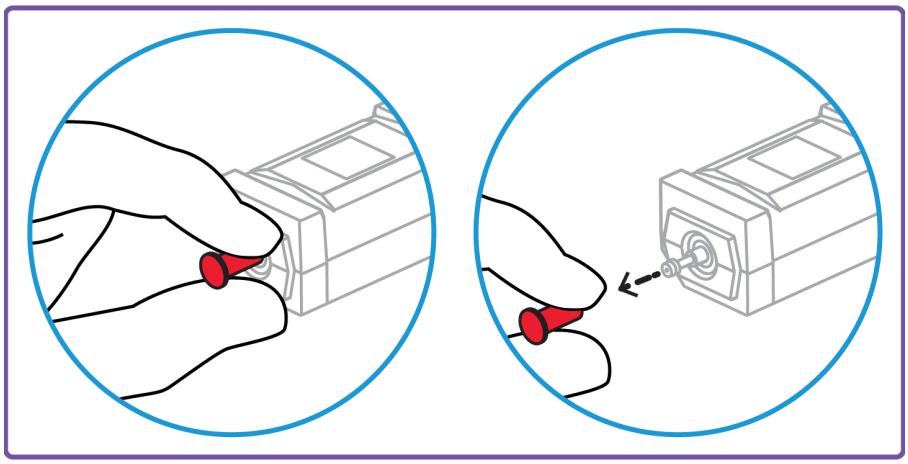
5. Unscrew the shipping plug in the cathode block.



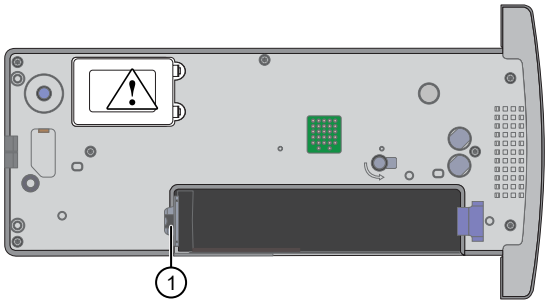
6. Turn the shipping plug in the check valve 90° counterclockwise, then remove it.



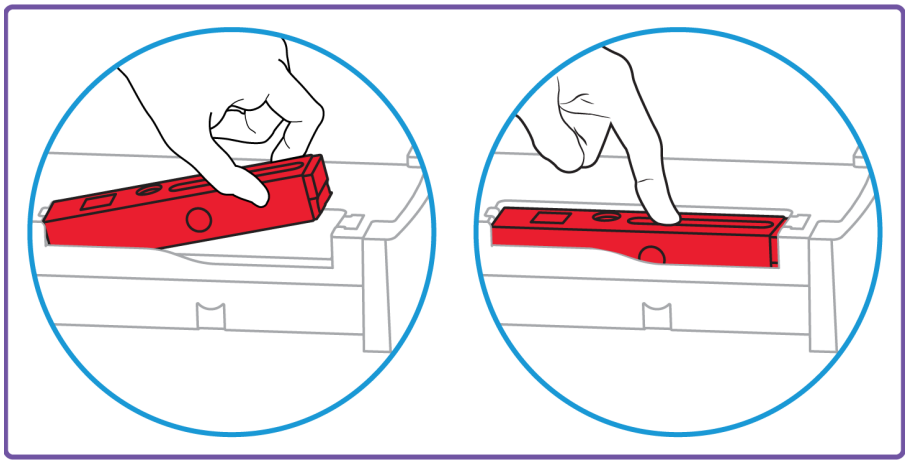
7. Gel cartridge: Remove the shipping plug from the gel cartridge inlet by pulling the plug straight out of the inlet. Do not twist the plug when removing.



8. Insert the gel cartridge into the primary cartridge with the tip of the gel cartridge facing the gel cartridge inlet and the square marker on the top. Do not twist the gel cartridge.



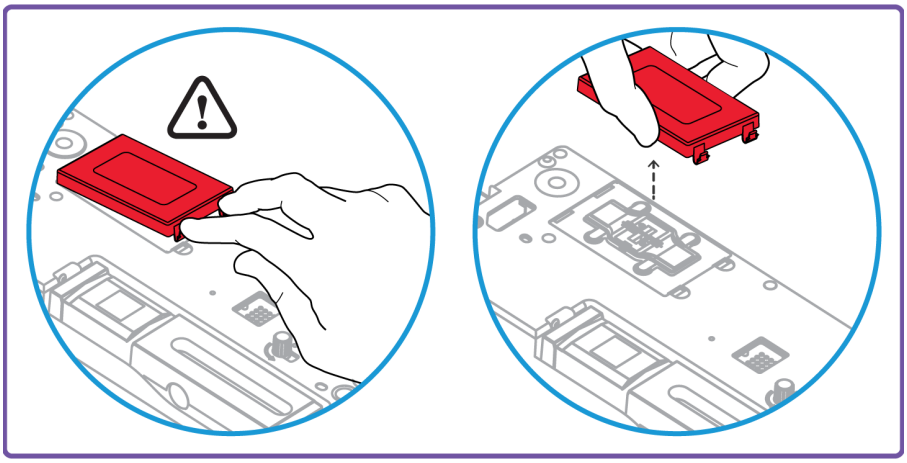
① Gel cartridge inlet



The gel cartridge clicks when it is fully inserted into the primary cartridge.

9. Remove the shipping cover from the capillary by pressing the brackets toward the cover, then swinging the cover up and away from the capillary.

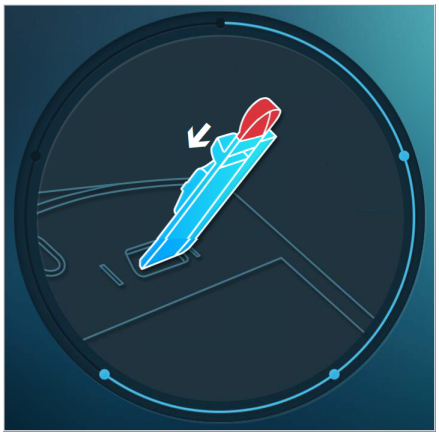
CAUTION! The capillary is fragile. Handle the primary cartridge with care after you remove the capillary cover.



10. Tap **Done** at the bottom of the primary cartridge screen.



11. When the insert utility cartridge screen is displayed, insert the utility cartridge into the instrument. The utility cartridge has a red label and is provided with the new primary cartridge.



A countdown timer starts at 9 minutes while the instrument disengages the primary cartridge.

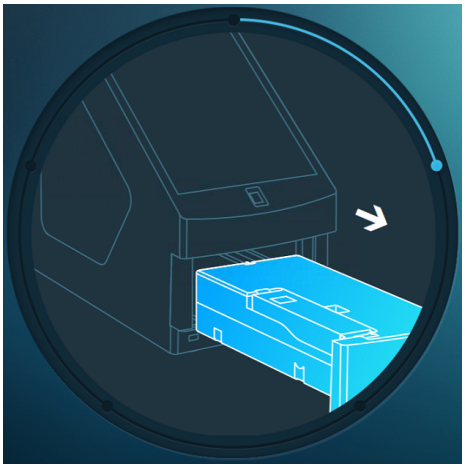
Immediately proceed to “Insert the prepared primary cartridge” on page 54.

Insert the prepared primary cartridge

The following components are required for this procedure:

- Prepared primary cartridge
- Allelic ladder cartridge that is provided with the new primary cartridge
- *(Optional)* Positive control cartridge that is provided with the new primary cartridge

1. When the remove primary cartridge screen is displayed (~3 minutes after you insert the utility cartridge in the instrument, step 11 on page 53), pull the used primary cartridge out of the instrument.



2. When the insert primary cartridge screen is displayed, insert the new prepared primary cartridge into the instrument.



CAUTION! The capillary is fragile. Do not let the capillary contact the instrument when you insert the prepared primary cartridge.

A countdown timer is displayed. The instrument performs a full 90-minute run.

3. When the remove utility cartridge screen is displayed, remove the utility cartridge from the instrument.



4. *(Recommended)* Run the allelic ladder cartridge: Insert and remove the allelic ladder cartridge when the associated screen is displayed.
   - The software displays a Green ✓ for a ladder profile that contains the expected number of alleles.
   - The software displays a Red ✗ for a ladder profile that does not contain the expected number of alleles. Before running any samples, proceed with the positive control cartridge run. See “Run a control cartridge” on page 56.

   Note: An allelic ladder run is recommended based on cartridge availability. Allelic ladder cartridges can be ordered separately through your local Thermo Fisher Scientific sales representative. See “(Optional) Control and utility cartridges” on page 15.

5. Discard the used primary and allelic ladder cartridges. Use proper precautions for disposal. Follow local, state, provincial and federal regulations for disposal.

(Optional) Run a positive and/or negative control cartridge; see “Run a control cartridge” on page 56.

Note: The software does not prompt you to run the positive or negative control cartridges.

Run a control cartridge

Note: For RapidINTEL™ Sample Cartridge applications, use a RapidHIT™ ID ACE GlobalFiler™ Express Control Cartridge (allelic ladder cartridge).

1. If the sample identification screen is not displayed, tap ← until it is displayed.
2. Insert the control cartridge in the instrument.

   The instrument reads the label on the cartridge and automatically assigns the sample identification as "POSCTRL", "NEGCTRL", or "LADDER".

   Note: If you enter names other than the names listed above, the names are overwritten in the RapidLINK™ Software with the names listed above.
3. When the run is complete, remove the cartridge from the instrument.
4. Review the status and take the appropriate action. See Table 9.
5. Tap **Done**. The instrument automatically signs out the user and displays the lock screen.

Table 9 Control cartridge results

Results for positive, negative, and/or allelic ladder control cartridges

Status	DNA profile was generated	Meaning	Action
Green 	Yes	• The DNA profile does not contain quality flags. • Expected alleles were called in the positive control or allelic ladder profile. • No alleles were called in the negative control profile. **Note:** Passing allelic ladder profiles are added to the allelic ladder library on the instrument.	No further action is needed on the instrument.
Red 	No	• The DNA profile was not generated, or the results were not as expected. • Expected alleles *were not* called in the positive control or allelic ladder profile. • Alleles *were* called in the negative control profile.	Follow your standard operating protocol for how to proceed.

Restore run data (administrator only)

The restore function overwrites the run data on the instrument with the run data from the USB device.

1. Sign in as an administrator.
2. Insert a USB device into the USB port on the front of the instrument.
   The **Removable Drives** field displays the name of the USB device.
3. Tap **(Menu)**.
4. Tap **(Settings)**.
5. In the settings screen, tap **Start Restore**.
   The button changes to **Abort** and a progress bar is displayed. When the procedure is complete, a message is displayed.



① Drive selection; if there is no USB device in the USB port, the screen displays **None Found**
② **Start Restore** button

Recover the instrument (administrator or supervisor only)

The recovery function resets the sample cartridge hardware in the instrument. You can use the recovery function if a cartridge is locked in the sample cartridge port.

This function does not affect run data, user profiles, or settings.

1. Tap **(Menu)**.
2. Tap **(Settings)**.
3. In the settings screen, tap **Start Recovery**.
4. Follow the prompts that are displayed on the screen.

Prepare the instrument for shipping

Contact Thermo Fisher Scientific Support for requirements.

A Troubleshooting

- Check the RapidLINK™ Software connection 58
- Troubleshooting: Symptoms, causes, and actions 58
- Error codes 61
- Perform the SCCL recovery procedure 63

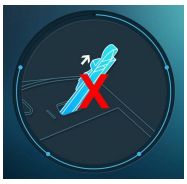
Check the RapidLINK™ Software connection

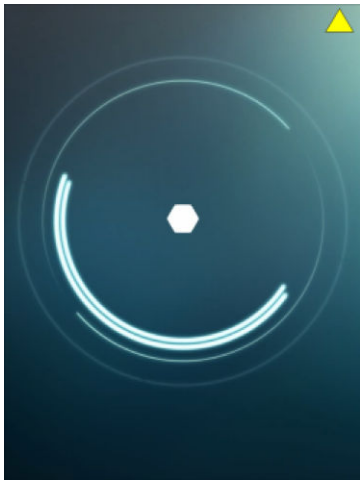
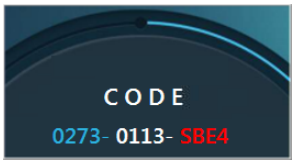
1. In the sample identification screen, check the top of the screen.

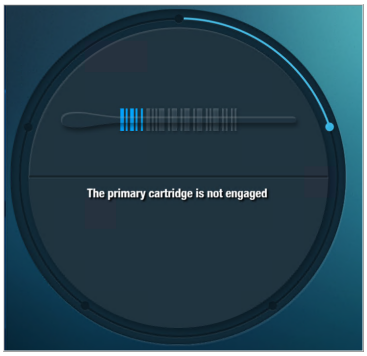
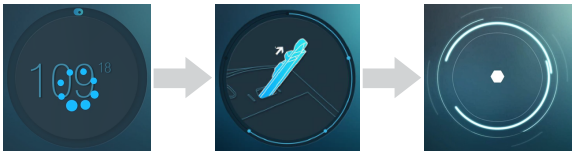
   If is displayed, the instrument is not connected to the software.

2. Ensure that the **RapidLINK** setting is enabled. See “Configure instrument settings (administrator or supervisor only)” on page 36.

Troubleshooting: Symptoms, causes, and actions

Observation	Possible cause	Recommended action
Red X on cartridge screen 	Sample cartridge is expired.	Obtain a new swab and prepare a new cartridge.
	Sample cartridge is inserted improperly in instrument.	Remove the sample cartridge from the instrument and re-insert.

Observation	Possible cause	Recommended action
A yellow triangle is displayed on the lock screen 	An error code has been generated.	Tap the lock screen to obtain the error code. Contact Thermo Fisher Scientific Support.
An error code is displayed when you sign in 	There is a problem with the instrument.	See “Error codes” on page 61.
		Open the **DX Code Troubleshooting Guide** within the RapidLINK™ Software. For more information, see the *RapidLINK™ Software v1.3.3 User Guide* (Pub. No. MAN0018939).
is displayed on the sample identification screen	Indicates that the instrument is not connected to the RapidLINK™ Software.	See “Check the RapidLINK™ Software connection” on page 58.
The sample cartridge cannot be easily removed from the instrument	The sample cartridge is locked in the sample cartridge port.	Run the recovery function (see “Recover the instrument (administrator or supervisor only)” on page 57).

Observation	Possible cause	Recommended action
A power failure occurs during a run	Various causes.	If the power failure occurs early in the run, the run will stop and it will not be listed in the run log. Obtain a new swab and rerun the sample.
		If the power failure occurs late in the run but before analysis starts, the run will be listed in the run log with a red ✗.
		If the power failure occurs after analysis is complete, the run will be listed in the run log with a green ✓ if it passes or a red ✗ if it fails.
"The primary cartridge is not engaged" message is displayed on the sample identification screen 	There is a problem with the instrument or the primary cartridge.	Contact Thermo Fisher Scientific Support.
You are required to remove a sample cartridge before the run is complete **Details:** You will see one of the following screens, depending on when you are looking at the instrument. 	An SCCL error has occurred.	Perform the SCCL recovery procedure on page 63.

Error codes

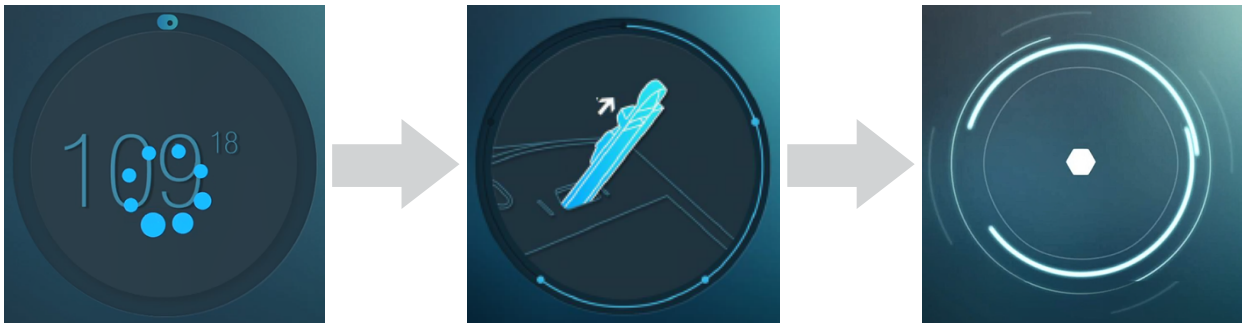
Error code (last 2 letters)	Restart required	Error description	Action
A1	Yes	The ambient temperature is not within specification for running the instrument.	Ensure that the ambient temperature is 15–30°C (59–86°F).
B1	No	The sample injection was not detected.	Contact Technical Support.
B2	No	The sample injection was delayed.	Contact Technical Support.
C4	Yes	The sample cartridge insertion was not detected.	Restart the instrument, then re-insert the sample cartridge.
D1	Yes	An internal heater did not reach the set temperature.	Contact Technical Support.
E3	Yes	The gel has expired.	Replace the primary cartridge.
E4	Yes	The gel syringe is empty.	Replace the primary cartridge.
F3	Yes	During primary cartridge installation, the cartridge was not able to prime.	Restart the instrument, then re-insert the sample cartridge. A new utility cartridge will be required.
G1	Yes	Gel refrigeration did not reach the set temperature.	Contact Technical Support.
G2	Yes	The gel temperature probe malfunctioned.	Contact Technical Support.
G3	Yes	Gel refrigeration was not able to maintain the set temperature.	Contact Technical Support.
H3	Yes	An internal heater did not reach the set temperature.	Press the rear button to restart the instrument, then run another blank or control cartridge.
K1	No	The amount of sample lysate transferred was low.	Ensure that the substrate did not absorb lysate. For example, a frayed swab or sponge is more likely to absorb lysate.
K2	Yes	The sample lysate transfer did not occur.	Ensure that the substrate did not absorb lysate. For example, a frayed swab or sponge is more likely to absorb lysate.
L3	Yes	An internal heater did not reach the set temperature.	Contact Technical Support.

(continued)

Error code (last 2 letters)	Restart required	Error description	Action
M6	Yes	During primary cartridge installation, the motor did not move as expected.	Press the rear button to restart the instrument, then try to re-insert the primary cartridge. A new utility cartridge will be required.
U1	Yes	The current is too high or too low.	Contact Technical Support.
U3	Yes	The current is too high or too low.	Contact Technical Support.
U4	Yes	The current is too high or too low.	Contact Technical Support.
ABV3	Yes	In the primary cartridge, the gel reprime was lower than expected.	Restart the instrument, then re-insert the primary cartridge. A new utility cartridge will be required.
SBV3	Yes	In the sample cartridge, the gel reprime was lower than expected.	Restart the instrument, then insert a new sample cartridge.
CL	Yes	A cartridge was compromised.	Perform the SCCL recovery procedure on page 63.
CP	Yes	A cartridge was compromised (after SCCL recovery).	Restart the instrument, then perform another run using a blank cartridge or blank swab.

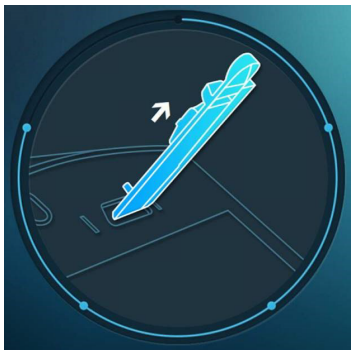
Perform the SCCL recovery procedure

After an SCCL error occurs, you will see one of the following screens, depending on when you are looking at the instrument.

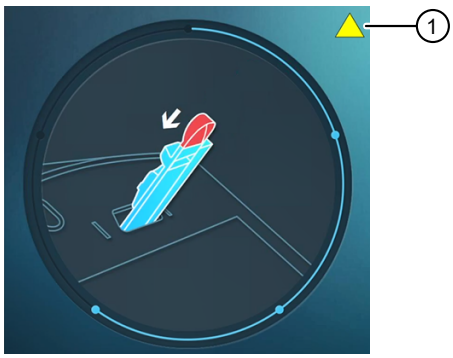


1. When the remove sample cartridge screen is displayed, remove the sample cartridge from the instrument.

   Note: If the lock screen is displayed, tap the lock screen, then sign in.

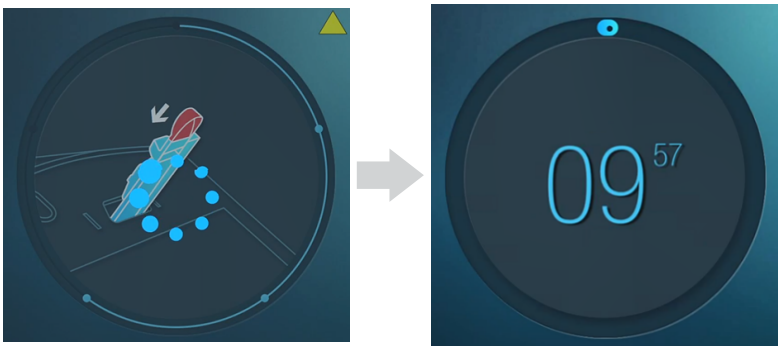
   

2. When the insert utility cartridge screen is displayed, insert a new utility cartridge into the instrument.

   ① The insert utility cartridge screen includes a yellow triangle to indicate that an error has occurred.

   The instrument prepares for the recovery steps, then displays the countdown timer. The recovery can take ~5–10 minutes.



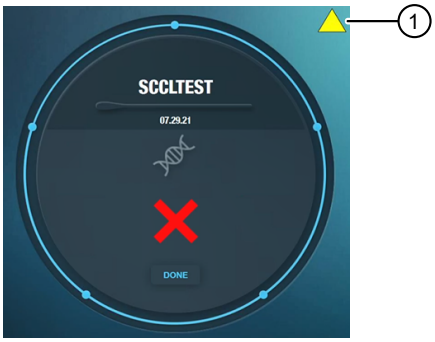
3. When the lock screen is displayed, tap the lock screen, then sign in.



4. When the remove utility cartridge screen is displayed, remove the utility cartridge from the instrument.



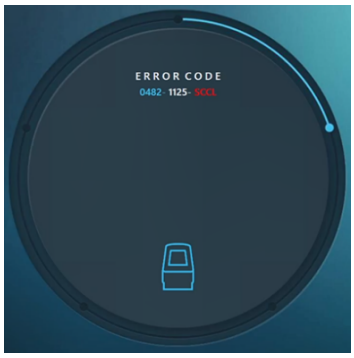
The result screen is displayed with a run status of ✕, indicating that the run failed.



① The result screen includes a yellow triangle to indicate that an error has occurred.

5. Tap **Done** to dismiss the status.
   The instrument automatically signs out the user.

6. When the lock screen is displayed, tap the lock screen, then sign in.
   The SCCL error code is displayed.



7. Restart the instrument to complete the recovery.

After restarting the instrument, you can run a new sample cartridge. See Chapter 3, “Run the instrument to generate a DNA profile”.

B

Analysis settings

- Threshold settings 66
- Display analysis settings in the GeneMarker™ HID STR Human Identity Software 68
- Analysis settings: GlobalFiler™ Express sample cartridges 69
- Analysis settings: RapidINTEL™ sample cartridges 70
- Analysis settings: NGM SElect™ Express sample cartridges 72

Threshold settings

Table 10 Threshold settings (contained in panel)

System Threshold	RapidHIT™ ID ACE GlobalFiler™ Express Sample Cartridge	RapidINTEL™ Sample Cartridge	RapidHIT™ ID ACE NGM SElect™ *Express* Sample Cartridge
Analytical threshold	35 RFU	50 RFU	All loci 50, except: SE33 = 35
Stochastic threshold (inconclusive Homozygous (IHO flag) threshold)	All loci 91 RFU, except: • TPOX = 105 • Y indel = 35 • DYS391 = 35 • TH01 = 140 • SE33 = 105 • D12S391 = 105 • D2S1338 = 105	1,600 RFU 50 RFU for Y indel and DYS391	All loci 150, except: • D22S1045 = 200 • TH01 = 200 • D2S441 = 100 • D1S1658 = 100 • SE33 = 105
Minimum peak height ratio threshold (Heterozygote Imbalance (IMB flag) threshold)	40%	40% 99% for Y indel and DYS391	40%
Minimum heterozygous peak intensity threshold (inconclusive Heterozygous (IHE flag) threshold)	—	640 RFU	—
Stutter filters	20%	Locus-specific[1]	20%

Table 10 Threshold settings (contained in panel) *(continued)*

System Threshold	RapidHIT™ ID ACE GlobalFiler™ Express Sample Cartridge	RapidINTEL™ Sample Cartridge	RapidHIT™ ID ACE NGM SElect™ *Express* Sample Cartridge
Locus-specific filter	20%	21%	20%
		30% for the positive control	
Ploidy (PL flag) threshold (maximum number of expected peaks)	2	2	2
Global filter (between loci)	20%	21%	20%
		30% for the positive control	
Minimum off-ladder (OL) intensity	30 RFU	30 RFU	30 RFU

[1] See Table 11.

Table 11 RapidINTEL™ Sample Cartridge Stutter thresholds. When not indicated, the stutter threshold is –4 nucleotides from the allele peak.

Dye	Marker	Avg Stutter + 4SD (%)
6-FAM™	D3S1358	27
	vWA	25
	D16S539	25
	CSF1PO	22
	TPOX	16
VIC™	Y indel (stutter –2)[1]	21
	AMEL (stutter –9)[1]	21
	D8S1179	20
	D21S11	25
	D18S51	28
	DYS391	18
NED™	D2S441	16
	D19S433	29
	TH01	18
	FGA	27

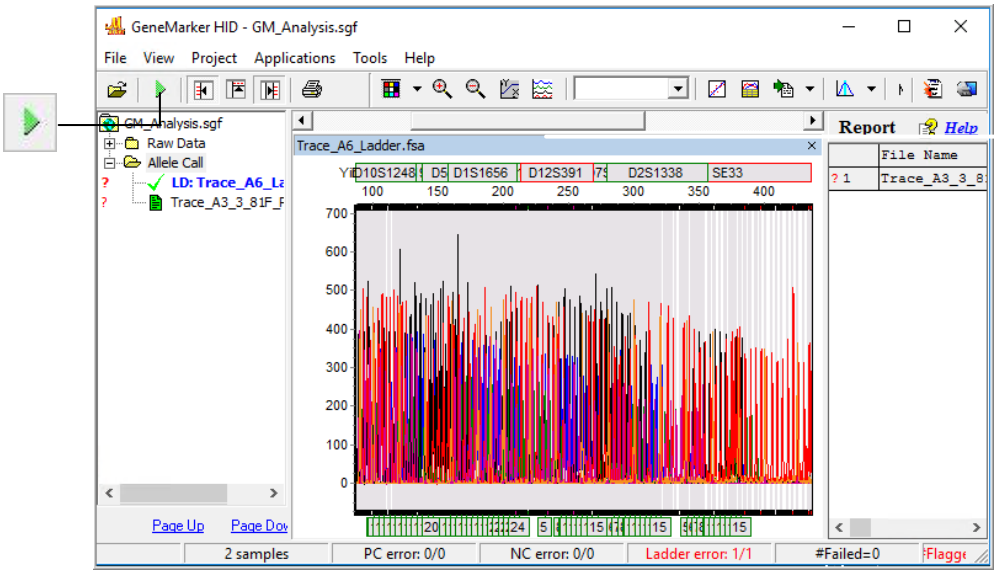
Table 11 RapidINTEL Sample Cartridge Stutter thresholds. When not indicated, the stutter threshold is –4 nucleotides from the allele peak. *(continued)*

Dye	Marker	Avg Stutter + 4SD (%)
TAZ™	D22S1045 (stutter –3)	34
	D5S818	26
	D13S317	18
	D7S820	18
	SE33	30
SID™	D10S1248	29
	D1S1656	26
	D12S391	30
	D2S1338	31

[1] Stutter threshold is set to the same value as the locus-specific filter values and will have no effect on filtering.

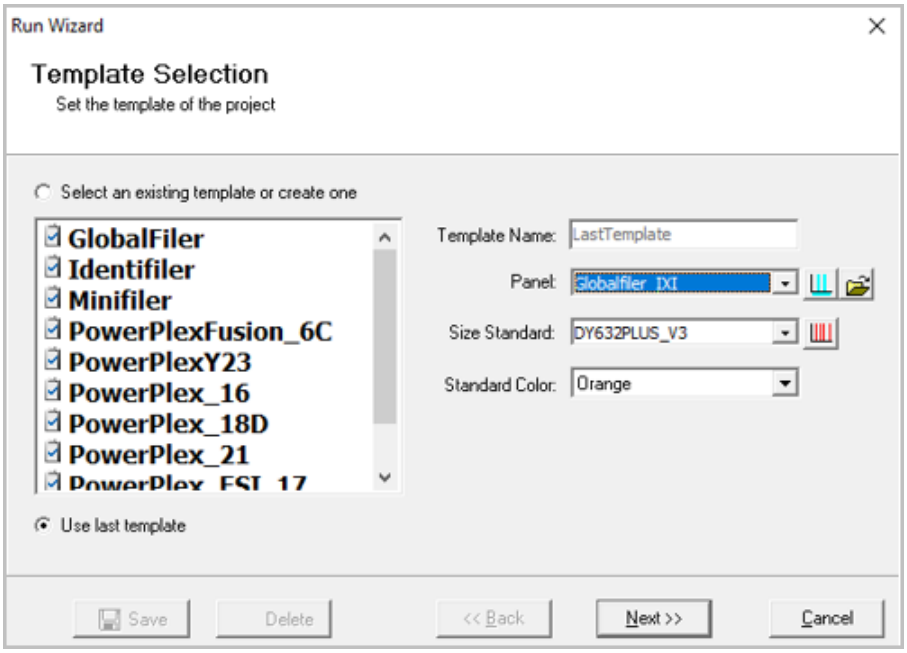
Display analysis settings in the GeneMarker™ HID STR Human Identity Software

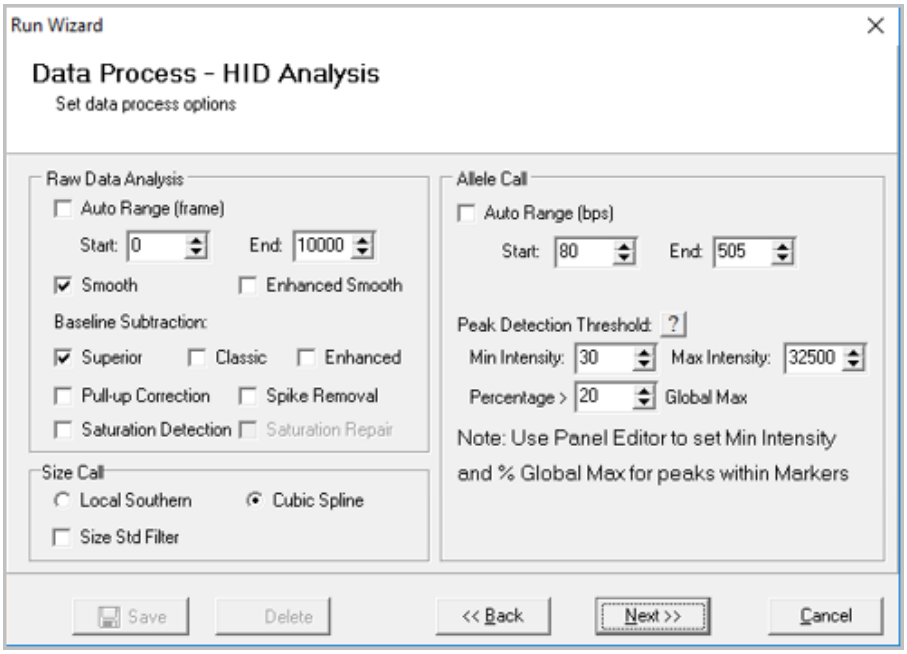
1. Open a sample in the RapidLINK™ Software.
2. Double-click the sample to open the GeneMarker™ HID Software.
3. Click .

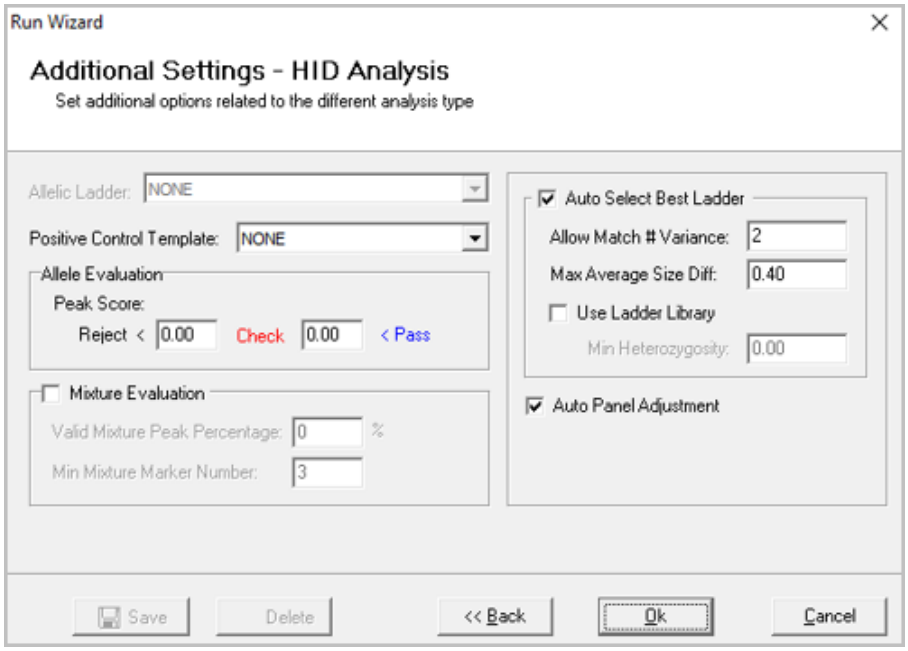


The **Run Wizard** screens are displayed with analysis settings. Click **Next** to display additional settings screens.

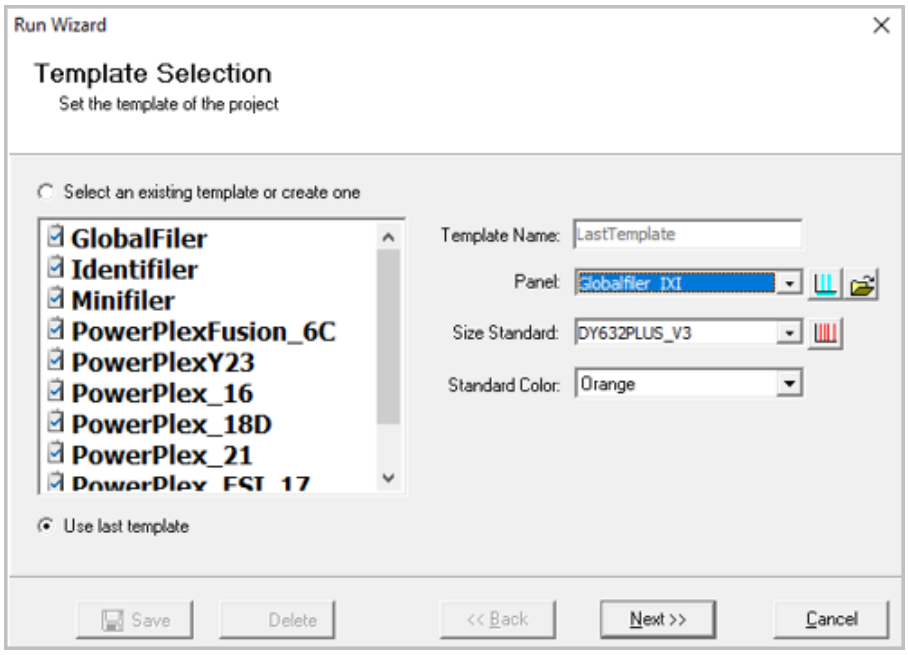
Analysis settings: GlobalFiler™ Express sample cartridges

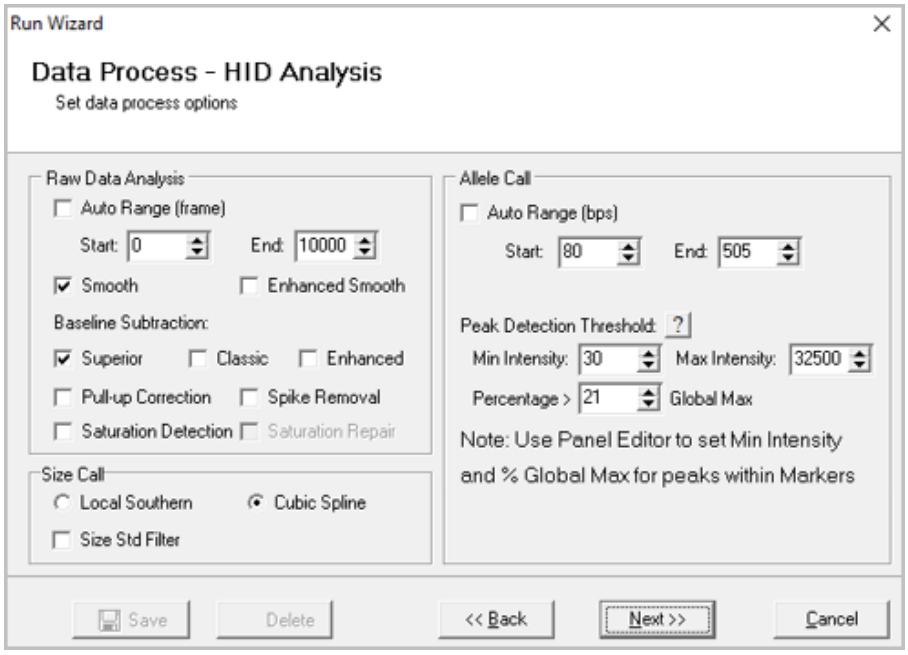




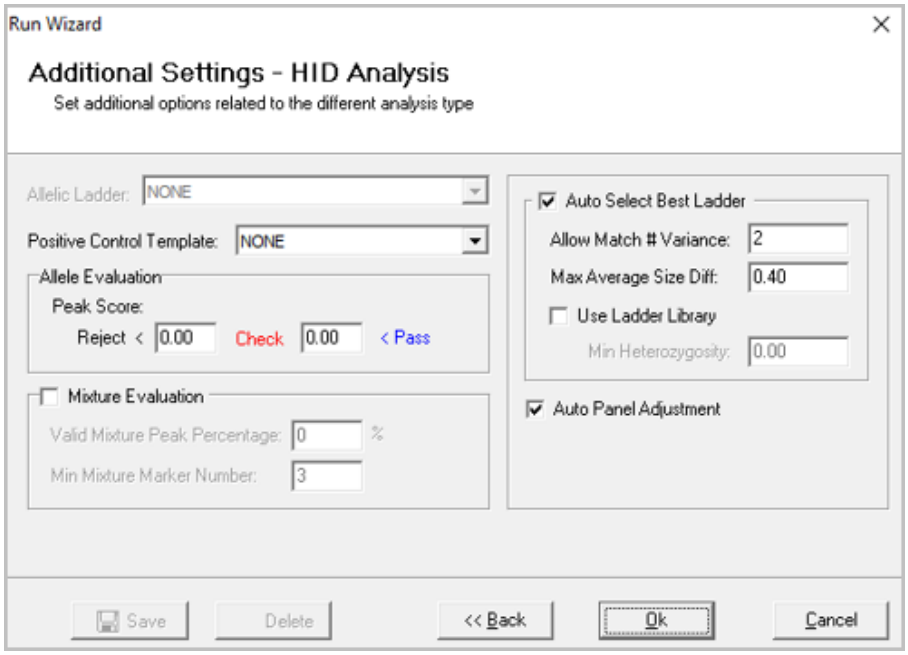


Analysis settings: RapidINTEL™ sample cartridges

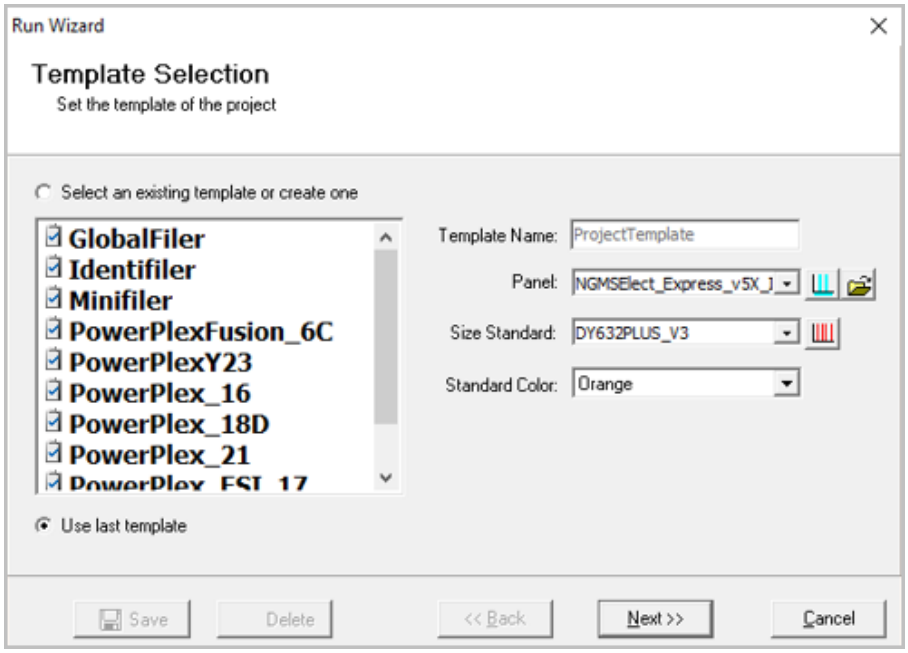


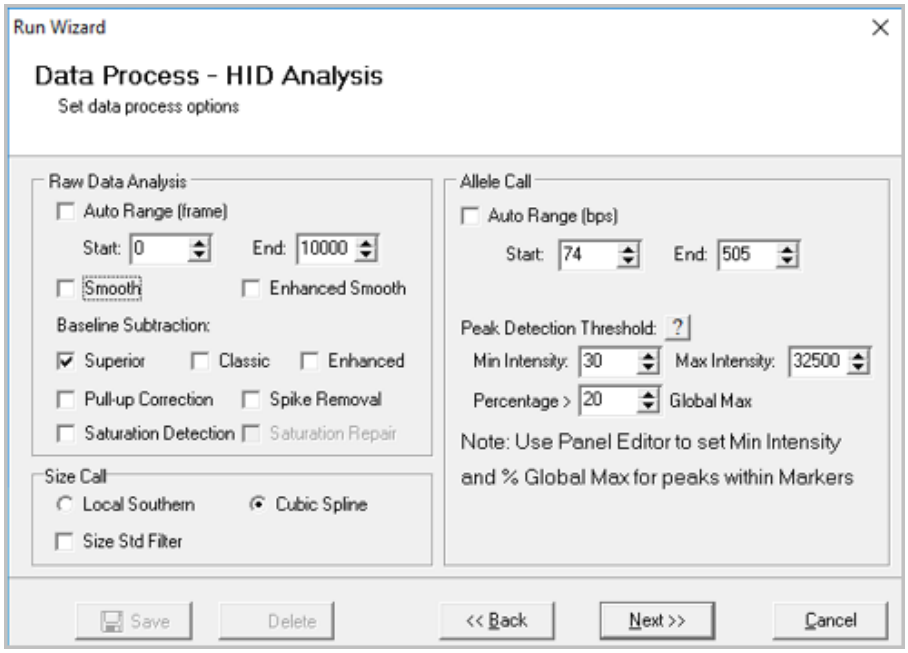


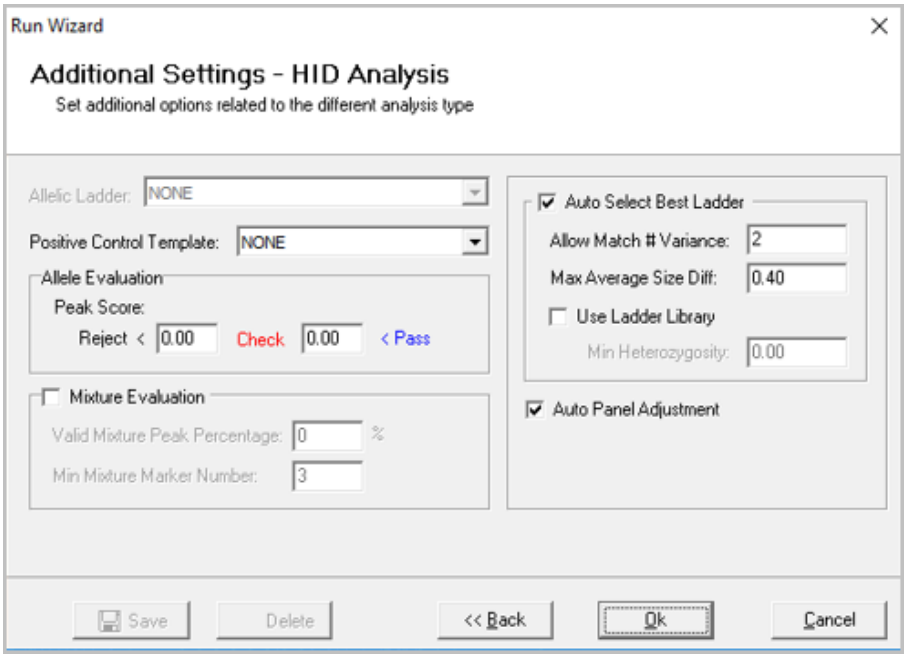
Note: The screen above shows the **Global Max Percentage** for samples. For the RapidINTEL™ positive control, the setting is <30% .



Analysis settings: NGM SElect™ *Express* sample cartridges






Run Wizard
Additional Settings - HID Analysis
Set additional options related to the different analysis type
Allelic Ladder:
NONE
Positive Control Template:
NONE
Allele Evaluation
Peak Score:
Reject <
0.00
Check
0.00
< Pass
Mixture Evaluation
Valid Mixture Peak Percentage:
0
%
Min Mixture Marker Number:
3
Auto Select Best Ladder
Allow Match # Variance:
2
Max Average Size Diff:
0.40
Use Ladder Library
Min Heterozygosity:
0.00
Auto Panel Adjustment
Save
Delete
<< Back
Ok
Cancel

C

Instrument specifications

- Dimensions, clearance, and weight 74
- Instrument layout and connections 75
- Environmental requirements 77
- Electrical requirements 78

Dimensions, clearance, and weight

Specification	Height	Length (depth)	Width	Weight
Dimension	48 cm (19 in.)	53 cm (21 in.)	27 cm (10.5 in.)	• 28.4 kg (62.6 lb.)—With primary cartridge • 25.4 kg (56.0 lb.)—Instrument only • 3.0 kg (6.6 lb.)—Primary cartridge only
Additional clearance	25 cm (10 in.)	28 cm (11 in.) Minimum rear clearance is 10 cm (4 inches). Clearance is required to allow air flow around the instrument and to allow the power cord to be easily removed without moving the instrument.	13 cm (5 in.) 64 cm (25 in.) required on one side for computer	N/A

Instrument layout and connections



Figure 7 Dimensions and clearances

① Height 48 cm (19 in.)
② Height clearance 25 cm (10 in.)
③ Length (depth) 53 cm (21 in.)
④ Length (depth) clearance 28 cm (11 in.)
Length (depth) clearance when replacing the primary cartridge114 cm (45 in.)
⑤ Width 27 cm (10.5 in.)
⑥ Width clearance 13 cm (5 in.)
64 cm (25 in.) required on one side for computer

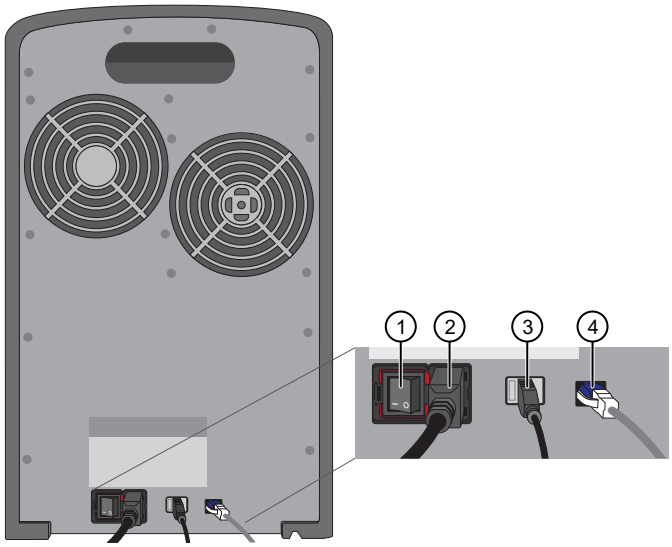


Figure 8 Rear panel

① Main power switch
② Power connection
③ USB port for optional barcode scanner connection
④ Ethernet port for computer connection

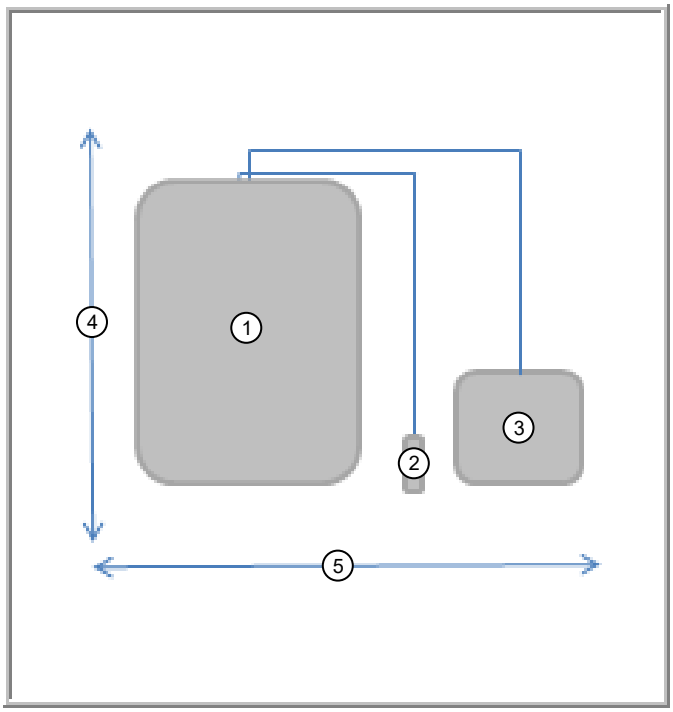


Figure 9 Component connections

① Instrument
② Optional barcode scanner
③ Computer
④ Total length (depth) space required 109 cm (43 in.)
⑤ Total width space required 116 cm (45 in.)

Environmental requirements

Condition	Requirement
Installation site	Indoor use only
Altitude	Safety tested up to 2,600 meters (8,500 feet)
Electromagnetic interference	Do not use this instrument near sources of strong electromagnetic radiation (for example, unshielded intentional RF sources). Strong electromagnetic radiation can interfere with the proper operation of the device.
Transient/overvoltage category	Installation categories II
Vibration	Do not install the instrument near strong vibration sources, such as a centrifuge, pump, or compressor. Excessive vibration affects instrument performance.

(continued)

Condition	Requirement
Pollution degree	II Do not install the instrument in an environment that has nonconductive pollutants such as dust particles or wood chips. Typical environments with a Pollution Degree II rating are laboratories and sales and commercial areas.
Operating conditions	• 15°C–30°C (59°F–86°F) (Room temperature should not fluctuate ±2°C during an instrument run) • 20–80% relative humidity, noncondensing
Storage conditions	For instrument only without primary cartridge installed: 4°C to 40°C (39°F to +104°F) Minimum 20% relative humidity, maximum 80% (non-condensing) **IMPORTANT!** • The primary cartridge requires storage at 15°C to 30°C (59°F to +86°F). • The gel cartridge requires storage at 4°C to 10°C (39°F to +50°F).
Transport conditions	–20°C to +60°C (–4°F to +140°F) Minimum 20% relative humidity, maximum 80% (non-condensing)
Other conditions	Ensure that the room is away from any vents that could expel particulate material on the components. Avoid placing the instrument and computer near heaters, cooling ducts, or in direct sunlight.

Electrical requirements

CAUTION! Do not unpack or plug in any components until they are configured for the proper operating voltage by the service representative.

WARNING! For safety, the power outlet for the instrument must be accessible at all times. In case of emergency, you must be able to immediately disconnect the main power supply to all the equipment. Allow sufficient space between the wall and the equipment so that the power cords can be disconnected in case of emergency.

- Dedicated line and ground between the instrument and the main electrical service
- Maximum power dissipation: 230 W, 370 VA (not including computer)
- Mains AC line voltage tolerances should not exceed ±10 percent of nominal voltage
- Maximum current: 3 A
- Fuse rating: 5 A, 250 V, Slow-blow, 5 × 20 mm

Device	Rated voltage	Circuit required	Rated frequency	Rated power
Instrument	100–240 ±10% VAC[1]	≥10 A	50/60 Hz	600 W

[1] If the supplied power fluctuates beyond the rated voltage, a power line regulator may be required. High or low voltages can adversely affect the electronic components of the instrument.

D System modification study (RapidHIT™ ID System v1.3.2)

- About the system modification studies 80
- Gel polymer priming study 80
- Primary cartridge stability testing 83
- Systematic Allelic Ladder Library 85
- Automated maintenance check 91
- Conclusion 92

About the system modification studies

RapidHIT™ ID System v1.3.2 was developed to expand customer usability under a variety of conditions, ranging from low-to-high throughput. Emphasis was placed on enhancing gel polymer management to maximize the on-instrument life of the primary cartridge.

To ensure robust data quality that is consistent with past results, the impact of the gel polymer management modifications was assessed in the studies described in this appendix. The measurement of electrical current and resolution were the focus of most studies because of their direct correlation to gel polymer age and the length of time gel polymer remains at room temperature. If current values rise significantly, the potential exists for DNA fragment migration to be negatively affected. These studies were performed to ensure that resolution requirements are maintained with the changes implemented in RapidHIT™ ID System v1.3.2.

Gel polymer priming study

A central aspect to gel polymer management is the time between gel polymer priming events. To accommodate a wider variety of instrument usage patterns, the amount of time between gel polymer re-priming events was increased. Because of this modification, the instrument consumes less gel polymer between idle periods when no runs are being performed.

Feasibility data demonstrated that re-priming the gel polymer following a rise in current after an idle period effectively lowers conductivity for the immediate and successive runs. Figure 10 demonstrates conductivity values in 3 usage patterns using the modified gel polymer re-priming method.

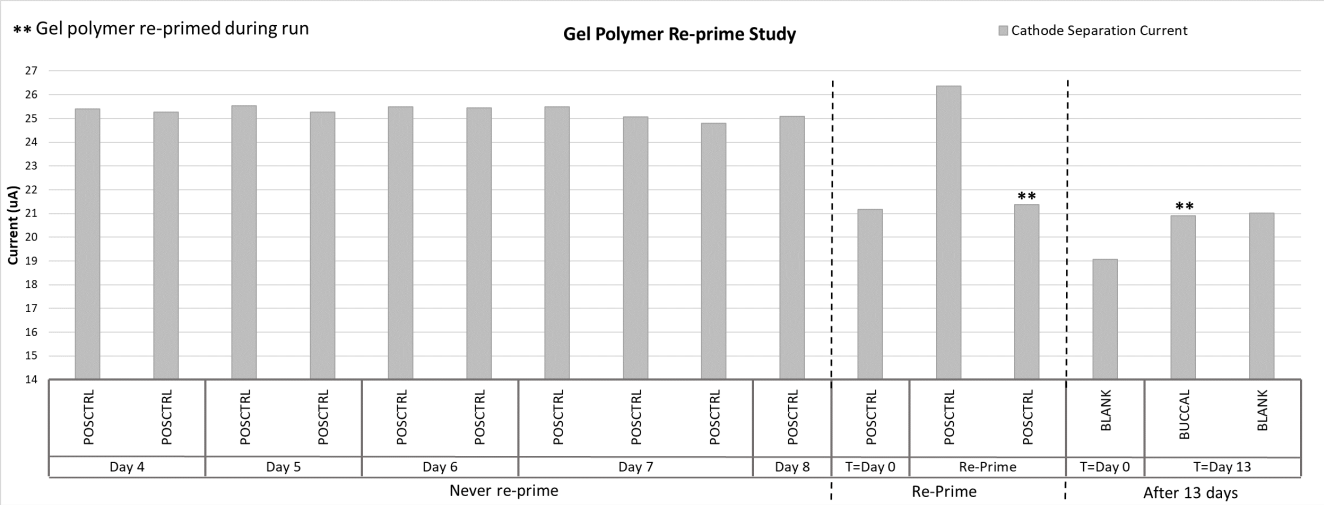


Figure 10 *Monitoring of cathode separation current for 3 usage patterns, separated by a dashed line*

In Figure 10, the first data set (left) represents a scenario where the instrument is run consistently on consecutive days. There is no idle period, and the system consumes and re-primes gel polymer as needed to perform the runs. The first data set shows that the gel polymer current measured during fragment separation remains consistent for the data labeled "Never re-prime".

The second data set (middle) demonstrates a rise in cathode separation current as the instrument sits idle. The first positive control run was run exactly before the re-prime was scheduled to take place. As expected, a rise in separation current is observed because the gel polymer is exposed to ambient conditions. The second run in the second data set occurs following a gel polymer re-priming event; the separation current decreases as a result of the re-prime event.

The third data set (right) represents an instrument that has been idle for 13 days. The first run on day 13 does not show an increase in separation current because the instrument conducted re-primes on the prescribed basis during the idle period. The 2 time points in the 13-day idle period demonstrate the effectiveness of the re-prime event in reducing conductivity in a run following an immediate gel polymer re-prime and in a subsequent run.

Similar usage pattern testing yielded comparable results when repeated on another instrument and primary cartridge.

To ensure that the system continues to perform as expected, current conductivity and crossover values were assessed for the samples described in Table 12.

Table 12 Samples assessed as part of the gel polymer priming study

	ACE GFE	ACE NGM	AB RI
Sample	383	236	0
Allelic ladder	34	37	—
Positive control	94	63	23
Negative control	28	14	3

As shown in Figure 11, the data were evaluated and compared in 2 subsets:

- Unmodified data—Represent conditions before the modification
- Modified data—Represent the new gel polymer management change

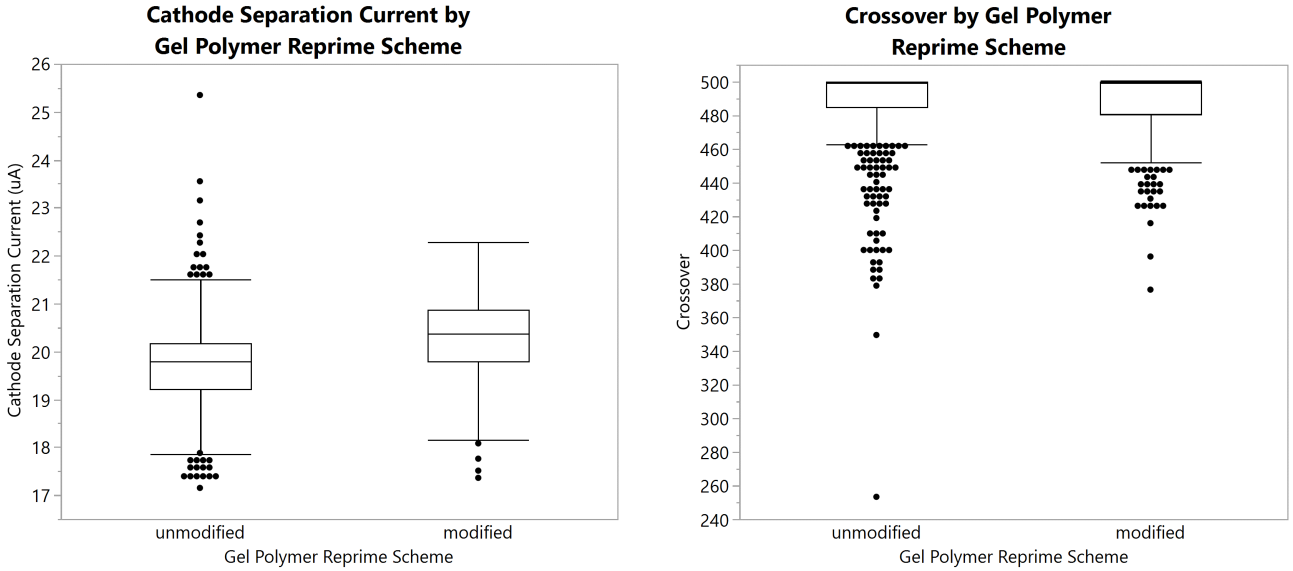


Figure 11 Separation current (left) and crossover (right) values compared by the amount of time between gel polymer priming events

The results demonstrate comparable performance between the unmodified and modified data sets when current and crossover values were evaluated. The outlier data points in the crossover graph are described in “Buffer and water stability testing” on page 83.

Concordance, accuracy, peak heights, and peak height ratio data points were also evaluated. Accuracy data is summarized in “Systematic Allelic Ladder Library” on page 85. All allele calls were determined to be concordant to previously typed, known genotypes. Figure 12 shows the peak height and peak height ratios for ACE GFE (left) and ACE NGM (right). The values are in line with expectations for the presently validated system.

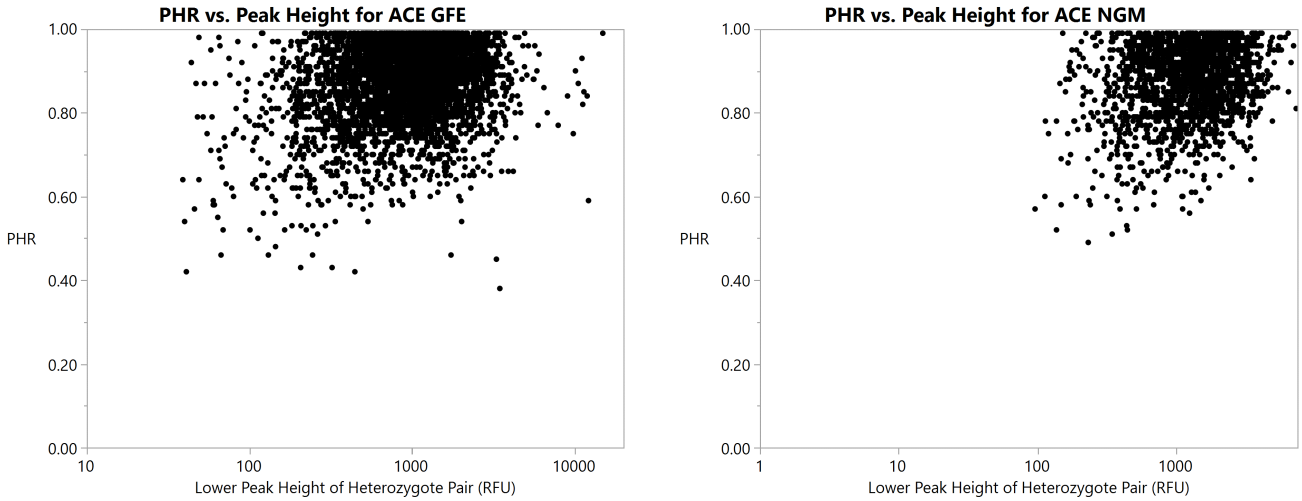


Figure 12 Distribution of the peak height ratio (PHR) for ACE GFE (left) and ACE NGM (right)

Graphed by the lower peak height of the 2 heterozygote alleles on the X-axis vs. the PHR on the Y-axis.

Primary cartridge stability testing

Before RapidHIT™ ID System v1.3.2, primary cartridges were dated for expiration 8 months after the date of manufacture. During development of the RapidHIT™ ID System v1.3.2, an assessment was performed to determine if the expiration date could be extended.

Gel polymer stability testing

Testing was performed using 3 primary cartridges loaded with gel polymer that was aged 41, 23, and 19 months; each cartridge was stored at 4°C until testing was performed. During fragment separation, current readings for the 41-month aged gel polymer were ~50% higher compared to the 19- and 23-month aged gel polymer measurements. Additionally, crossover values were exceptionally lower for nearly all 41-month aged gel polymer runs compared to the 19- and 23-month aged gel polymer runs. Current and crossover values were within acceptable, expected ranges for the 19- and 23-month aged gel polymer runs. Additionally, the expected results were observed for sample genotype concordance and peak heights (data not shown). The data associated with this study supports the implementation of an expiration date 12 months after the date of manufacture.

Buffer and water stability testing

Accelerated aging testing was performed through incubation at 40°C. Table 13 lists the incubation time points for the following:

- 10 primary cartridges from 2 different lots (5 cartridges per lot), without gel polymer that is normally stored separately at 4°C
- Buffer and water components from 2 different lots, stored outside the primary cartridge

Table 13 also lists the time equivalency that the 40°C storage represents for 20°C and 30°C storage

Table 13 Incubation times and storage length equivalencies for the tested components

Component	40℃ Incubation	30℃ Equivalence	30℃ Equivalence
Buffer and water	2.5 months	5 months	10 months
Primary cartridge	4.5 months	9 months	18 months
Buffer and water	5.5 months	11 months	22 months
Primary cartridge	6.5 months	13 months	26 months
Buffer and water	6.5 months	13 months	26 months

Following incubation, the components being tested were assembled into primary cartridges, as applicable, and unexpired gel polymer cartridges were loaded. The 10 primary cartridges were then run for 151 runs and the data was assessed. The results were in line with performance expectations of the presently validated system for concordance, accuracy, peak height, peak height ratio, and resolution (crossover) values (data not shown).

Non-accelerated, real-time testing was also performed using 3 primary cartridges that were stored for 13 months according to manufacturer recommendations. Before running, the primary cartridges were paired with unexpired gel polymer that was also stored according to manufacturer recommendations. Multiple samples and cartridge types were run, as listed in Table 14.

Table 14 Summary of 151 runs performed on each of the 3 primary cartridges stored at room temperature for a period of 13 months

Cartridge type		Primary Cartridge		
		1	2	3
ACE GFE	Sample	88	68	67
	Allelic ladder	1	1	2
	Positive control	12	11	18
	Negative control	5	11	4
ACE NGM	Sample	28	46	40
	Allelic ladder	1	1	1
	Positive control	10	8	9
	Negative control	2	4	4
AB RI	Positive control	4	1	6

Analysis of the results indicated separation current and crossover values in the expected ranges; see Figure 10.

The group of crossover outlier data points for Primary Cartridge 3 (inside the square) were the result of an instrument issue that was rectified by transferring the primary cartridge to a different RapidHIT™ ID instrument. Following that, performance was returned to expected levels.

Crossover outlier data points for Primary Cartridge 2 were the result of inefficient gel polymer filling as evidenced by current readings taken during the gel polymer fill process. For the purposes of scale, 17 data points were omitted from the crossover graph. Sixteen of the 17 data points had values of 0 because of sample injection failure. The number of these failures do not fall outside normal operational expectation as the data generation rate exceeded 95%. The other omitted crossover data point value was ~210 because of primer peak bleed into the read region.

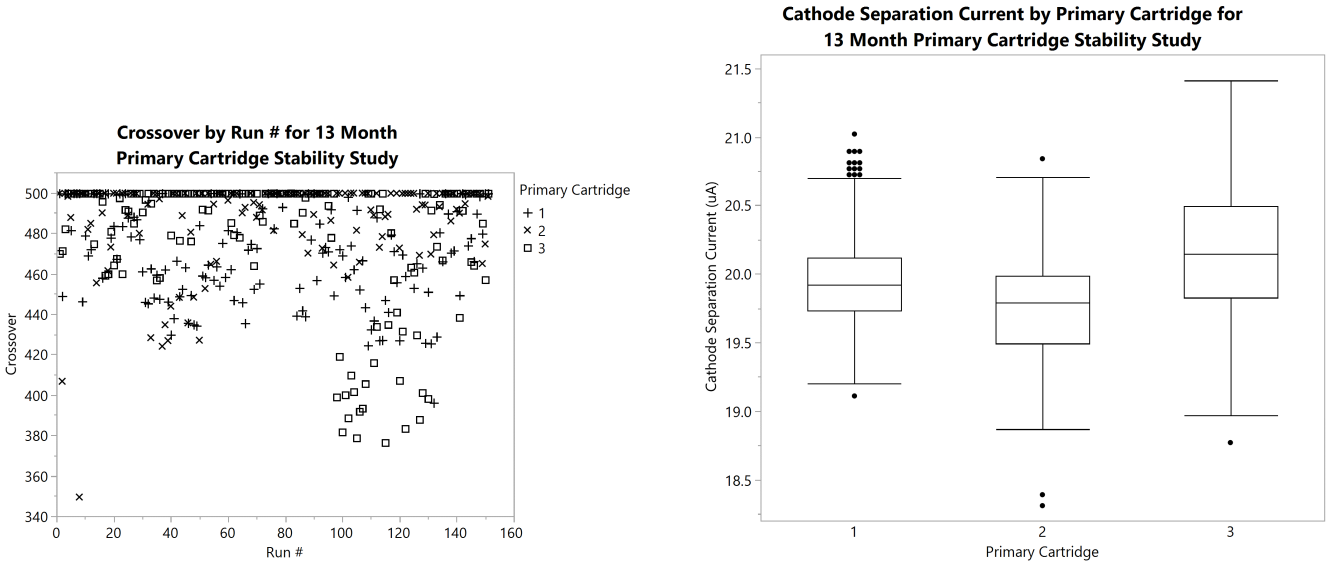


Figure 13 Crossover (left) and separation current (right) values for 3 primary cartridges stored for 13 months before use

Concordance, accuracy, peak heights, and peak height ratio data points were also evaluated. Accuracy data is summarized in "Systematic Allelic Ladder Library" on page 85. All allele calls were determined to be concordant to previously typed, known genotypes. Figure 12 shows the peak height and peak height ratios for ACE GFE (left) and ACE NGM (right). The values are in line with expectations for the presently validated system.

Based on the findings of the stability testing, 12 months after the date of manufacture was deemed a suitable expiration date for primary cartridges.

Systematic Allelic Ladder Library

Background

The size of a sample peak is calculated based on its relation to peaks in a co-migrating size standard. After sizing, the sample peak is assigned an allelic designation by matching the sample peak to a similarly sized allelic-ladder-derived bin. The migration of sample and allelic ladder peaks depends on electrophoresis conditions such as run temperature and gel polymer age, which can result in peak migration differences from run-to-run. If the migration difference between sample peaks and corresponding allelic ladder peaks is too large, allele matching can result in off-ladder (OL) or out-of-bin (OB) calls.

When using a multi-capillary system, it is possible to run allelic ladder and samples simultaneously to reduce run-to-run migration variation for proper sample genotyping. The RapidHIT™ ID System, a single-capillary system, cannot run an allelic ladder alongside the sample in an injection. Therefore, the

use of an Empirical Allelic Ladder Library that contains various ladders run under different migration conditions was applied to address variation.

A new allelic ladder library model, the Systematic Allelic Ladder Library, was introduced in RapidHIT™ ID System v1.3.1 for the RapidHIT™ ID ACE NGM SElect™ *Express* Sample Cartridge. In RapidHIT™ ID System v1.3.2, the Systematic Allelic Ladder Library has been updated for ACE NGM and expanded to include a systematic allelic ladder for the RapidHIT™ ID ACE GlobalFiler™ Express Sample Cartridge to improve genotyping accuracy and reproducibility. The model predicts the behavior of allelic ladders run under various electrophoretic conditions and is based on characterized differences for each ladder allele as a function of temperature and gel age. The Systematic Allelic Ladder Library enables the system software to predict the sizing of a ladder at any combination of the 2 run conditions.

We evaluated an expansive dataset to validate the Systematic Allelic Ladder Library model. The dataset consisted of the following:

- >2,000 ACE NGM runs
- >3,500 ACE GFE runs

Allelic ladders and samples were run across the recommended temperature range and gel lifespan.

Our evaluation of the model confirmed that environmental temperature and gel age were the 2 variables that most significantly affect sizing on the RapidHIT™ ID System. From the model, the Systematic Allelic Ladder Library was developed to represent prescribed combinations of temperature and gel age, taking into consideration the modified gel polymer priming scheme and adjusted expiration dates. Figure 14 and Figure 15 represent comparisons between the ladder libraries in RapidHIT™ ID System v1.3.1 and v1.3.2 along with the validation samples listed in Table 15.

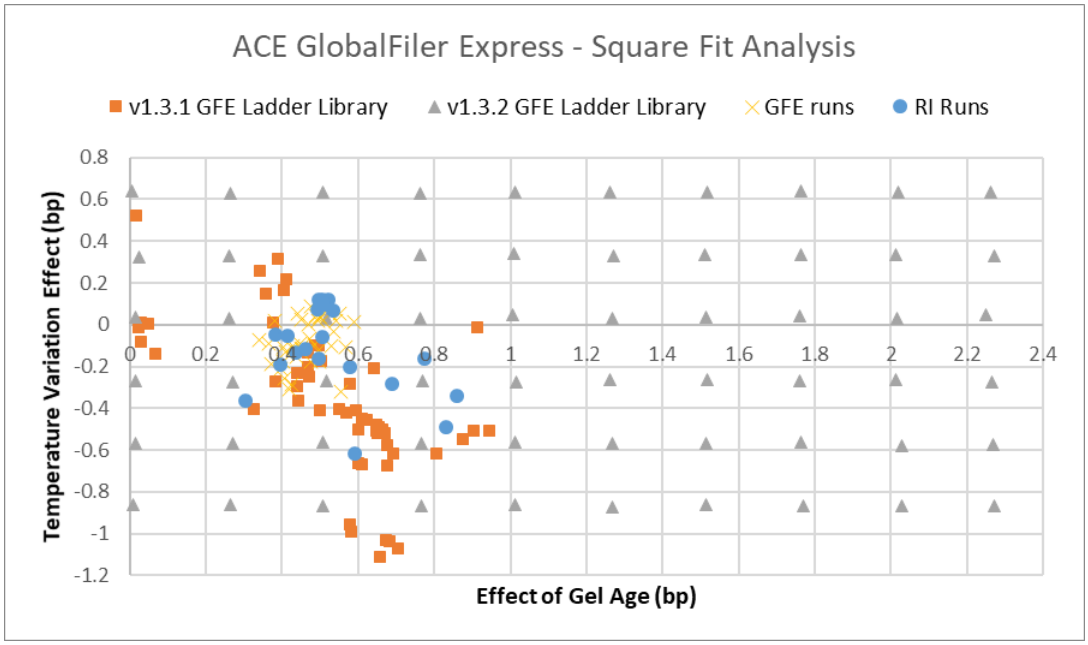


Figure 14 Square fit analysis for ACE GFE

The square fit analysis for ACE GFE demonstrates the position of the validation samples relative to the Empirical (RapidHIT™ ID System v1.3.1) and Systematic (RapidHIT™ ID System v1.3.2) Allelic Ladder Libraries.

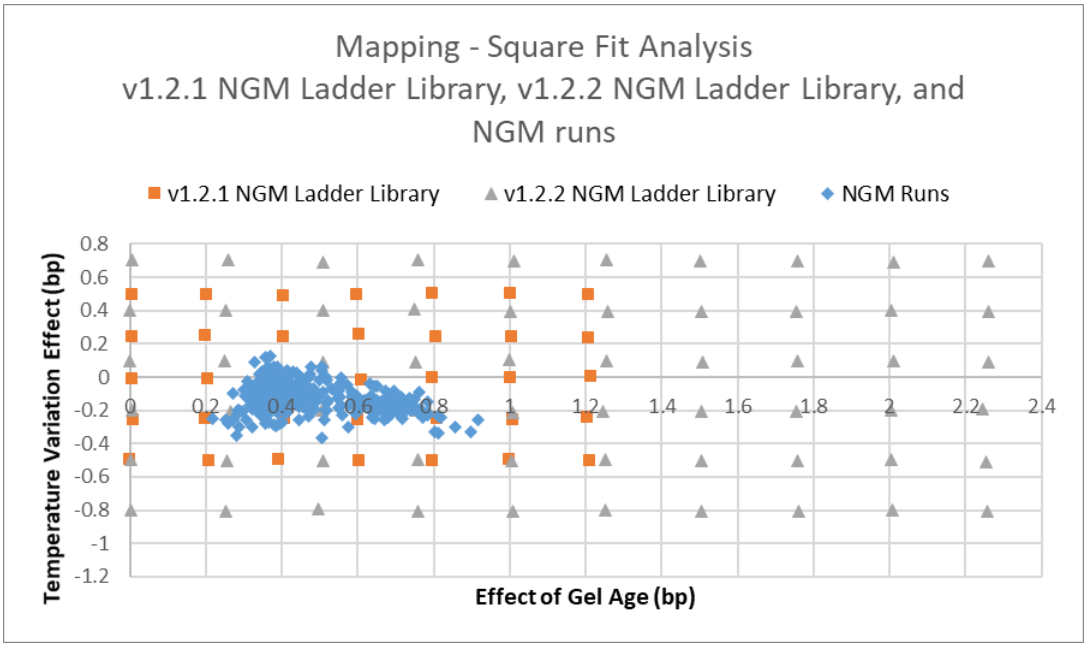


Figure 15 Square fit analysis for ACE NGM

The square fit analysis for ACE NGM demonstrates the position of the validation samples relative to the Systematic Allelic Ladder Libraries in RapidHIT™ ID System v1.3.1 and v1.3.2.

Note: The figure is from RapidHIT™ ID System v1.2.x, which is equivalent to RapidHIT™ ID System v.1.3.x.

The performance of the Systematic Allelic Ladder Library was assessed for concordance and accuracy by evaluating 468 ACE GFE and 286 ACE NGM runs, as listed in Table 15.

Table 15 Samples and positive controls analyzed during the Systematic Allelic Ladder Library validation

Cartridge type	No. of sample cartridges	No. of positive control cartridges
ACE GFE	404	64
ACE NGM	224	62

Concordance

Concordance was evaluated using expected sample genotypes from previous 3130*x*l Genetic Analyzer or 3500xL Genetic Analyzer runs. These genotypes were compared to the data obtained using RapidHIT™ ID System v1.3.2. Analysis using the Systematic Allelic Ladder Library resulted in 100% concordance for the expected genotypes.

Accuracy

The Systematic Allelic Ladder Library performance was evaluated for accuracy by comparing performance of the Empirical Allelic Ladder Library in RapidHIT™ ID System v1.3.1 to the Systematic Allelic Ladder Library in RapidHIT™ ID System v1.3.2 for ACE GFE and ACE NGM runs. For the comparison, each allele designation in the validation samples was evaluated. The alleles were measured for size differences between the sample and each selected allelic ladder from the ladder libraries. The smaller the difference between the observed allele size in the sample and the allele from the selected ladder indicates a closer match.

Figure 16 shows that the Systematic Allelic Ladder Library (v1.3.2) analysis resulted in a comparable performance upon the evaluation of the sample and selected ladder peaks when compared to the Empirical Allelic Ladder Library (v1.3.1). Figure 17 shows comparable performance between the Systematic Allelic Ladder Libraries used in RapidHIT™ ID System v1.3.1 and v1.3.2.

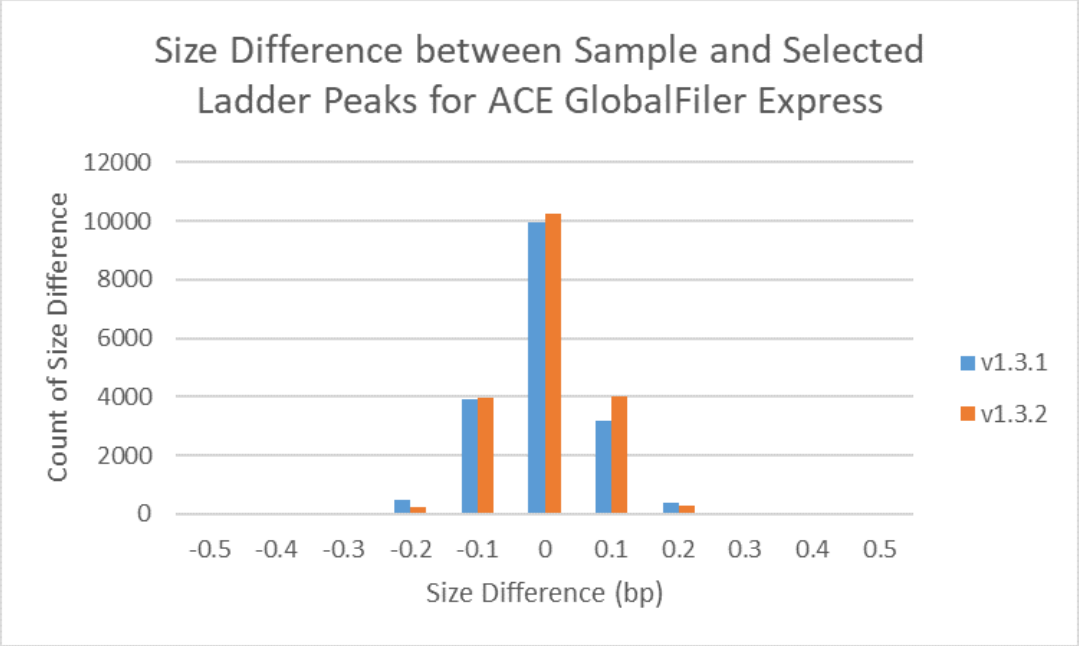


Figure 16 Size differences between sample and selected ladder peaks for ACE GFE

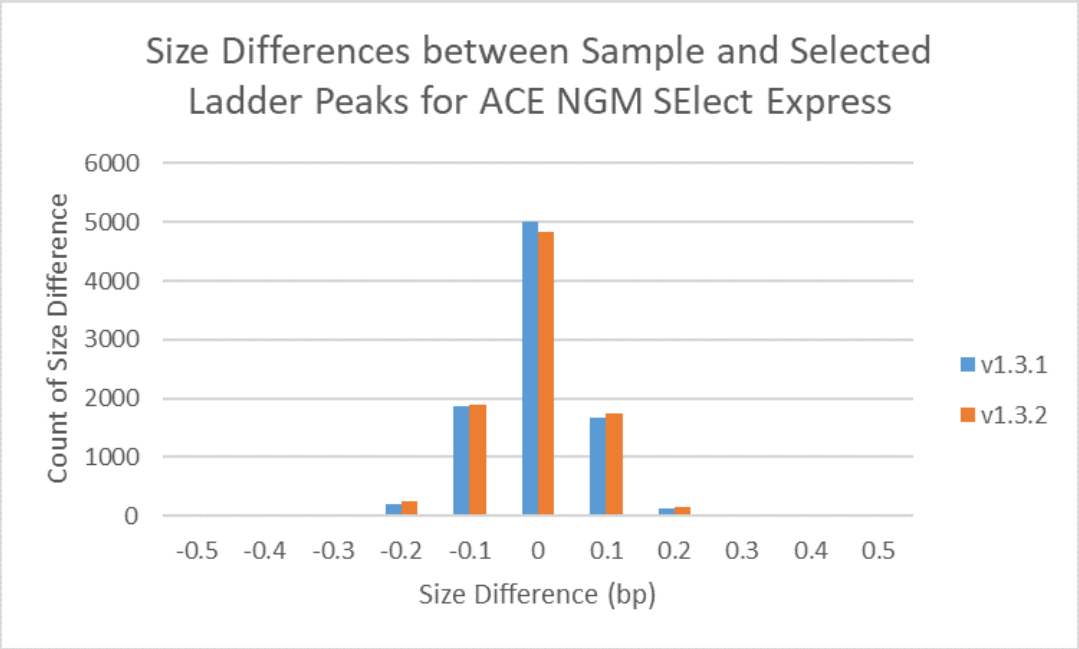


Figure 17 Size differences between sample and selected ladder peaks for ACE NGM

Absolute size difference

Because allelic ladder selection is performed on a per-sample basis, the aggregate size difference between sample peaks and selected allelic ladder peaks was another method used to compare performance. For each ladder library, the total size difference between 18,830 ACE GFE and 8,871 ACE NGM sample peaks and selected allelic ladder peaks was calculated on a per-sample basis. Because sample genotype concordance was already confirmed, absolute values of the observed size differences were used for this comparison.

Figure 16 and Figure 19 show the overall trend that allelic ladders selected from the Systematic Allelic Ladder Library in RapidHIT™ ID System v1.3.2 perform comparably to the allelic ladders selected from the ladder libraries in RapidHIT™ ID System v1.3.1.

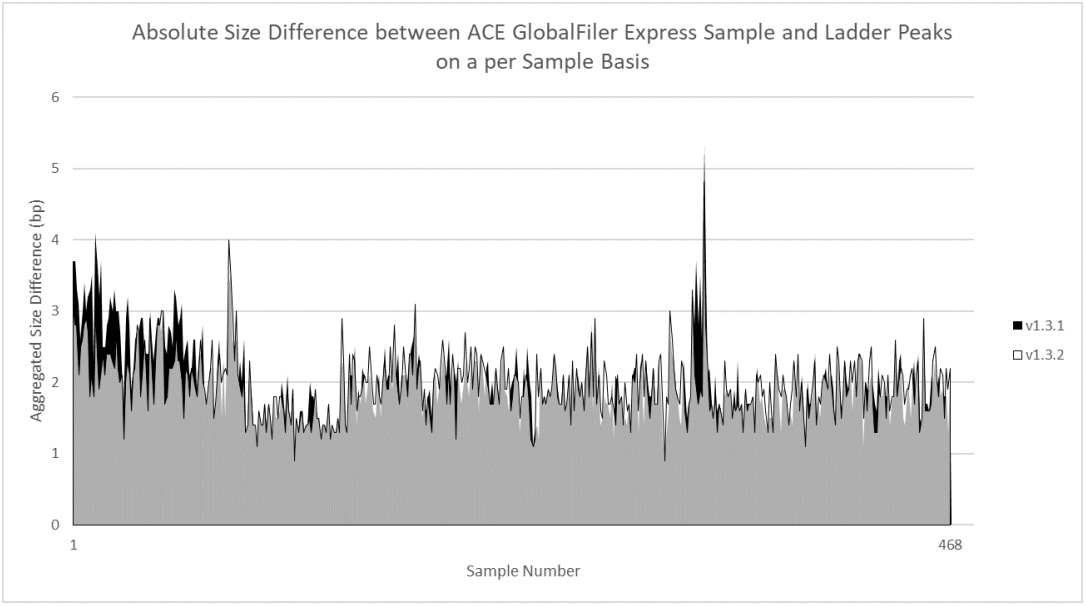


Figure 18 Absolute size difference for the ACE GFE ladder libraries

Calculated absolute size difference between sample and selected allelic ladder peaks on a per-sample basis for the ACE GFE ladder libraries. The gray areas in the plot demonstrate overlap.

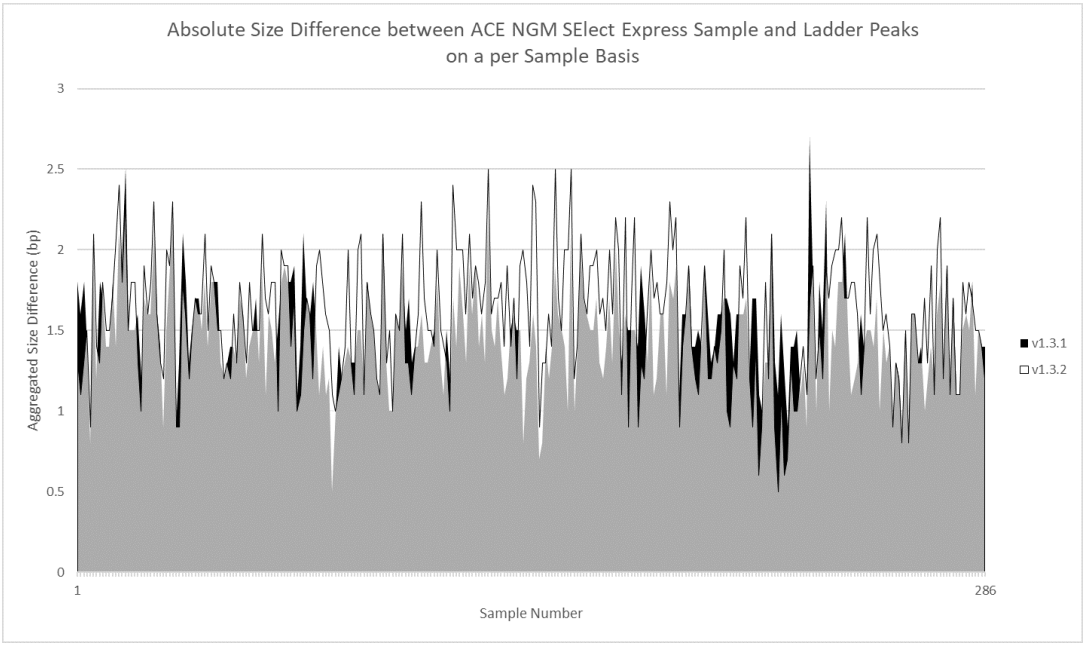


Figure 19 Absolute size difference for the ACE NGM ladder libraries

Calculated absolute size difference between sample and selected allelic ladder peaks on a per-sample basis for the ACE NGM ladder libraries. The gray areas in the plot demonstrate overlap.

Automated maintenance check

To provide additional flexibility while addressing variation in system usage patterns, RapidHIT™ ID System v1.3.2 introduces an automated maintenance check that removes the requirement for performing a weekly run. The effectiveness of the maintenance check was evaluated by testing 3 primary cartridges with system idle times ≤2.5 months. Each primary cartridge was run ~151 times to ensure that the idle time did not negatively affect sample performance. Figure 14 shows the idle time points and crossover results for each primary cartridge.

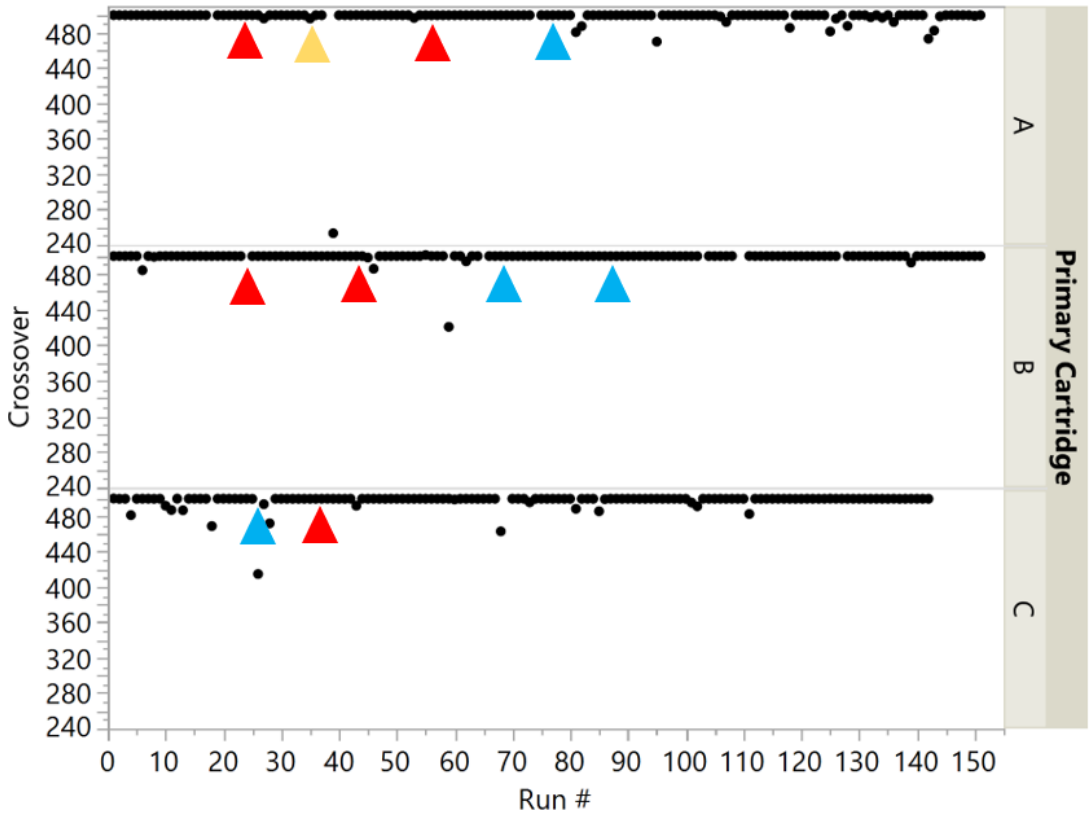


Figure 20 Maintenance mode study—Crossover vs. run number

Crossover values for 3 primary cartridges before and after periods of 1 and 2.5 months of inactivity. The red triangles indicate where usage was discontinued for a period of 1 month. The yellow triangle indicates an inactive period of 2 months.

The crossover values before and after the periods of inactivity demonstrate that the automated maintenance check does not negatively affect resolution immediately following a period of activity or over the useful life of the primary cartridge. The results for other key data points were in line with performance expectations of the presently validated system for concordance, accuracy, peak height, locus (color) balance, and peak height ratio. This study demonstrates that the maintenance procedure introduced as part of RapidHIT™ ID System v1.3.2 is effective at preserving system performance following extended periods of run inactivity.

Conclusion

The changes introduced in RapidHIT™ ID System v1.3.2 preserve robust data quality while providing increased flexibility to support a wider variety of system usage patterns. The studies performed confirm that resolution requirements are maintained with the system modifications. Additionally, the system achieves data quality consistent with past performance when evaluated for concordance, accuracy, peak height, and peak height ratio.

Software validation (RapidHIT™ ID System v1.3.2)

- Software description 93
- Materials and methods 93
- Studies, standards, and samples 94
- Functionality testing results 95
- Regression testing results 101
- Accuracy testing results 101
- Reliability testing results 105
- Conclusion 106

Software description

RapidHIT™ ID System v1.3.2 builds upon the foundation of RapidHIT™ ID System v1.3.1. Changes include a new allelic ladder library, increasing the time between gel polymer repriming events, and extending the gel polymer cartridge expiration date to 12 months.

RapidHIT™ ID System v1.3.1 is the NDIS-approved system software for laboratory analysis of single-source, buccal, reference samples using the RapidHIT™ ID ACE GlobalFiler™ Express Sample Cartridge. RapidHIT™ ID System v1.3.2 contains the same Expert System software settings as RapidHIT™ ID System v1.3.1, with the exception of a new allelic ladder library for processing RapidHIT™ ID ACE GlobalFiler™ Express buccal samples.

In addition to the RapidHIT™ ID ACE GlobalFiler™ Express Sample Cartridge, RapidHIT™ ID System v1.3.2 continues to support the RapidINTEL™ Sample Cartridge andRapidHIT™ ID ACE NGM SElect™ *Express* Sample Cartridge.

Materials and methods

To ensure that the software modification for RapidHIT™ ID System v1.3.2 is fit for purpose, an evaluation was performed to include functionality, regression, accuracy, and reliability testing. New sensitivity and precision studies are not included in this evaluation; the software functionality for v1.3.2 has not been altered in a way that impacts sensitivity or precision.

Specificity at the chemistry level was tested in the developmental validations for the GlobalFiler™ Express PCR Amplification Kit and AmpFℓSTR™ NGM SElect™ Express PCR Amplification Kit, and is not applicable to the current software modification. For validation information, see the

GlobalFiler™ Express PCR Amplification Kit User Guide (Pub. No. 4477672) and *AmpFℓSTR™ NGM SElect™ Express PCR Amplification Kit User Guide* (Pub. No. 4474109).

Cartridge types are abbreviated throughout this appendix, as shown in Table 16.

Table 16 Cartridge type abbreviations used in this appendix

Type	Abbreviation
RapidHIT™ ID ACE GlobalFiler™ Express	ACE GFE sample, allelic ladder, or control cartridge
RapidINTEL™	AB RI sample or control cartridge (uses ACE GFE allelic ladder cartridge)
RapidHIT™ ID ACE NGM SElect™ *Express*	ACE NGM sample, allelic ladder, or control cartridge

Studies, standards, and samples

Table 17 describes the samples for the evaluation. The samples are applicable to the specific cartridge being tested.

Table 17 Overview of the RapidHIT™ ID System v1.3.2 evaluation

Test parameter	QAS reference	Samples
Functionality	8.9.3.1 8.9.3.2	96 total runs were performed on RapidHIT™ ID System v1.3.2 across 3 instruments. For each instrument, the distribution of controls, allelic ladders, and samples was as follows: • 6 positive control cartridges—1 passing and 1 failing for ACE GFE, ACE NGM, and AB RI control cartridges • 6 negative control cartridges—1 passing and 1 failing for ACE GFE, ACE NGM, and AB RI control cartridges • 1 ACE GFE allelic ladder cartridge • 1 ACE NGM allelic ladder cartridge • 7 ACE GFE sample cartridges—Buccal samples • 7 AB RI sample cartridges—Saliva diluted 1:1 with PBS; 4 µL of the 1:1 dilution were pipetted directly onto the swab and dried before processing • 4 ACE NGM sample cartridges—Buccal samples
Regression	8.9.3.1 8.9.3.2	The following sample cartridges were processed with RapidHIT™ ID System v1.3.1 and v1.3.2 on 1 instrument: • ACE GFE sample cartridges—20 buccal reference samples • AB RI sample cartridges—10 blood and 10 saliva samples of varying sample input volumes • ACE NGM sample cartridges—20 buccal reference samples

Table 17 Overview of the RapidHIT ID System v1.3.2 evaluation *(continued)*

Test parameter	QAS reference	Samples
Accuracy	8.9.3.2	57 total ACE GFE runs were performed on RapidHIT™ ID System v1.3.2 across 3 instruments. For each instrument, the distribution of controls and samples was as follows: • 1 positive control cartridge—Passing • 7 ACE GFE sample cartridges—Buccal samples • 7 AB RI sample cartridges—Saliva diluted 1:1 with PBS; 4 µL of the 1:1 dilution were pipetted directly onto the swab and dried before processing • 4 ACE NGM sample cartridges—Buccal samples
Reliability	8.9.3.1 8.9.3.2	The following were processed with RapidHIT™ ID System v1.3.2: • Sample runs: – 8 controls from the functionality study – 10 samples from the functionality study • Data reprocessing runs—20 passing samples for each sample cartridge type
Precision	8.9.3.2	Not applicable
Sensitivity	8.9.3.2	
Specificity	8.9.3.2	

Functionality testing results

To confirm that the software performed sample processing and analysis as expected, 96 runs consisting of samples, allelic ladders, and controls (using ACE GFE, ACE NGM, and AB RI cartridges) were run on 3 instruments, as shown in Table 18.

Each cartridge type was correctly identified, the appropriate run protocol was selected, and the applicable data analysis thresholds were applied.

Control and allelic ladder testing

- **Positive control cartridges (6 per instrument)**—For each cartridge type, a sample swab was inserted into a positive control cartridge. In all instances, the software correctly identified a discordant profile for the positive control and reported a failing result (✗). All other positive controls passed with the expected outcomes.
- **Negative control cartridges (6 per instrument)**—For each cartridge type, a positive control swab was inserted into a negative control cartridge. In all instances, the software correctly reported a failing result (✗) because of the presence of DNA. All other negative controls passed with the expected outcomes, except the ACE NGM cartridges that failed because of the presence of dye artifacts.
- **Allelic ladder cartridges (2 per instrument)**—1 ACE GFE and 1 ACE NGM allelic ladder cartridge was run appropriately. The software reported passing results (✓) for each instrument.

Note: The same allelic ladder cartridge is used to size ACE GFE and AB RI samples.

Sample testing

- **ACE GFE sample cartridges (7 per instrument)**—20 of 21 buccal samples met the defined thresholds and produced passing results (✓). One sample was flagged (✗) because of split peaks.
- **AB RI sample cartridges (7 per instrument)**—20 of 21 saliva samples generated appropriate flags, based on the data and analysis parameters. One sample failed to generate allele data. Flags tested included IHE, OB, IMB, and IHO. All samples resulted in concordant genotype results compared to known reference data.
- **ACE NGM sample cartridges (4 per instrument)**—10 of 12 buccal samples met the defined thresholds and produced passing results (✓). Two samples were flagged (✗) because of dye artifacts.

Table 18 Functionality tests

Runs performed to evaluate the functionality of RapidHIT™ ID System v1.3.2, using ACE GFE, ACE NGM, and AB RI sample and control cartridges.

Sample number	Cartridge type	RHID-1 result	RHID-2 result	RHID-3 result	Flag details (listed by instrument)
1	ACE GFE sample	✓	✓	✗	RHID-3: OB (SE33) peaks called because of split peaks
2		✓	✓	✓	—
3		✓	✓	✓	—
4		✓	✓	✓	—
5		✓	✓	✓	—
6		✓	✓	✓	—
7		✓	✓	✓	—
8	ACE GFE positive control	✓	✓	✓	—
9		✗	✗	✗	RHID-1, RHID-2, RHID-3: Control swab was removed and replaced with sample swab
10	ACE GFE negative control	✓	✓	✓	—
11		✗	✗	✗	RHID-1, RHID-2, RHID-3: Positive control swab processed in cartridge
12	ACE GFE allelic ladder	✓	✓	✓	—
13	AB RI positive control	✓	✓	✓	—
14		✗	✗	✗	RHID-1, RHID-2, RHID-3: Control swab was removed and replaced with sample swab
15	AB RI negative control	✓	✓	✓	—
16		✗	✗	✗	RHID-1, RHID-2, RHID-3: Positive control swab processed in cartridge

Table 18 Functionality tests *(continued)*

Sample number	Cartridge type	RHID-1 result	RHID-2 result	RHID-3 result	Flag details (listed by instrument)
17	AB RI saliva sample	✗	✗	✗	RHID-1: IHO (D16S539, CSF1PO, D8S1179, TH01) IMB (D13S317), IMB/IHE (D12S391), unfiltered stutter IMB/IHE/PL (D2S1338) RHID-2: IHE (D3S1358, AMEL, D2S441, D19S433, D5S818, D13S317, SE33, D10S1248, D12S391), IHO (vWA, D16S539, CSF1PO, TPOX, D8S1179, D21S11, TH01, FGA, D7S820, D1S1656), IMB/IHE (D18S51) RHID-3: IHO (D16S539, CSF1PO, TPOX, D8S1179, D21S11, TH01, FGA, D7S820, D12S391), IHE (AMEL, D18S51, D5S818), IMB/IHE (SE33, D2S1338)
18		✗	✗	✗	RHID-1: IHO (CSF1PO, TPOX, D8S1179, D2S1338), IHE (D21S11, SE33, D12S391), IMB/IHE (TH01) RHID-2: IMB/IHE (D3S1358, AMEL, D18S51), IHE (vWA, D21S11, D2S441, D19S433, TH01, D5S818, D12S391), IHO (D16S539, CSF1PO, TPOX, D8S1179, FGA, D7S820, SE33, D1S1656, D2S1338) RHID-3: IHE (D3S1358, AMEL, D21S11, D18S51, D2S441, D19S433, D22S1045, D13S317, D10S1248, D12S391, D2S1338, D13S317), IHO (D16S539, CSF1PO, D8S1179, FGA, D7S820, SE33, D1S1656), IMB/IHE (TH01)
19		✗	✗	✗	RHID-1: No profile detected RHID-2: IHE (D3S1358, AMEL, D21S11, D18S51, TH01, D12S391), IHO (D16S539, CSF1PO, TPOX, D8S1179, D7S820, SE33, D2S1338), IMB (D5S818) RHID-3: IHE (D3S1358, vWA, AMEL, D19S433, TH01, D12S391, D2S1338), IHO (CSF1PO, TPOX, D8S1179, D7S820, SE33, D1S1656), IMB/IHE (D21S11, D5S818)

Table 18 Functionality tests *(continued)*

Sample number	Cartridge type	RHID-1 result	RHID-2 result	RHID-3 result	Flag details (listed by instrument)
20	AB RI saliva sample	✗	✗	✗	RHID-1: IMB/IHE (D3S1358, D21S11, D18S51), IHE (vWA, AMEL, D19S433, D2S1338), IHO (D16S539, CSF1PO, TPOX, D8S1179, TH01, FGA, D7S820, D16S1656), unfiltered stutter IMB/IHE/PL (D12S391) RHID-2: IHE (D3S1358, vWA, AMEL, D18S51, D19S433, TH01, D5S818, SE33), IHO (D16S539, D8S1179, D21S11, FGA, D1S1656, D2S1338), IMB/IHE (D13S317, D10S1248, D12S391) RHID-3: IMB/IHE (D3S1358, D21S11, D13S317, D10S1248), IHO (D16S539, CSF1PO, TPOX, D8S1179, D7S820, D1S1656), IHE (AMEL, D18S51, TH01, D12S391, D2S1338)
21		✗	✗	✗	RHID-1: IHE (D3S1358, TH01), IHO (TPOX) RHID-2: IHO (D3S1358, D16S539, CSF1PO, TP0X, D8S1179, FGA, D7S820, D1S1656), IHE (vWA, AMEL, D21S11, D18S51, D2S441, D19S433, TH01, D22S1045, D5S818, SE33, D12S391, D2S1338) RHID-3: IMB/IHE (D3S1358, AMEL, D18S51, D22S1045, SE33), IHO (D16S539, CSF1PO, TPOX, D8S1179, TH01, FGA, D7S820, D1S1656), IHE (D21S11, D2S441, D19S433, D5S818, D13S317, D10S1248, D12S391, D2S1338)

Table 18 Functionality tests *(continued)*

Sample number	Cartridge type	RHID-1 result	RHID-2 result	RHID-3 result	Flag details (listed by instrument)
22	AB RI saliva sample	✗	✗	✗	RHID-1: Unfiltered stutter IHE/PL (D3S1358), IMB/IHE (vWA), IHE (D16S539, CSF1PO, D21S11, D18S51, TH01, SE33, D12S391), IHO (TPOX, D8S1179, D7S820, D1S656), unfiltered stutter IMB/IHE/PL (D2S441, D2S1338), drop-in IMB/IHE/PL (D13S317) RHID-2: IHE (D3S1358, vWA, AMEL, D21S11, D18S51, D2S441, D19S433, TH01, D22S1045, D13S317, SE33, D10S1248, D12S391), IHO (D16S539, CSF1PO, TPOX, D8S1179, FGA, D5S818, D1S1656, D2S1338), IMB/IHE (D7S820) RHID-3: IHE (vWA, AMEL, D5S818, SE33, D10S1248, D12S391, D2S1338), IHO (D16S539, CSF1PO, TPOX, D8S1179, FGA, D7S820), IMB/IHE (D18S51, TH01)
23		✗	✗	✗	RHID-1: IHO (CSF1PO, TPOX, D8S1179, D18S51, D7S820, D1S1656), IHE (D21S11, TH01, D12S391, D2S1338), IMB/IHE (SE33) RHID-2: IHE (D3S1358, D2S441, D19S433, D22S1045, D5S818), IHO (vWA, D16S539, CSF1PO, D8S1179, D21S11, D18S51, FGA, D13S317, D7S820, D1S1656, D12S391, D2S1338), unfiltered stutter IMB/IHE/PL (D10S1248) RHID-3: IHE (D3S1358, vWA, AMEL, D2S441, D19S433, TH01, SE33, D10S1248, D12S391, D2S1338), IHO (CSF1PO, TPOX, D7S820, D1S1656), IMB/IHE (D21S11, D18S51, D5S818)
24	ACE NGM positive control	✓	✓	✓	—
25		✗	✗	✗	RHID-1, RHID-2, RHID-3: Control swab was removed and replaced with sample swab
26	ACE NGM negative control	✗	✗	✗	RHID-1, RHID-2, RHID-3: Unfiltered dye artifact
27		✗	✗	✗	RHID-1, RHID-2, RHID-3: Positive control swab processed in cartridge

Table 18 Functionality tests *(continued)*

Sample number	Cartridge type	RHID-1 result	RHID-2 result	RHID-3 result	Flag details (listed by instrument)
28	ACE NGM allelic ladder	✓	✓	✓	—
29	ACE NGM sample	✓	✓	✓	—
30		✗	✓	✓	RHID-1: Unfiltered dye artifact
31		✓	✓	✗	RHID-3: Unfiltered dye artifact
32		✓	✓	✓	—

Regression testing results

Regression testing was performed to confirm that the updates included in RapidHIT™ ID System v1.3.2 do not impact the software functionality. Twenty passing samples were selected to evaluate performance. The samples were processed with RapidHIT™ ID System v1.3.1, then the same data set was reprocessed with RapidHIT™ ID System v1.3.2.

The results from all samples were concordant for allele call, peak height, and base pair (bp) size between the software versions, as shown in Table 19.

Table 19 Allele call, peak height, and base pair size concordance

Allele call, peak height, and bp size concordance evaluated for 20 samples per sample cartridge type in previous (v1.3.1) and updated (v1.3.2) software versions

Sample cartridge type	Previous software version	Allele call, peak height, and size concordance to v1.3.2
ACE GFE	v1.3.1	100%
AB RI	v1.3.1	100%
ACE NGM	v1.3.1	100%

Accuracy testing results

Sizing accuracy was measured by calculating the difference, in base pair (bp), of each sample allele compared to the equivalent allelic ladder allele. This calculation was performed with the processed sample runs, then evaluated by reprocessing the same data in RapidHIT™ ID System v1.3.1 and comparing the accuracy results for regression analysis.

Sample runs

Each allele in the functionality test samples was evaluated for precision. Samples that did not generate data, such as negative controls and modified (swab removed) positive controls, were not included.

The size difference between a sample allele and the associated ladder allele was evaluated for the ACE GFE, AB RI, and ACE NGM cartridges. Size differences <0.4 bp promote accurate genotyping. The maximum difference for each allele compared to the allelic ladder is ≤0.3 bp for all samples and controls tested with RapidHIT™ ID System v1.3.2. See Figure 21 to Figure 23.

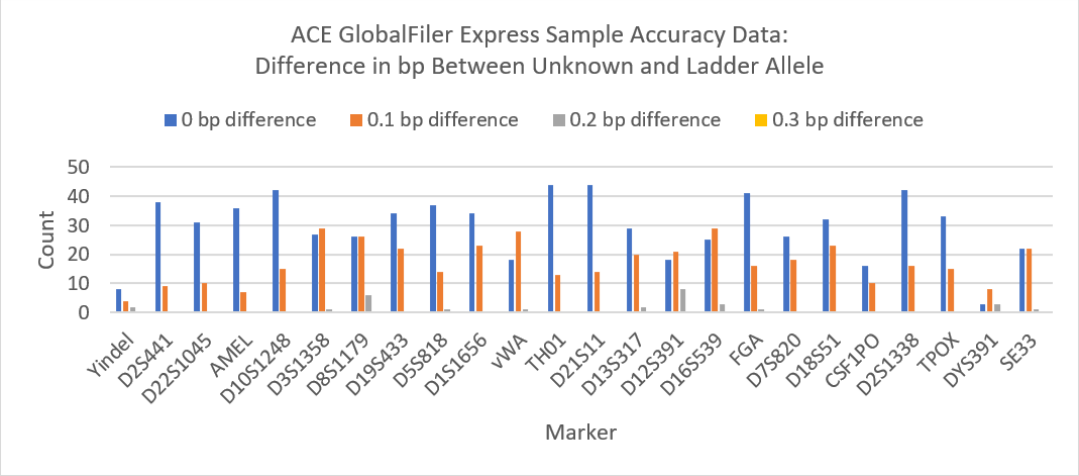


Figure 21 Sample accuracy data—ACE GFE cartridges

Samples and positive controls processed with ACE GFE cartridges across 3 RapidHIT™ ID instruments. The data is organized per marker and runs from the smallest to the largest marker in terms of bp size.

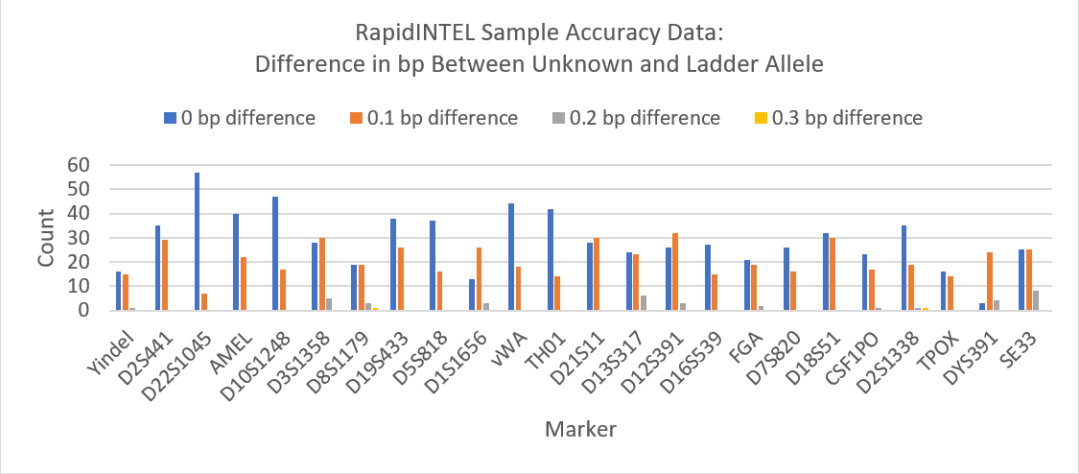


Figure 22 Sample accuracy data—AB RI cartridges

Samples and positive controls processed with AB RI cartridges across 3 RapidHIT™ ID instruments. The data is organized per marker and runs from the smallest to the largest marker in terms of bp size.

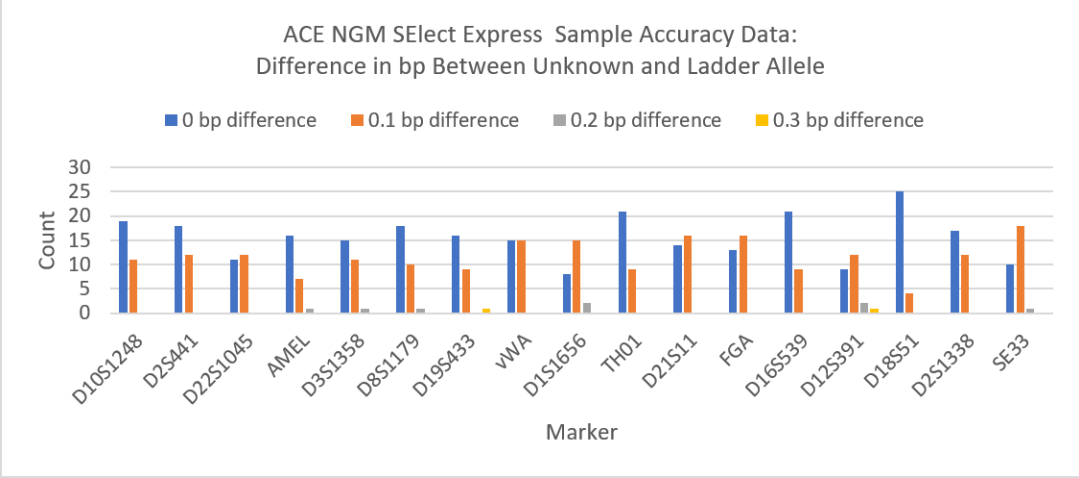


Figure 23 Sample accuracy data—ACE NGM cartridges

Samples and positive controls processed with ACE NGM cartridges across 3 RapidHIT™ ID instruments. The data is organized per marker and runs from the smallest to the largest marker in terms of bp size.

Data reprocessing

The regression test data for each cartridge type analyzed with the v1.3.1 and v1.3.2 software versions were evaluated for accuracy. Size differences were measured between the sample and each selected allelic ladder from the ladder libraries in each software version. Because the software performs allelic ladder selection on a per-sample basis, the aggregate size difference between sample peaks and selected allelic ladder peaks was used to compare performance. For each ladder library, the total size difference between sample peaks and selected allelic ladder peaks was calculated on a per-sample basis. Absolute values of the observed size differences were compared.

The results demonstrate that accuracy remains consistent when comparing the software modification from RapidHIT™ ID v1.3.1 to v1.3.2, including the introduction of the new allelic ladder library. See Figure 24 to Figure 26.

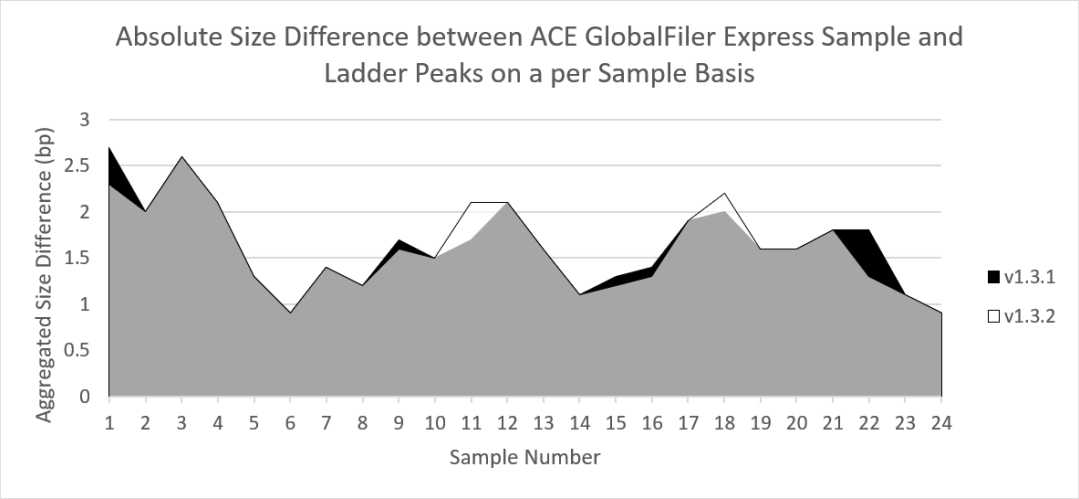


Figure 24 Absolute size difference—ACE GFE cartridges

Samples and positive controls processed with ACE GFE cartridges across 3 RapidHIT™ ID instruments. Aggregate absolute size differences are shown on a per-sample basis between the v1.3.1 and v1.3.2 software versions. The gray area in the plot demonstrates overlap.

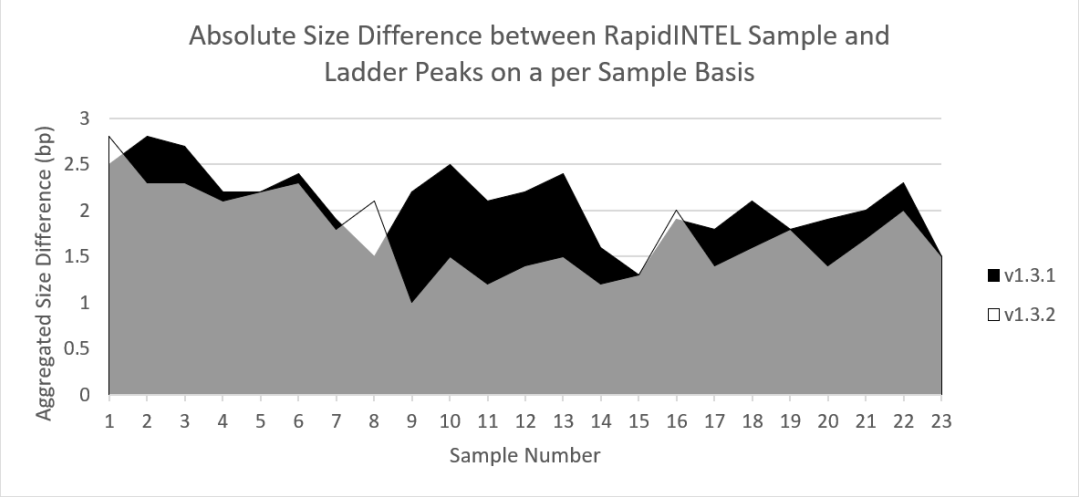


Figure 25 Absolute size difference—AB RI cartridges

Samples and positive controls processed with AB RI cartridges across 3 RapidHIT™ ID instruments. Aggregate absolute size differences are shown on a per-sample basis between the v1.3.1 and v1.3.2 software versions. The gray area in the plot demonstrates overlap.

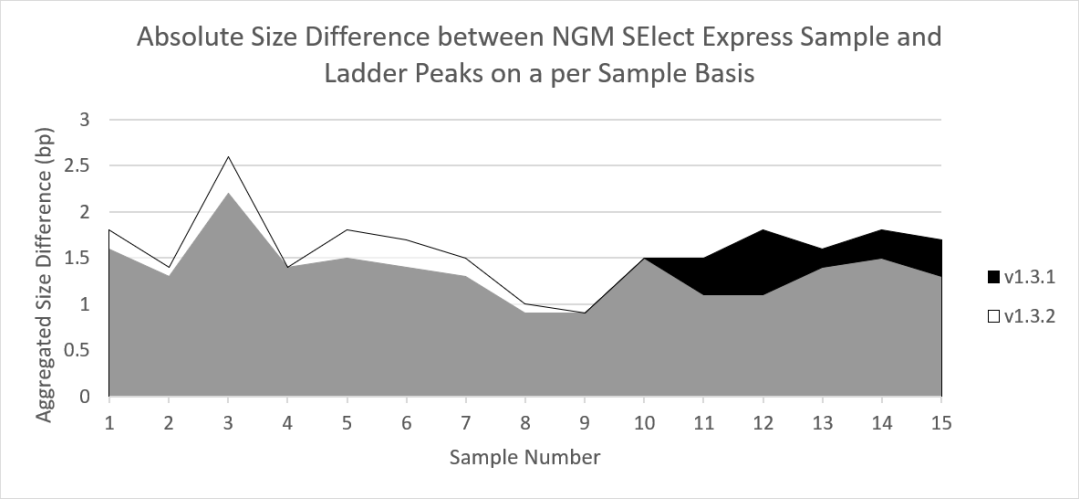


Figure 26 Absolute size difference—ACE NGM cartridges

Samples and positive controls processed with ACE NGM cartridges across 3 RapidHIT™ ID instruments. Aggregate absolute size differences are shown on a per-sample basis between the v1.3.1 and v1.3.2 software versions. The gray area in the plot demonstrates overlap.

Reliability testing results

To ensure that the updated software is performing appropriately, a variety of user scenarios were tested to evaluate the software within and beyond functional aspects; see Table 20. Functional and non-functional tasks from instrument sign-in through secondary software integration performed as expected.

Table 20 Test cases to evaluate the reliability of RapidHIT™ ID System v1.3.2

Functional and non-functional testing evaluated critical aspects of the software performance.

Category	Test	Performed as expected (Y/N)
Instrument sign-in	The system verifies biometric (fingerprint) sign-in information against a database and grants access appropriately.	Y
	If two-factor authentication is enabled, the system requires a second form of authentication for sign-in.	Y
	The instrument does not run without identity authentication and the user is tracked by the instrument.	Y
	The security roles and permissions are fully functional.	Y
	The user is prompted if a sample cartridge is inserted without entering or scanning the barcode. The control and allelic ladder cartridges can be processed without scanning the barcode.	Y
Non-functional testing	The user can install and/or upgrade RapidHIT™ ID System v1.3.2 with Expert System v1.1.	Y

Table 20 Test cases to evaluate the reliability of RapidHIT ID System v1.3.2 *(continued)*

Category	Test	Performed as expected (Y/N)
Non-functional testing	The user cannot start a run if space is not available to store the run on RapidHIT™ ID System v1.3.2.	Y
Functional testing (2 instruments)	The following functions were evaluated to ensure that software changes did not affect the workflow: • Instrument settings • User management • Authorization workflow • Consumables chart • Workflow management chart • Run reports • Table reports to PDF or CSV format • Consistency of key table and button functionalities across the software UI • Data analysis review pipeline using GeneMarker™ HID STR Human Identity Software	Y
Run results	A sample generates one of the following results when a run is complete:  ,  , or  .	Y
Installation	The RapidHIT™ server/client installation is successful.	Y

Conclusion

Following the thorough evaluation of RapidHIT™ ID System v1.3.2, the modified software demonstrated the ability to maintain the functionality, accuracy, and reliability of RapidHIT™ ID System v1.3.1 while integrating a new allelic ladder library and gel polymer management capabilities.

Upon consideration of the data described here, RapidHIT™ ID System v1.3.2 with Expert System v1.1 is deemed to generate data similar to RapidHIT™ ID System v1.3.1, which was previously approved by NDIS as an Expert System.

Software validation (RapidHIT™ ID System v1.3.3)

- Software description 107
- Cartridge type abbreviations 108
- Validation overview 108
- Functionality testing results 110
- Regression testing results 114
- Accuracy testing results 115
- Reliability testing results 116
- Amelogenin verification 117
- Conclusion 117

Software description

RapidHIT™ ID System v1.3.3 builds upon the foundation of RapidHIT™ ID System v1.3.2. The updates to the software address usability improvements and field-reported issues described in the RapidHIT™ ID System v1.3.3 Release Notes and this user guide.

Based on customer feedback, we reintroduced Amelogenin as a locus used for profile evaluation and inclusion in the CMF file. Additionally, a Y-only allele at Amelogenin that meets peak height thresholds will result in a yellow X () on the instrument and an empty CMF file in the run folder. Amelogenin was removed from evaluation and inclusion in the CMF file in RapidHIT™ ID System v1.3.2 because the logic was not built to flag a Y-only allele that resulted in a green checkmark () on the instrument (and CMF file) if no other marker flags were present at the 20 core loci or SE33.

There is also an update to the instrument run method that repositions the sample before electrokinetic injection. This change minimizes a common failure mode encountered during sample injection.

RapidHIT™ ID System v1.3.2 is the last NDIS-approved system software for laboratory analysis of single-source, buccal, reference samples using the RapidHIT™ ID ACE GlobalFiler™ Express Sample Cartridge. RapidHIT™ ID System v1.3.3 contains the same Expert System software settings as RapidHIT™ ID System v1.3.2.

Cartridge type abbreviations

Cartridge types are abbreviated throughout this appendix, as shown in Table 21.

Table 21 Cartridge type abbreviations used in this appendix

Type	Abbreviation
RapidHIT™ ID ACE GlobalFiler™ Express	ACE GFE sample, allelic ladder, or control cartridge
RapidINTEL™	AB RI sample or control cartridge (uses ACE GFE allelic ladder cartridge)
RapidHIT™ ID ACE NGM SElect™ *Express*	ACE NGM sample, allelic ladder, or control cartridge

Validation overview

Functionality, regression, accuracy, and reliability testing were completed to ensure that the software update for RapidHIT™ ID System v1.3.3 performs as expected. Verification testing was performed at the conclusion of the validation testing to confirm the expected performance of the Amelogenin-related update.

Test parameters, QAS section reference, and samples used during the software validation are summarized in Table 22.

Software functionality for v1.3.3 has not been altered to impact sensitivity, precision, or specificity; therefore, these studies are not included in this evaluation.

- Sensitivity and precision testing were performed during sample cartridge developmental validation for RapidHIT™ ID ACE GlobalFiler™ Express (Barican et al. 2017)[1] and RapidINTEL™ cartridges (Pub. No. MAN0018979) and for RapidHIT™ ID ACE NGM SElect™ *Express* cartridges (Pub. No. MAN0018471).
- Specificity testing was performed at the chemistry level during developmental validation for the GlobalFiler™ Express PCR Amplification Kit (Pub. No. 4477672) and *AmpFℓSTR™ NGM SElect™ Express PCR Amplification Kit User Guide* (Pub. No. 4474109).

[1] Barican A, Buscaino J, Salceda S, et al. (2017) Validation of a rapid DNA process with the RapidHIT™ ID System using GlobalFiler™ Express chemistry, a platform optimized for decentralized testing environments. *Forensic Sci Int Genet* 28:21–34.

Table 22 Software validation study and sample overview

Test parameter	QAS reference	Samples
Functionality	8.9.3.1 8.9.3.2	90 total runs were performed on RapidHIT™ ID System v1.3.3 across 2 instruments. For each instrument, the distribution of controls, allelic ladders, and samples was as follows: **Controls** • 3 ACE GFE positive control cartridges (including 1 modified failure[1]) • 3 ACE GFE negative control cartridges (including 1 modified failure[1]) • 3 AB RI positive control cartridges (including 1 modified failure[1]) • 3 AB RI negative control cartridges (including 1 modified failure[1]) • 2 ACE NGM positive control cartridges (including 1 modified failure[1]) • 2 ACE NGM negative control cartridges (including 1 modified failure[1]) **Ladders** • 2 ACE GFE allelic ladder cartridges • 2 ACE NGM allelic ladder cartridges **Samples** • ACE GFE sample cartridges—10 buccal samples • AB RI sample cartridges – 5 blood samples: 2 µL of blood spotted on a glass slide, dried, swabbed with a wet cotton swab, dried before processing – 5 saliva samples: 2 µL of saliva diluted 1:1 with PBS, 4 µL of the dilution pipetted directly onto the swab, dried before processing • ACE NGM sample cartridges—5 buccal samples
Regression	8.9.3.1 8.9.3.2	The following sample cartridges were processed with the previous (v1.3.2) and updated (v1.3.3) software versions: • ACE GFE sample cartridges—20 buccal reference samples • AB RI sample cartridges—20 saliva samples of varying sample input volumes • ACE NGM sample cartridges—5 buccal reference samples
Accuracy	8.9.3.2	Control, allelic ladder, and sample runs from the functional study (excluding negative controls and modified positive controls)

Table 22 Software validation study and sample overview *(continued)*

Test parameter	QAS reference	Samples
Reliability	8.9.3.1 8.9.3.2	Runs from the functionality, regression, and accuracy testing Additional user scenarios: • 2 RapidHIT™ ID v1.3.3 instruments • Runs were performed with ACE GFE and AB RI cartridges
Amelogenin verification	Not applicable	End-to-end testing and user scenarios were performed on 1 RapidHIT™ ID v1.3.3 instrument with ACE GFE cartridges.

[1] For descriptions of the modified controls, see "Control and allelic ladder testing" on page 110 in the "Functionality testing results" section.

Functionality testing results

A total of 90 controls, allelic ladders, and samples were processed with ACE GFE, ACE NGM, and AB RI cartridges to confirm that the software performed sample processing and analysis as expected. Two RapidHIT™ ID v1.3.3 instruments were used for testing. Each cartridge was correctly identified, the appropriate protocol selected, and applicable thresholds applied. For a summary of the run results, see Table 23.

Control and allelic ladder testing

- **Positive control cartridges**—One positive control of each cartridge type was modified, and the control swab was removed or replaced with a buccal sample swab. The software identified that the resulting profiles did not match the expected control profiles, and a red X () was generated. This test provided verification that the software could correctly identify a discordant profile for the positive control. All other positive controls passed with the expected outcomes.
- **Negative control cartridges**—For negative control testing, artifacts above the calling threshold resulted in failed runs for all 3 cartridge types. A red X () was generated in these cases, verifying negative control failure because of the presence of DNA. Other ACE GFE and AB RI negative controls passed with the expected outcomes. Both ACE NGM cartridges failed because of artifacts.
- **Allelic ladder cartridges**—ACE GFE and ACE NGM allelic ladder cartridges passed with expected outcomes, except for 1 ACE GFE ladder that failed because of an extra peak above the threshold in the size standard (ILS).

Note: The same allelic ladder cartridge is used to genotype ACE GFE and AB RI samples.

Sample testing

- **ACE GFE sample cartridges (2 instruments, 10 samples per instrument)**—Of the 20 ACE GFE samples, all met the defined thresholds and produced green checkmarks ( ), except for 1 run that was flagged with a red X ( ) because of size standard failure.
- **AB RI sample cartridges (2 instruments, 10 samples per instrument)**—Of the 20 AB RI samples, 19 generated appropriate flags, based on the data and analysis parameters. One AB RI run failed to generate allelic data.
- **ACE NGM sample cartridges (2 instruments, 5 samples per instrument)**—Of the 10 ACE NGM samples, 5 met the defined thresholds and produced green checkmarks ( ). Three runs were flagged with a yellow X ( ) because of artifacts. One sample was appropriately flagged with a yellow X ( ) on both runs because of a known microvariant.

All samples resulted in concordant genotype results as compared to known reference data.

Quality flags, including inconclusive heterozygous (IHE), inconclusive homozygous (IHO), out-of-bin (OB), and off-ladder (OL), were observed and were appropriately triggered based on the data and analysis parameters.

Table 23 Functionality tests

Summary of the runs performed and run results to evaluate the functionality of RapidHIT™ ID System v1.3.3

Sample number	Cartridge type	RHID-1 result	RHID-2 result	Yellow or red X run status and marker quality flag details
1	ACE GFE sample	✓	✓	—
2		✓	✓	—
3		✓	✓	—
4		✗	✓	RHID-1: ILS peaks 475 bp, 490 bp, and 500 bp dropped out
5		✓	✓	—
6		✓	✓	—
7		✓	✓	—
8		✓	✓	—
9		✓	✓	—
10		✓	✓	—
11	ACE GFE positive control	✓	✓	—
12		✓	✓	—
13		✗	✗	RHID-1 and RHID-2: Modified positives, expected result
14	ACE GFE negative control	✗	✗	RHID-1 and RHID-2: FAM™ 80 dye artifact
15		✗	✓	RHID-1: FAM™ 80 dye artifact
16		✓	✓	—
17	ACE GFE allelic ladder	✓	✗	RHID-2: Extra peak in ILS at 339.3 bp
18		✓	✓	—
19	AB RI sample	✓	✓	RHID-2: IHE (TH01, D2S1338)
20		✓	✓	—
21		✓	✓	RHID-1: IHO (D3S1358, D8S1179), IHE (vWA, CSF1PO, TPOX, AMEL, D21S11, D18S51, D22S1045, D5S818, D13S317, D7S820, SE33, D1S1656, D12S391, D2S1338), OB- minus stutter (D10S1248, D12S391) RHID-2: IHE (D22S1248)
22		✓	✓	—

Table 23 Functionality tests *(continued)*

Sample number	Cartridge type	RHID-1 result	RHID-2 result	Yellow or red X run status and marker quality flag details
23	AB RI sample	✓	✗	RHID-2: IHE (TH01)
24		✓	✓	—
25		✓	✓	—
26		✓	✓	—
27		✓	✗	RHID-2: Failed injection
28		✗	✓	RHID-1: 87 bp dye artifact (Yindel), IHO (AMEL, D8S1179, D21S11, D18S51, D7S820, SE33, D10S1248, D1S1656, D12S391, D2S1338), OB-minus stutter (D10S1248, D12S391)
29	AB RI positive control	✓	✓	—
30		✓	✓	—
31		✗	✗	RHID-1 and RHID-2: Modified positives, expected result
32	AB RI negative control	✓	✓	—
33		✓	✓	—
34		✗	✗	RHID-1: FAM™ 80 dye artifact RHID-2: FAM™ 82 dye artifact, D2S441 artifacts 86 bp, 90 bp, 93 bp, TAZ™ 82 dye artifact, D22S1045 dye artifact
35	ACE NGM sample	✗	✗	RHID-1 and RHID-2: OB- microvariant concordant with known reference (FGA)
36		✓	✓	—
37		✗	✓	RHID-1: OL- artifact 99.9 bp (AMEL)
38		✗	✗	RHID-1 and RHID-2: OL pull-up artifact 99.4 bp (AMEL)
39		✓	✓	—
40	ACE NGM positive control	✓	✓	—
41		✗	✗	RHID-1 and RHID-2: Modified positives, expected result

Table 23 Functionality tests *(continued)*

Sample number	Cartridge type	RHID-1 result	RHID-2 result	Yellow or red X run status and marker quality flag details
42	ACE NGM negative control	✗	✗	RHID-1: 82.3 bp dye artifact (D10S1248), 84 bp and 86.9 dye artifacts (D22S1045), 75.5 dye artifact (D2S441) RHID-2: 83 bp dye artifact (D10S1248), 81.6 bp dye artifact (D22S1045), 75.5 dye artifact (D2S441)
43		✗	✗	RHID-1: 82.2 bp dye artifact (D10S1248), 82 bp and 86.9 dye artifacts (D22S1045) RHID-2: 82.8 bp dye artifact (D10S1248), 81.7 bp dye artifact (D22S1045), 75.5 dye artifact (D2S441)
44	ACE NGM allelic ladder	✓	✓	—
45		✓	✓	—

Regression testing results

A total of 45 reference samples run on RapidHIT™ ID System v1.3.2 were reprocessed with RapidHIT™ ID System v1.3.3 to confirm that updates included in v1.3.3 do not impact proper functionality seen in v1.3.2.

The results from all samples were concordant for allele call, peak height, and base pair (bp) size between the software versions, as shown in Table 24.

Table 24 Summary of the regression testing results between v1.3.2 and v1.3.3

Cartridge type	Number of samples	Previous software version	Allele call, peak height, and size concordance to v1.3.3
ACE GFE	20	v1.3.2	100%
AB RI	20	v1.3.2	100%
ACE NGM	5	v1.3.2	100%

Accuracy testing results

Sizing accuracy was measured by calculating the base pair (bp) difference of each sample allele to the equivalent allelic ladder allele, for controls and samples run in the functionality test. Samples that did not generate data, such as negative controls and modified positive controls, were not evaluated.

Results showed that the maximum difference for each allele compared to the allelic ladder is ≤0.3 bp. Size differences <0.4 bp promote accurate genotyping. For ACE GFE, AB RI, and ACE NGM accuracy results, see Figure 27 to Figure 29.

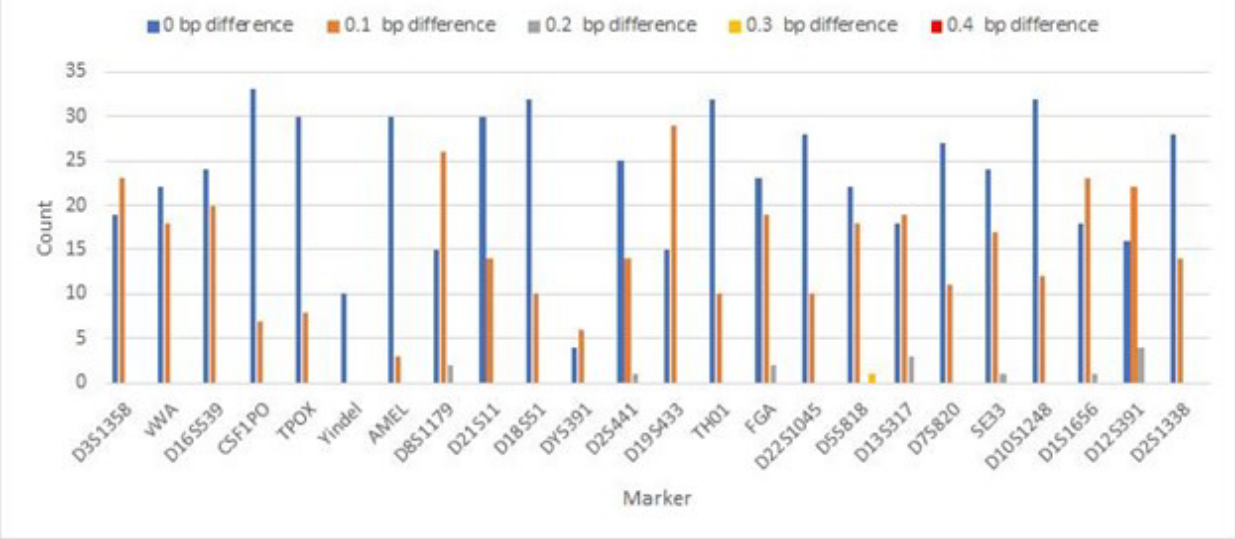


Figure 27 ACE GFE sample accuracy data: difference in bp between unknown and ladder alleles

Samples and positive controls processed with ACE GFE cartridges across 2 RapidHIT™ ID v1.3.3 instruments. The size difference in bp is shown per marker, running from the smallest to largest markers in terms of bp size

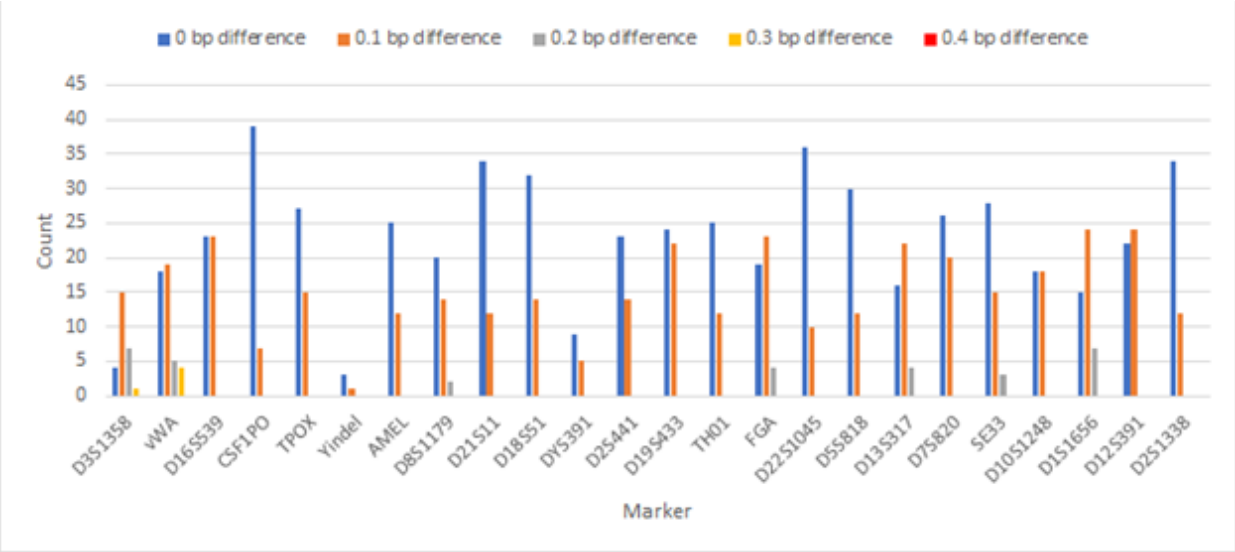


Figure 28 AB RI sample accuracy data: difference in bp between unknown and ladder alleles

Samples and positive controls processed with AB RI cartridges across 2 RapidHIT™ ID v1.3.3 instruments. The size difference in bp is shown per marker, running from the smallest to largest markers in terms of bp size

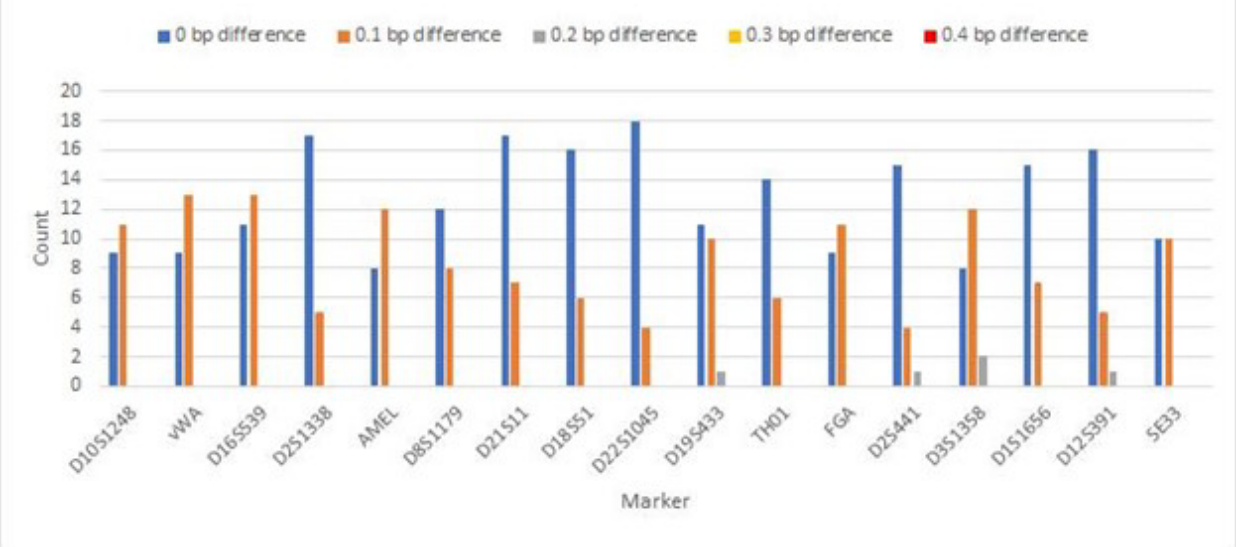


Figure 29 ACE NGM sample accuracy data: difference in bp between unknown and ladder alleles

Samples and positive controls processed with ACE NGM cartridges across 2 RapidHIT™ ID v1.3.3 instruments. The size difference in bp is shown per marker, running from the smallest to largest markers in terms of bp size.

Reliability testing results

All runs from the functionality, regression, and accuracy testing were checked to confirm the correct script version, supported kit version, and concordant RapidLINK™ Software run status. Results were as expected.

Various user scenarios were tested to evaluate critical aspects of the software's performance; see Table 25. Functional and non-functional tasks from software installation through secondary software integration performed as expected.

Table 25 Test cases to evaluate the reliability of RapidHIT™ ID System v1.3.3

Category	Test	Performed as expected (Y/N)
Installation	User can update RapidHIT™ ID System software v1.3.1, v1.3.2, and v1.3.2.2 to v1.3.3	Y
Sign-in	All user roles can sign in to the system with a single authentication method	Y
User roles and permissions	User roles and permissions are restricted where applicable, including account management, primary cartridge replacement, access to run logs, and instrument settings	Y
Maintenance	User prompts and maintenance runs are initiated as expected	Y
Run	Verify that the run started and run completed files display cartridge information	Y
	User can export files for troubleshooting from the instrument	Y

Table 25 Test cases to evaluate the reliability of RapidHIT ID System v1.3.3 *(continued)*

Category	Test	Performed as expected (Y/N)
RapidLINK™ Software	Instrument syncs with instrument/user management on the RapidLINK™ Software	Y
	An administrator can enable or disable the RapidLINK™ server connection to the instrument	Y
	When the RapidLINK™ server connection is established, run data is transferred from the instrument	Y
	If the RapidLINK™ server connection is disabled, users receive a warning message that data is in the transfer queue	Y
	When the RapidLINK™ Software is integrated with an external database, the instrument prompts for a sample category for all sample cartridge types before the run	Y

Amelogenin verification

We confirmed the following scenarios:

- Amelogenin is listed in the configuration XML file for the instrument application.
- When the RapidLINK™ Software is integrated with an external database, the CMF file correctly includes Amelogenin.
- The user sees a green checkmark (✔) on the instrument when Amelogenin results are X, X or X, Y. A CMF file in the run folder contains Amelogenin.
- The user sees a yellow X (✖) on the instrument when Amelogenin results are Y or null. A CMF file in the run folder is empty.
- The user can perform end-to-end testing with RapidHIT™ ID System v1.3.3.

Conclusion

Following the thorough evaluation of RapidHIT™ ID System v1.3.3, the updated software demonstrated the ability to maintain the functionality, accuracy, and reliability of RapidHIT™ ID System v1.3.2. Upon considering the data described here, RapidHIT™ ID System v1.3.3 software (Expert System v1.1) generates data similar to the NDIS-approved RapidHIT™ ID System v1.3.2 software (Expert System v1.1).

Safety

WARNING! GENERAL SAFETY. Using this product in a manner not specified in the user documentation may result in personal injury or damage to the instrument or device. Ensure that anyone using this product has received instructions in general safety practices for laboratories and the safety information provided in this document.

- Before using an instrument or device, read and understand the safety information provided in the user documentation provided by the manufacturer of the instrument or device.
- Before handling chemicals, read and understand all applicable Safety Data Sheets (SDSs) and use appropriate personal protective equipment (gloves, gowns, eye protection, and so on). To obtain SDSs, visit **thermofisher.com/support**.

Symbols on this instrument

Symbols may be found on the instrument to warn against potential hazards or convey important safety information. In this document, the hazard symbol is used along with one of the following user attention words.

- **CAUTION!**—Indicates a potentially hazardous situation that, if not avoided, may result in minor or moderate injury. It may also be used to alert against unsafe practices.
- **WARNING!**—Indicates a potentially hazardous situation that, if not avoided, could result in death or serious injury.
- **DANGER!**—Indicates an imminently hazardous situation that, if not avoided, will result in death or serious injury.

Standard safety symbols

Symbol and description
CAUTION! Risk of danger. Consult the manual for further safety information.

Location of safety labels

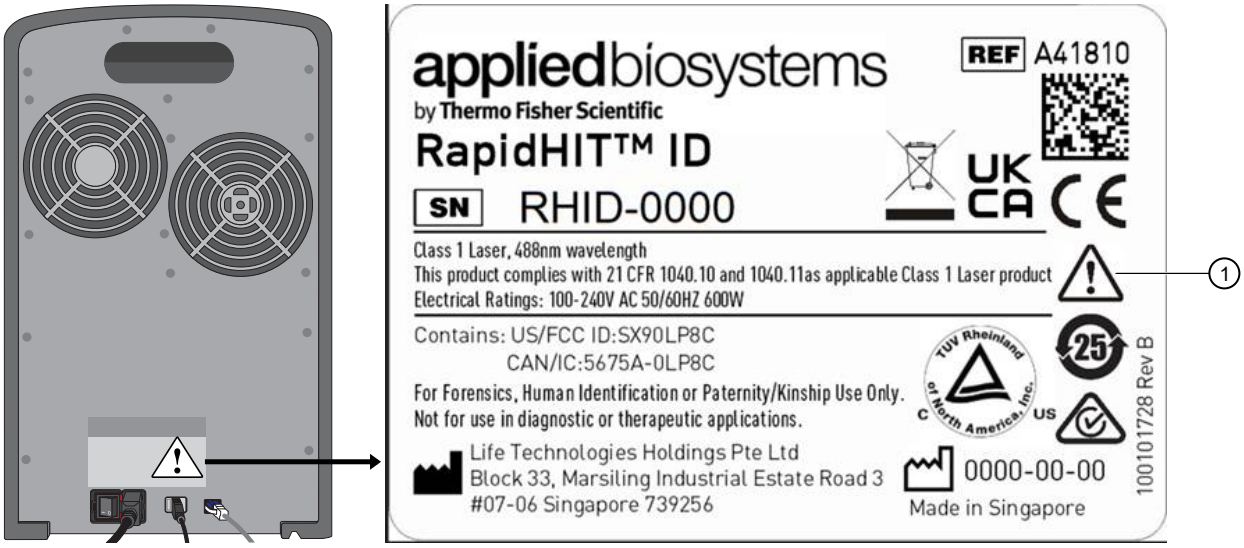


Figure 30 Rear panel

① ⚠ **CAUTION!** Risk of danger. Consult the manual for further safety information.

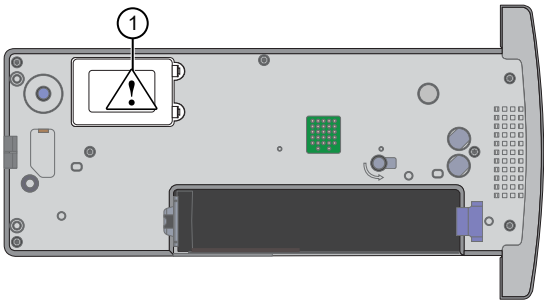


Figure 31 Primary cartridge

① ⚠ **CAUTION!** The capillary is fragile. Handle the primary cartridge with care after you remove the capillary cover.

Control and connection symbols

Symbols and descriptions	
	On (Power)
	Off (Power)
	Earth (ground) terminal

(continued)

Symbols and descriptions	
	Protective conductor terminal (main ground)
	Direct current
	Alternating current
	Both direct and alternating current

Conformity symbols

Conformity mark	Description
	Indicates conformity with safety requirements for Canada and U.S.A.
	Indicates conformity with European Union requirements.
	INDICATES CONFORMITY WITH UNITED KINGDOM REQUIREMENTS
	Indicates conformity with the WEEE Directive 2012/19/EU. **CAUTION!** To minimize negative environmental impact from disposal of electronic waste, do not dispose of electronic waste in unsorted municipal waste. Follow local municipal waste ordinances for proper disposal provision and contact customer service for information about responsible disposal options.

Safety information for instruments not manufactured by Thermo Fisher Scientific

Some of the accessories provided as part of the instrument system are not designed or built by Thermo Fisher Scientific. Consult the manufacturer's documentation for the information needed for the safe use of these products.

Instrument safety

General



CAUTION! Do not remove instrument protective covers. If you remove the protective instrument panels or disable interlock devices, you may be exposed to serious hazards including, but not limited to, severe electrical shock, laser exposure, crushing, or chemical exposure.

Physical injury



CAUTION! Moving Parts. Moving parts can crush, pinch and cut. Keep hands clear of moving parts while operating the instrument. Disconnect power before servicing.

Electrical safety



WARNING! Ensure appropriate electrical supply. For safe operation of the instrument:

- Plug the system into a properly grounded receptacle with adequate current capacity.
- Ensure the electrical supply is of suitable voltage.
- Never operate the instrument with the ground disconnected. Grounding continuity is required for safe operation of the instrument.



WARNING! Power Supply Line Cords. Use properly configured and approved line cords for the power supply in your facility.



WARNING! Disconnecting Power. To fully disconnect power either detach or unplug the power cord, positioning the instrument such that the power cord is accessible.

Cleaning and decontamination

CAUTION! Cleaning and Decontamination. Use only the cleaning and decontamination methods that are specified in the manufacturer user documentation. It is the responsibility of the operator (or other responsible person) to ensure that the following requirements are met:

- No decontamination or cleaning agents are used that can react with parts of the equipment or with material that is contained in the equipment. Use of such agents could cause a HAZARD condition.
- The instrument is properly decontaminated a) if hazardous material is spilled onto or into the equipment, and/or b) before the instrument is serviced at your facility or is sent for repair, maintenance, trade-in, disposal, or termination of a loan. Request decontamination forms from customer service.
- Before using any cleaning or decontamination methods (except methods that are recommended by the manufacturer), confirm with the manufacturer that the proposed method will not damage the equipment.

Instrument component and accessory disposal

To minimize negative environmental impact from disposal of electronic waste, do not dispose of electronic waste in unsorted municipal waste. Follow local municipal waste ordinances for proper disposal provision and contact customer service for information about responsible disposal options.

Laser safety

WARNING! LASER HAZARD. Under normal operating conditions, the RapidHIT™ ID Instrument is categorized as a Class 1 laser product. However, removing the protective covers and (when applicable) defeating the interlocks can result in exposure to the internal Class 3Bv laser. Lasers can burn the retina, causing permanent blind spots. To ensure safe laser operation:

- Never look directly into the laser beam.
- Do not remove safety labels, instrument protective panels, or defeat safety interlocks.
- The system must be installed and maintained by a Thermo Fisher Scientific Technical Representative.
- Remove jewelry and other items that can reflect a laser beam into your eyes or the eyes of other people.
- Wear proper eye protection and post a laser warning sign at the entrance to the laboratory if the laser protection is defeated for servicing.
- DO NOT operate the laser when the laser cooling fan is off. When the laser is overheated, it can cause severe burns on contact.

The following table lists laser safety symbols and alerts that may be present on the instrument.

Alert
DANGER! Class 3b (III) visible and/or invisible laser radiation present when open and interlocks defeated. Avoid exposure to beam.

Safety and electromagnetic compatibility (EMC) standards

The instrument design and manufacture complies with the following standards and requirements for safety and electromagnetic compatibility.

Safety standards

Reference	Description
EU Directive 2014/35/EU	European Union “Low Voltage Directive”
IEC 61010-1 EN 61010-1 UL 61010-1 CAN/CSA C22.2 No. 61010-1	*Safety requirements for electrical equipment for measurement, control, and laboratory use – Part 1: General requirements*
IEC 61010-2-010 EN 61010-2-010	*Safety requirements for electrical equipment for measurement, control and laboratory use – Part 2-010: Particular requirements for laboratory equipment for the heating of materials*
IEC 61010-2-081 EN 61010-2-081	*Safety requirements for electrical equipment for measurement, control and laboratory use – Part 2-081: Particular requirements for automatic and semi-automatic laboratory equipment for analysis and other purposes*

EMC standards

Reference	Description
EU Directive 2014/30/EU	European Union "EMC Directive"
EN 61326-1 IEC 61326-1	*Electrical Equipment for Measurement, Control and Laboratory Use – EMC Requirements – Part 1: General Requirements*
FCC Part 15.225 (47 CFR)	*Operation within the band 13.110-14.010 MHz.*
AS/NZS CISPR 11	*Limits and Methods of Measurement of Electromagnetic Disturbance Characteristics of Industrial, Scientific, and Medical (ISM) Radiofrequency Equipment*

(continued)

Reference	Description
ICES-001, Issue 4	*Industrial, Scientific and Medical (ISM) Radio Frequency Generators*
FCC Part 15 Subpart B (47 CFR)	*U.S. Standard Radio Frequency Devices* This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

Environmental design standards

Reference	Description
Directive 2012/19/EU	European Union "WEEE Directive"—Waste electrical and electronic equipment
Directive 2011/65/EU	European Union "RoHS Directive"—Restriction of hazardous substances in electrical and electronic equipment

Radio compliance standards

Reference	Description
Directive 2014/53/EU	European Union "RE Directive"—Radio equipment
RFID	FCC Notice (for U.S. Customers): This device complies with Part 15 of the FCC Rules: Operation is subject to the following conditions: 1. This device may not cause harmful interference, and 2. This device must accept any interference received, including interference that may cause undesired operation. Changes and modifications not expressly approved by Thermo Fisher Scientific can void your authority to operate this equipment under Federal Communications Commissions rules.

(continued)

Reference	Description
RFID	Canada (English) CAN ICES-3(A) / NMB-3(A): This device complies with Industry Canada license-exempt RSS standards. Operation is subject to the following two conditions: (1) this device may not cause interference, and (2) this device must accept any interference, including interference that may cause undesired operation of the device.
RFID	Canada (Français) CAN ICES-3(A) / NMB-3(A): Le présent appareil est conforme aux CNR d'Industrie Canada applicables aux appareils radio exempts de licence. L'exploitation est autorisée aux deux conditions suivantes : (1) l'appareil ne doit pas produire de brouillage, et (2) l'utilisateur de l'appareil doit accepter tout brouillage adioélectrique subi, même si le brouillage est susceptible d'en compromettre le fonctionnement.
Radio regulatory approval information	Singapore—Complies with IMDA Standards DA105282
	Japan—Complies with MIC Radio Laws 本装置は、総務省指定第 AC-19117 号の型式指定を受けた誘導式読み書き通信設備を内蔵しています。 This instrument has a built-in the inductive read/write communication equipment specified by the Ministry of Internal Affairs and Communications designated number AC-19117. 

Chemical safety



WARNING! GENERAL CHEMICAL HANDLING. To minimize hazards, ensure laboratory personnel read and practice the general safety guidelines for chemical usage, storage, and waste provided below. Consult the relevant SDS for specific precautions and instructions:

- Read and understand the Safety Data Sheets (SDSs) provided by the chemical manufacturer before you store, handle, or work with any chemicals or hazardous materials. To obtain SDSs, see the "Documentation and Support" section in this document.
- Minimize contact with chemicals. Wear appropriate personal protective equipment when handling chemicals (for example, safety glasses, gloves, or protective clothing).
- Minimize the inhalation of chemicals. Do not leave chemical containers open. Use only with sufficient ventilation (for example, fume hood).
- Check regularly for chemical leaks or spills. If a leak or spill occurs, follow the manufacturer cleanup procedures as recommended in the SDS.
- Handle chemical wastes in a fume hood.
- Ensure use of primary and secondary waste containers. (A primary waste container holds the immediate waste. A secondary container contains spills or leaks from the primary container. Both containers must be compatible with the waste material and meet federal, state, and local requirements for container storage.)
- After emptying a waste container, seal it with the cap provided.
- Characterize (by analysis if needed) the waste generated by the particular applications, reagents, and substrates used in your laboratory.
- Ensure that the waste is stored, transferred, transported, and disposed of according to all local, state/provincial, and/or national regulations.
- **IMPORTANT!** Radioactive or biohazardous materials may require special handling, and disposal limitations may apply.

Biological hazard safety



WARNING! BIOHAZARD. Biological samples such as tissues, body fluids, infectious agents, and blood of humans and other animals have the potential to transmit infectious diseases. Conduct all work in properly equipped facilities with the appropriate safety equipment (for example, physical containment devices). Safety equipment can also include items for personal protection, such as gloves, coats, gowns, shoe covers, boots, respirators, face shields, safety glasses, or goggles. Individuals should be trained according to applicable regulatory and company/ institution requirements before working with potentially biohazardous materials. Follow all applicable local, state/provincial, and/or national regulations. The following references provide general guidelines when handling biological samples in laboratory environment.

- U.S. Department of Health and Human Services, *Biosafety in Microbiological and Biomedical Laboratories (BMBL)*, 6th Edition, HHS Publication No. (CDC) 300859, Revised June 2020
  https://www.cdc.gov/labs/pdf/CDC-BiosafetymicrobiologicalBiomedicalLaboratories-2020-P.pdf
- Laboratory biosafety manual, fourth edition. Geneva: World Health Organization; 2020 (Laboratory biosafety manual, fourth edition and associated monographs)
  www.who.int/publications/i/item/9789240011311

Documentation and support

Related documentation

Document	Publication number
RapidLINK™ Software v1.3.3 User Guide	MAN0018939
RapidHIT™ ID System v1.3.3 User Guide	MAN0025746
RapidHIT™ ID System v1.3.1 User Guide	MAN0018938
RapidINTEL™ Sample Cartridge for Blood and Saliva Samples Validation User Bulletin	MAN0018979
GlobalFiler™ Express PCR Amplification Kit User Guide	4477672
AmpFℓSTR™ NGM SElect™ Express PCR Amplification Kit User Guide	4474109
Application Note: Bone sample processing on the RapidHIT™ ID system with RapidINTEL™ cartridges	COL33625 0620
Applied Biosystems RapidHIT™ ID System v1.3 with RapidHIT™ ID ACE GlobalFiler™ Express Sample Cartridges Validation Technical Note	**https://assets.thermofisher.com/TFS-Assets/GSD/Technical-Notes/rapidhit-id-validation-technical-note.pdf**

Customer and technical support

For support:

- **In North America**—Send an email to **HIDTechSupport@thermofisher.com**, or call **888-821-4443 option 1**.
- **Outside North America**—Contact your local support office.

For the latest services and support information for all locations, go to **thermofisher.com/support** to obtain the following information.

- Worldwide contact telephone numbers
- Product support
- Order and web support
- Safety Data Sheets (SDSs; also known as MSDSs)

Additional product documentation, including user guides and Certificates of Analysis, are available by contacting Customer Support.

Limited product warranty

Life Technologies Corporation and/or its affiliate(s) warrant their products as set forth in the Life Technologies' General Terms and Conditions of Sale at **www.thermofisher.com/us/en/home/global/terms-and-conditions.html**. If you have any questions, please contact Life Technologies at **www.thermofisher.com/support**.